

PERICARDIUM

The pericardium contains the heart and the juxtacardiac parts of its great vessels. It consists of two components – the fibrous and the serous pericardia – with a normal combined thickness of 1–2 mm, as frequently seen on cross-sectional imaging (Figs 57.1–57.2). The fibrous pericardium is a sac made of tough connective tissue, completely surrounding and unattached to the heart. It develops through a sequential process of cavitation of the embryonic body wall by expansion of the secondary pleural cavity; its lateral walls are thus clothed by parietal mediastinal pleura. The serous pericardium consists of two layers, one inside the other; the inner (visceral) serosal layer adheres to the heart and forms its outer covering, known as the epicardium, whereas the outer (parietal) serosal layer adheres to the internal surface of the fibrous pericardium. These two serosal surfaces are apposed and separated by a film of fluid, which allows independent movement of the inner membrane and its adherent heart, constituting two parietovisceral lines of serosal reflection. The separation of the two membranes of the serous pericardium creates a narrow pericardial cavity that provides complete cleavage between the heart and its surroundings, allowing freedom of cardiac motion and shape change.

The multifunctionality of the pericardium has been suggested to include facilitation of secure intrathoracic cardiac stability (via the sternopericardial ligaments and from the fusion of fibrous pericardium with the central diaphragmatic tendon and great vessel adventitia; see below); friction reduction between the heart and the mediastinal structures; limitation of cardiac distension and chamber dilation with prevention of hypertrophy; uniform cardiac distribution of pericardial cavity hydrostatic pressure, preventing ventriculoatrial regurgitations



Fig. 57.1 The effect of section thickness on visualization of normal pericardium. Multidetector CT image of the chest obtained from a dual-source scanner (Somatom Definition Flash; Siemens, Erlangen, Germany) with 1.0 mm collimation. Non-ionic contrast material (120 ml) was injected at a rate of 5 ml/sec. In this image with a section thickness of 10 mm, the pericardium appears as a thin, curvilinear structure (arrows). The pericardium is typically best visualized along the right side of the heart because of surrounding fat. With increasing section thickness, maximal pericardial thickness (measured along the right ventricle) increases from 0.9 mm to 1.0 and 1.6 mm by using section thicknesses of 3 mm and 10 mm, respectively. (With permission from Bogaert J, Francone M. 2013. Pericardial disease: value of CT and MR imaging. *Radiology* 267:340–356.)

under increased ventricular end-diastolic pressure; formation of a barrier to spread of infection; facilitation of atrial filling by promoting negative pericardial pressure during ventricular systole; and, in association with the pleural fluid, the formation of a system that automatically applies compensating hydrostatic pressures to the cardiac periphery when accelerating forces act.

FIBROUS AND SEROUS PERICARDIUM**Fibrous pericardium**

The fibrous pericardium is a roughly conical, compact, collagenous cardiac cloak composed mainly of three collagen layers, cross-woven at 120° to each other, which limits stretch and ensures a physical barrier to disease.

A three-point 'cardiac seat belt' anchors the pericardium securely, maintaining the intrathoracic cardiac disposition. Pericardial anchoring is obtained by superior continuity with the great vessel adventitia and pretracheal fascia, inferior attachment to the central diaphragmatic tendon and a small adjacent muscular area to its left, together with anterior attachment to the posterior sternal surface via the superior and inferior sternopericardial ligaments (the extent of these 'ligaments' is very variable; the superior is often undetectable).

Anteriorly, the fibrous pericardium is separated from the thoracic wall by the lungs and pleural coverings. However, in a small area behind the lower left half of the body of the sternum and the sternal ends of left fourth and fifth costal cartilages, the pericardium is in direct contact with the thoracic wall. Until it regresses, the lower end of the thymus is also anterior to the upper pericardium. The principal bronchi, oesophagus, descending thoracic aorta and posterior parts of the mediastinal surface of both lungs are all posterior relations. Laterally are the pleural coverings of the mediastinal surface of the lungs. The phrenic neurovascular bundle descends bilaterally between the fibrous pericardium and mediastinal pleura. Inferiorly, the pericardium is separated

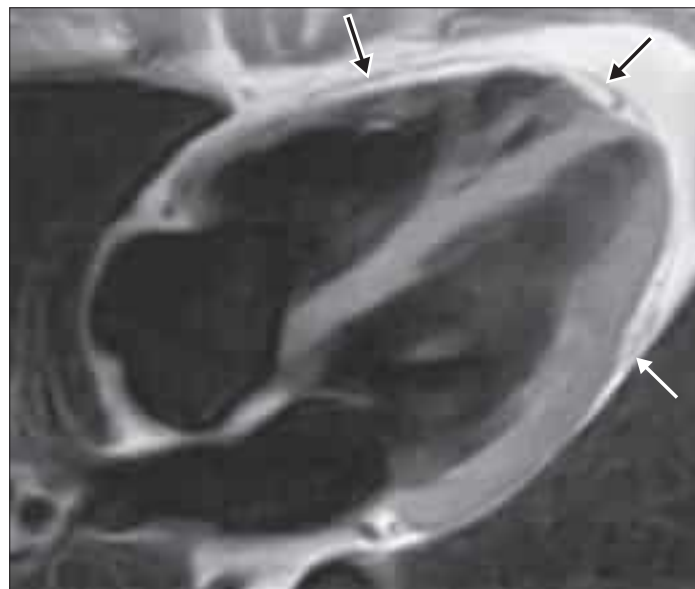


Fig. 57.2 Normal pericardium. T1-weighted fast spin-echo magnetic resonance image (MRI) (1.4×2.0 mm resolution) in horizontal long-axis plane, in which the pericardium appears as a thin, curvilinear hypointense structure (arrows), best visible when surrounded by fatty tissue. (With permission from: Bogaert J, Francone M. 2013. Pericardial disease: value of CT and MR imaging. *Radiology* 267:340–356.)

from the liver and gastric fundus by the diaphragm. The aorta, superior vena cava and the pulmonary arteries and veins all receive extensions of the fibrous pericardium but the inferior vena cava traversing the central tendon has no such covering.

Serous pericardium

The serous pericardium is a closed sac within the fibrous pericardium and has visceral and parietal layers. The parietal layer forms the internal surface of the fibrous pericardium. The visceral layer, or epicardium, covers the heart and great vessels, and is reflected into the parietal layer, which lines the internal surface of the fibrous pericardium. The visceral layer of the serous pericardium is a single layer composed of ciliated mesothelial cells on a thin subserosal layer of connective tissue abundant in elastic fibres, blending with the fibrous pericardium at the parietal membrane and with the interstitial myocardial tissue at the visceral membrane. The serosal cilia increase the surface area for fluid production and resorption, in addition to reducing friction between the opposing pericardial layers. The cardiac subserosal layer contains fat, especially along the ventricular side of the atrioventricular and inter-ventricular grooves and the inferior cardiac border. The main coronary vessels and their larger branches are embedded in this fat; the amount is related to the general extent of body fat and gradually increases with age. The amount of epicardial adipose tissue may be a cardiometabolic risk factor, with increasing thickness linked to metabolic syndrome and diabetes mellitus. A significant correlation has been demonstrated

between the thickness of this fat layer and obesity in children (Ozdemir et al 2010).

Pericardial reflections The reflections of the serosal layer are arranged as two complex 'tubes'; the aorta and pulmonary trunk are enclosed in one, and the venae cavae and four pulmonary veins lie in the other. The perivenous tube is an inverted J; the cul-de-sac within its curve posterior to the left atrium is termed the oblique sinus. The transverse sinus is a passage between the two pericardial 'tubes' (Fig. 57.3); the aorta and pulmonary trunk are anterior (Fig. 57.4A,C), and the atria and their great veins are posterior (Fig. 57.4B,D).

Pericardial space and fluid The pericardial cavity is the space formed between the parietal and visceral layers of the serous pericardium. The pericardial cavity covers the proximal part of the great vessels and the vast majority of the heart, extending between the left atrium and the descending thoracic aorta; the left pleural space extends posterior to the descending aorta. The pericardial cavity normally contains approximately 15–20 ml of serous plasma ultrafiltrate.

VASCULAR SUPPLY AND LYMPHATIC DRAINAGE

Approximately 80% of the pericardial blood supply is derived from the right and left internal thoracic arteries by way of their pericardiophrenic branches. Inferiorly, the pericardium is supplied by small

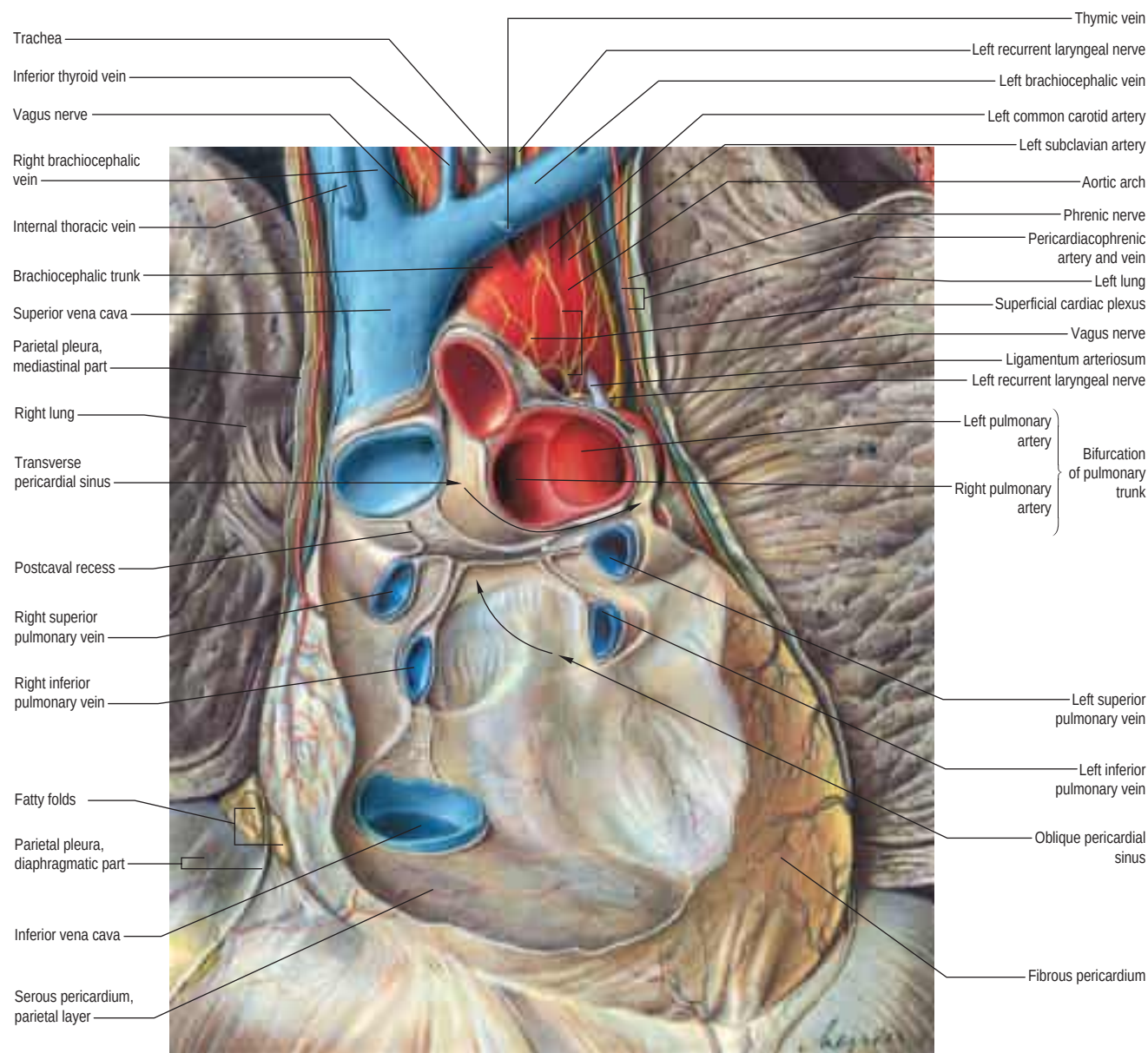


Fig. 57.3 Interior of the serosal pericardial sac after section of the large vessels at their cardiac origin and removal of the heart, anterior view. Refer to the text for additional named recesses of the general serous pericardial cavity and its transverse sinus. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

The arrangement of the oblique and transverse sinuses, along with that of the main 'principal' cavity, is further affected by the development of complex three-dimensional pericardial recesses between adjacent structures, grouped according to the siting of their orifices or 'mouths'. From the principal pericardial cavity, the postcaval recess projects towards the left, behind the atrial termination of the superior vena cava, limited superiorly by the right pulmonary artery and inferiorly by the superior right pulmonary vein, with its mouth opening superolaterally to the right. The right and left pulmonary venous recesses each project superomedially on the posterior surface of the left atrium between the superior and inferior pulmonary veins on each side, indenting the side walls of the oblique sinus. The superior aortic recess extends from the transverse sinus; via its inferior mouth, it ascends posterior to, then to the right of, the ascending aorta and ends at the level of the sternal angle. The inferior aortic recess, also extending from the transverse sinus, is a diverticulum that descends from a superiorly located mouth to run between the inferior ascending part of the aorta and the right atrium. The left pulmonary recess, with its mouth under the vena caval fold, passes to the left between the inferior aspect of the left pulmonary artery and the superior border of the superior left pulmonary vein. The right pulmonary recess lies between the inferior surface of the proximal part of the right pulmonary artery and the superior border of the left atrium.

A triangular fold of serous pericardium is reflected from the left pulmonary artery to the subjacent superior left pulmonary vein as the fold of the left superior vena cava. It contains a fibrous ligament, a remnant of the obliterated left common cardinal vein (left duct of Cuvier), which descends anterior to the left pulmonary hilum from the superior part of the left superior intercostal vein to the posterior surface of the left atrium, where it is continuous with the oblique vein of the left atrium. The left common cardinal vein may persist as a left superior vena cava, which then replaces the oblique vein of the left atrium and empties into the coronary sinus. When both common cardinal veins persist as a double superior vena cava, the transverse anastomosis between them, which normally forms the left brachiocephalic vein, may be small or absent. When there is a left superior vena cava, it is joined by the left superior intercostal vein. Knowledge of these recesses and sinuses is important in the avoidance of misdiagnosis as lymphadenopathy or aortic dissection and in surgical control of the great vessels.

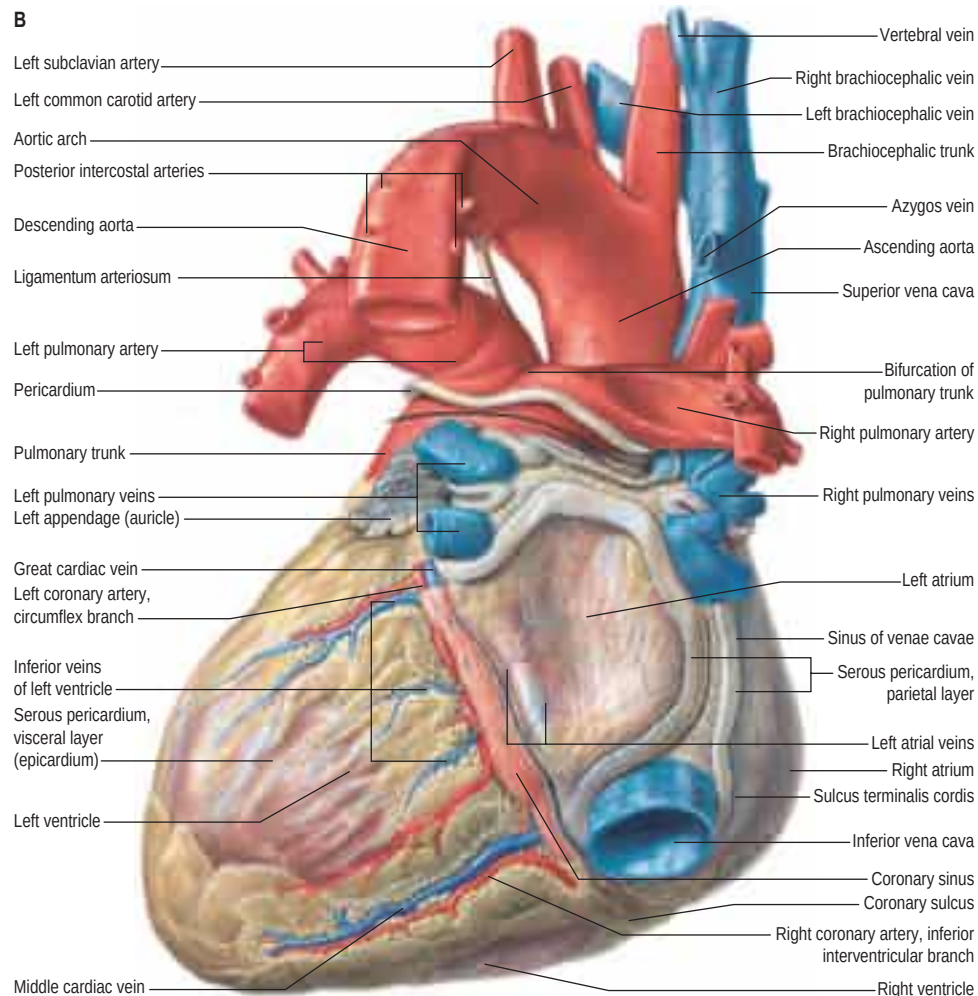
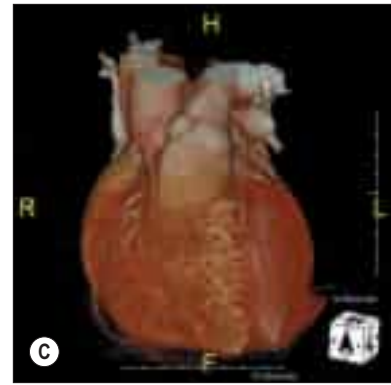
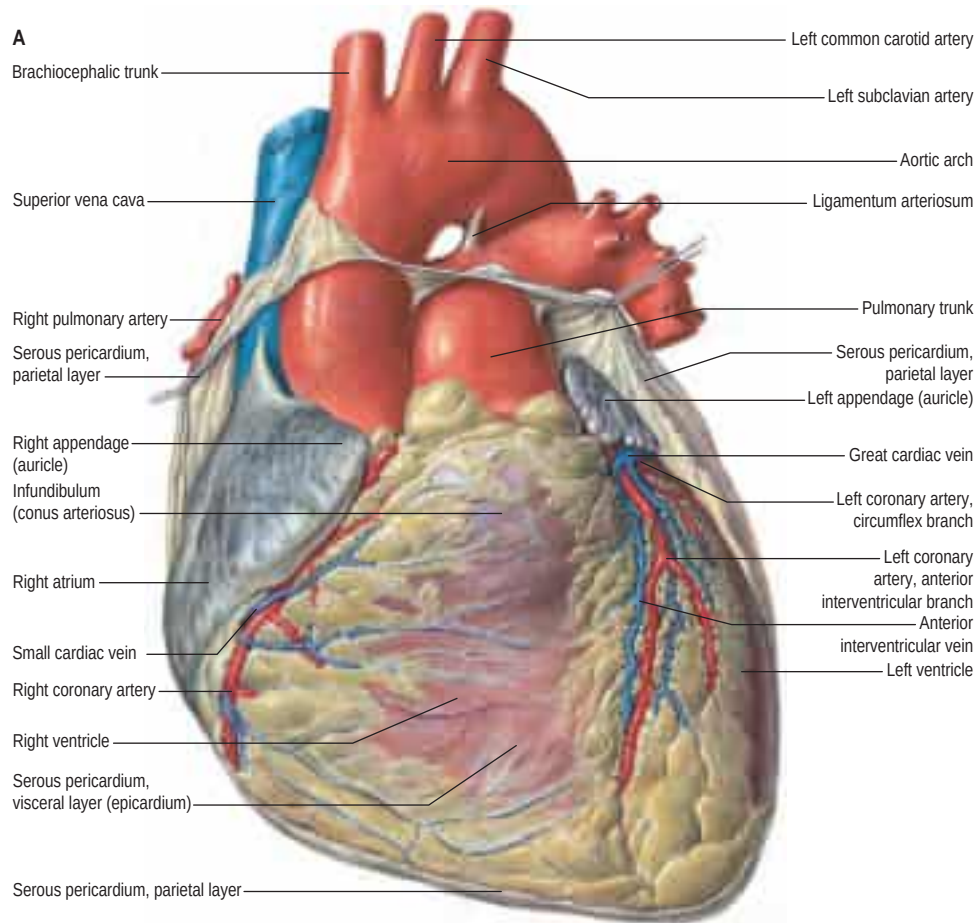


Fig. 57.4 The heart and great vessels. **A**, Anterior view. **B**, Posteroinferior view. **C,D**, Corresponding three-dimensional reconstructions from multislice computed tomography (CT) scanning. (A,B, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

branches from the intercostal and superior phrenic arteries. The posterior portion of the pericardium is supplied from small mediastinal branches that are derived either directly from the descending aorta or from its oesophageal or bronchial branches. The veins are tributaries of the azygos system, internal thoracic and superior phrenic veins.

Most lymph drainage of the pericardium is to the thoracic and the right lymphatic ducts; bilateral upper mediastinal and parasternal internal thoracic lymph nodal groups also receive efferents. Overall, a double layer of pericardial lymphatic vasculature exists: one contacting and surrounding the parietal pericardium, the other present in the fat and loose areolar tissue. The sternocostal pericardium either drains laterally towards the phrenic nerves as they enter the diaphragm, or travels along the ventral border of the pericardium to enter the prepericardial nodes located at the pericardio-diaphragmatic junction. Drainage from the lateral pericardium is more variable. Inferiorly, it drains towards the phrenic nerves and the pericardiophrenic vessels as they reach the diaphragm; superiorly, it drains into the tracheobronchial and paratracheal lymph nodes. The posterior pericardium, including the posterior aspect of the cupula and extending as far as the pulmonary veins, drains superiorly to the superior and inferior tracheobronchial lymph nodes. The portion of the pericardium that adheres to the diaphragm drains via short channels to the lymph nodes at the right border of the inferior vena caval opening in the central diaphragmatic tendon.

INNERVATION

Pain from parietal pericardium is mainly transmitted by the phrenic nerves, while pain from serous pericardium is mediated via sympathetic fibres that arise from the cervical, upper thoracic and stellate ganglia and travel to the aortic and cardiac plexuses. The pericardium also receives vagal fibres via the oesophageal plexus and left recurrent laryngeal nerve (see Fig. 57.61). Pericardial pain is typically sharp, severe and substernal, typically exacerbated by lying back or on the left side and relieved by leaning forwards, occasionally radiating to the superior border of trapezius.

CARDIAC TAMPONADE



Available with the *Gray's Anatomy* e-book

CONGENITAL ANOMALIES OF THE PERICARDIUM



Available with the *Gray's Anatomy* e-book

HEART

The heart has a complicated, spiral, three-dimensional organization that is markedly skewed when compared with the planes of the body. Established anatomical orientation terms have been historically applied to the heart based on early embryological development before axial rotation of the cardiovascular tube. This, together with the traditional study of isolated whole or dissected hearts outside the body, hinders intuitive understanding of the descriptive relations of *in vivo* and surface cardiac anatomy. The following description emphasizes such difficulties in order to circumvent certain misconceptions, before proceeding to an account of more detailed structure. The oblique position of the heart in the thorax may be conceptualized by comparing it to a rather deformed pyramid, with the base facing posteriorly and to the right, and the apex anteriorly and to the left. A line from the apex to the approximate centre of the base projected posterolaterally emerges near the right mid-scapular line. Some surfaces of the cardiac 'pyramid' are flat and others more or less convex, these aspects merging along rather ill-defined 'borders'. Precise definitions of surfaces and intervening 'borders' are therefore difficult. In the account that follows, the official *Terminologia Anatomica* (2011) and more generally used terms from clinical practice are given as alternatives.

The heart is described as having a base and apex; its surfaces are designated as sternocostal (anterior), diaphragmatic (inferior), and right and left (pulmonary) (Fig. 57.9). Its borders are termed upper, inferior ('acute' margin or border) and left ('obtuse' margin or border). The right surface is sometimes called a 'border', despite its extent. Although the heart is placed obliquely in the thorax, the atrial and ventricular septal

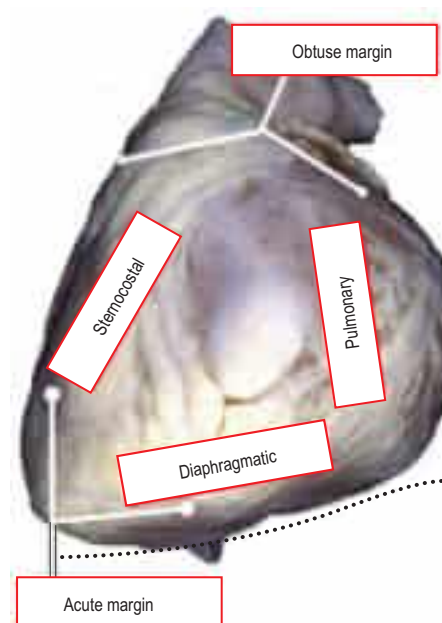


Fig. 57.9 A cadaveric heart, removed from the body and placed in an anatomically appropriate position, then viewed from the apex of the left ventricle. The ventricular mass takes the form of a squashed cone with anterior, inferior and posterior borders. The labels show the location of these borders relative to the thoracic structures, and illustrate the acute and obtuse angles between the borders. (With permission from Anderson R, Loukas M. 2009. The importance of attitudinally appropriate description of cardiac anatomy. *Clin Anat* 22:47–51.)

structures are virtually in line, but inclined forwards and to the left at 45° to the sagittal plane. The planes of the mitral and tricuspid valves are vertical and not precisely co-planar; they are broadly at right angles to the septal plane. The right atrium, therefore, is not only to the right, but also anterior and inferior to the left atrium, and it is partly anterior to the left ventricle. The right ventricle forms most of the anterior aspect of the ventricular mass (see Fig. 57.12); only its inferior end is to the right of the left ventricle, while its upper left extremity (pulmonary orifice) is to the left and superior relative to the aortic valve. The left atrium forms most of the posterior aspect of the heart, whereas the left ventricle is only prominent inferiorly, running along the left margin to reach the apex. The atria are essentially right of and posterior to their respective ventricles. These general dispositions are of the greatest importance in planning or interpreting radiographs, CT and MRI studies, cardiac angiograms and echocardiograms (Fig. 57.10).

The right heart, while forming the right aspect or 'border', follows a gentle curve and covers most of the anterior aspect of the left heart (except for a left-sided strip including the apex). Thus the right heart forms the largest part of the anterior surface, its outflow tract ascending until it terminates on the left side of the outflow tract from the left ventricle. The sites of the tricuspid and pulmonary valves are widely separated and on different planes, and the sectionally crescentic flat cavity of the right ventricle splays out between them. Conversely, the left heart (except the left-sided strip mentioned above) is largely posterior in position and, when viewed from the front, is obscured by the chambers of the right heart. The inlet to the left ventricle, which contains the mitral valve, is very close to its outlet (the aortic valve), the two being embraced by the wide tract linking inlet and outlet components of the right ventricle. The planes of the left ventricular orifices, although relatively inclined, are more nearly co-planar than those of the right. The left ventricular cavity is narrow and conical, and its tip occupies the cardiac apex.

GENERAL ORGANIZATION

The heart is a fused pair of valved muscular pumps circulating blood in series through systemic ('left' heart) and pulmonary ('right' heart) circulations (see Fig. 57.4). The fibromuscular framework and conduction tissues of these physiologically separate pumps are structurally interwoven. Despite the functional disposition in series, the two pumps are topographically described in parallel. Of the four cardiac chambers, the two atria receive venous blood as weakly contractile reservoirs for final filling of the two ventricles, which then provide the

Cardiac tamponade is external compression of the heart, usually caused by accumulation of fluid in the pericardial space (pericardial effusion) (Fig. 57.5). This causes compression of the right atrium, reducing venous return and therefore cardiac output. It may be secondary to trauma, proximal extension from a dissecting aortic aneurysm, cardiac surgery or central venous catheter insertion. Patients develop hypotension, chest tightness, dyspnoea and possibly eventual shock. The typical signs of a cardiac tamponade include three elements (Beck's triad): a small quiet heart (due to fluid within the pericardial cavity), falling atrial pressure (decreased stroke volume) and rising jugular venous pressure (impaired cardiac venous return). Emergency treatment involves first relieving the tamponade by percutaneous pericardial aspiration (pericardiocentesis; see below), possibly followed by surgery to address the underlying cause. The volume of fluid may not be proportional to the degree of physiological circulatory impairment. Echocardiography is an important tool in diagnosis and is also useful in guiding percutaneous pericardial aspiration. Surgery is via a subxiphoid incision or a left anterior thoracotomy.

Pericardiocentesis

Pericardiocentesis is the aspiration of fluid from the pericardial cavity in cases of cardiac tamponade and for relief of symptomatic pericardial effusion. In addition, this technique can be used for the diagnosis of neoplastic effusions and purulent pericarditis, and for introducing cytolytic agents into the pericardial space. Ultrasound-guided parasternal pericardiocentesis has become the preferred elective procedure. With acute traumatic cardiac tamponade, the subxiphoid approach is used and aspiration of as little as 10–20 ml of blood may greatly improve cardiac output. A complete and controlled parasternal drainage avoids the diaphragm and phrenic nerve. Apical and abdominal approaches may also be used (Fig. 57.6).

Congenital anomalies of the pericardium include pericardial cysts and diverticula, and absence of the pericardium.

Pericardial cysts and diverticula

A pericardial cyst is a congenitally encapsulated cyst with no pericardial communication, which forms when a portion of pericardium is pinched off during early development. Pericardial cysts are most often incidental and are located at the right cardiophrenic angle or sulcus, where they present as a well-defined ovoid expansion of the right inferolateral cardiac border on a chest film, usually round or elliptical, thin-walled and sharply defined with fluid attenuation on CT or homogeneously high signal on fluid-sensitive MR sequences.

A pericardial diverticulum is a rare condition that corresponds to a herniation through a defect in the parietal pericardium that communicates with the pericardial cavity. It typically occurs at the same place as a pericardial cyst and should be suspected when a complete wall cannot be identified in all parts of the lesion (Fig. 57.7).

Congenital absence of the pericardium

Congenital absence of the pericardium is rare. The condition encompasses defects ranging from a small foramen to absence of the entire pericardium. The majority of defects occur on the left side but they can

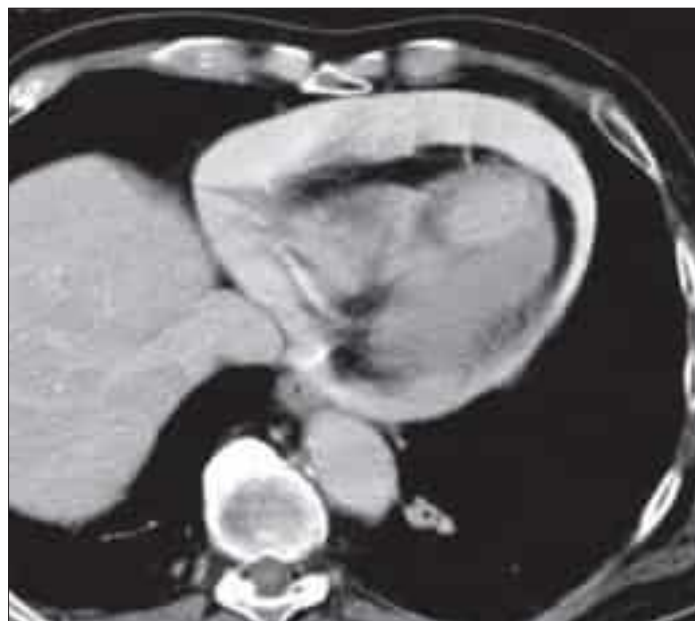


Fig. 57.5 Acute cardiac tamponade due to ruptured aortic dissection. An axial CT image acquired 60 seconds after contrast agent administration shows a large, high-attenuation pericardial effusion corresponding to massive haemopericardium. Weakening of the aortic wall with subsequent rupture has led to massive extravasation of contrast agent into the pericardial sac. (Image courtesy of GF Gualdi MD, and C Valentini MD, DEA Policlinico Umberto I, Rome, Italy, with permission from Bogaert J, Francone M 2013 Pericardial disease: value of CT and MR imaging. *Radiology* 267:340–356.)

be located anywhere in the pericardium, and are three times more common in males. It has been suggested that failure of pericardial development results from premature atrophy of the cardinal vein and hence poor nourishment of the left pleuropericardial membrane. The most common defect is complete absence of the left side of the pericardium. When this occurs, lung tissue is interposed between the main pulmonary artery and aorta. Bulging of the left atrial appendage through the defect will cause ischaemic necrosis and compression of the left coronary artery, and the heart rotates to the left. Complete absence of the pericardium is usually asymptomatic. Partial defects may cause complications such as herniation of cardiac structures (Fig. 57.8).

Congenital pericardial defects are associated with other congenital abnormalities such as bronchogenic cysts, ventricular and atrial septal defects, patent ductus arteriosus, mitral stenosis, bicuspid aortic valve, pulmonary arteriovenous malformations, diaphragmatic hernia, midline closure defects or bronchopulmonary sequestration (Hutchison 2009). Pentology of Cantrell is a rare congenital abnormality consisting of absence of the diaphragmatic portion of the pericardium; midline closure defects, including sternal and abdominal wall defects; anterior diaphragmatic defect; and congenital heart defects.

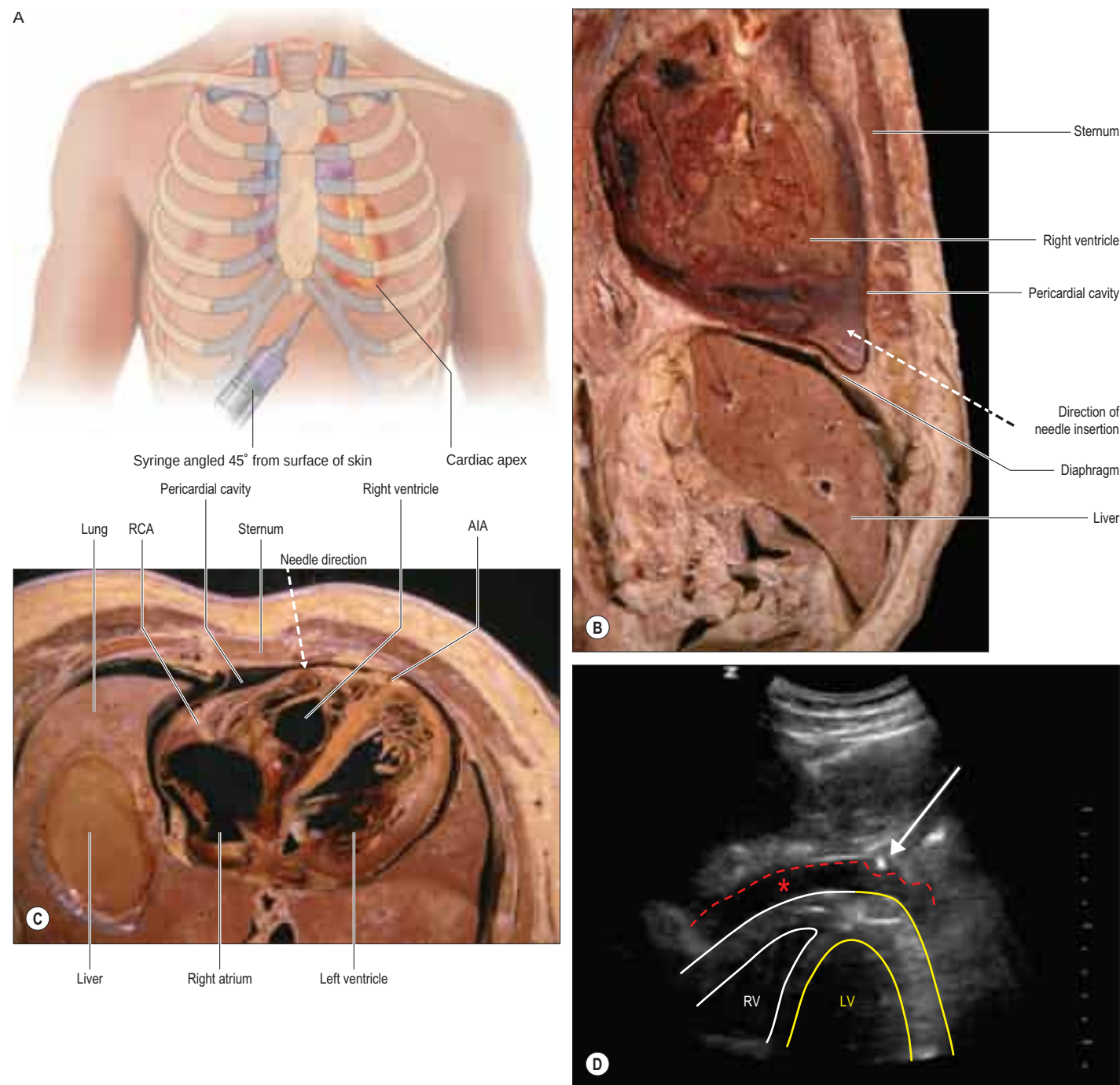
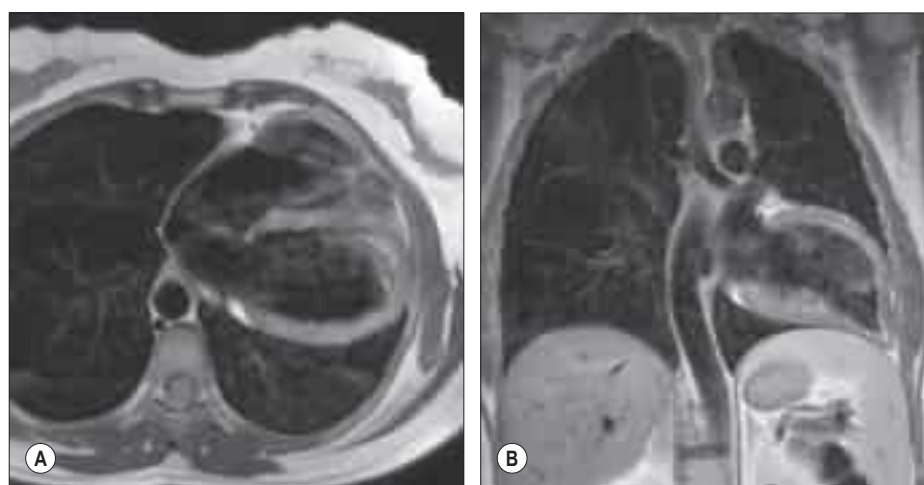
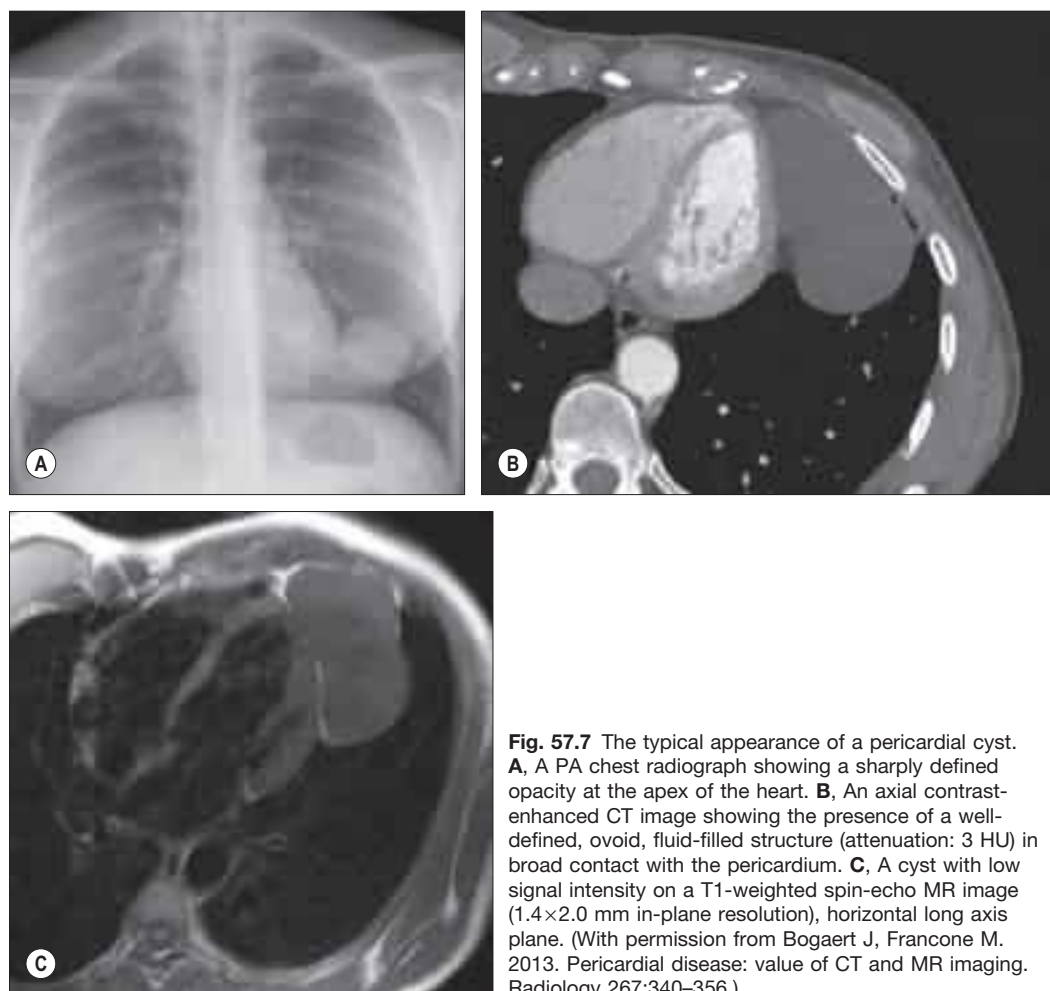


Fig. 57.6 **A**, The subxiphoid approach in pericardiocentesis. The needle is inserted into the angle between the xiphisternum and the left costal margin at about 45° to the anatomical transverse plane. **B**, A mid-sagittal section of a cadaver. The broken arrow shows the direction of the pericardiocentesis needle. **C**, A cross-section of the lower part of the thorax to show the relationships of the pericardial cavity with adjacent structures. The broken arrow shows the direction of the pericardiocentesis needle. Abbreviations: AIA, anterior interventricular artery; RCA, right coronary artery. **D**, Bedside cardiac ultrasound performed in a 62-year-old male, demonstrating a large pericardial effusion consistent with cardiac tamponade (red asterisk). The broken red line shows the extended pericardial cavity. An 18-gauge 3.5 inch needle was attached to a 60 ml syringe and advanced between the xiphoid process and left costal margin at a 45° angle, directed towards the left shoulder (arrow). The white line demarcates the borders of the right ventricular apical wall and right ventricular cavity (RV); the yellow line depicts the left ventricular apical wall and left ventricular cavity (LV). (With permission from Loukas M, Walters A, Boon JM, Welch TP, Meiring JH, Abrahams PH. 2012. Pericardiocentesis: a clinical anatomy review. Clin Anat 25:872–881.)



The basic rules of cardiac anatomy dictate that structures within the heart should be described relative to the bodily coordinates rather than to the heart itself. In other words, the structures within the heart should be described as the rest of the body when viewed in the anatomical position. An important concept to apply is recognition of the diaphragmatic surface as being inferior, and the left atrium as being posterior. A typical example of not applying this concept is the posterior interventricular artery. In cardiological and older anatomical texts, this artery is described as being posterior and descending, whereas in reality it is neither, since it is interventricular and inferior. Moreover, blockage of the artery is well recognized as producing inferior myocardial infarc-

tion. Another example is the papillary muscles that support the leaflets of the mitral valve. These were conventionally described as being anterolateral and posteroseptal, reflecting the fact that the heart was placed on its apex in order to describe its component parts. However, examination in left anterior oblique orientation, as for the inferior interventricular artery, shows that, in reality, the so-called posteroseptal muscle is anterior to its partner, despite the fact that the latter is currently described as being anterolateral. Similarly, the posterior interventricular sulcus is situated inferiorly rather than posteriorly (Fig. 57.11; see Figs 57.9, 57.10).

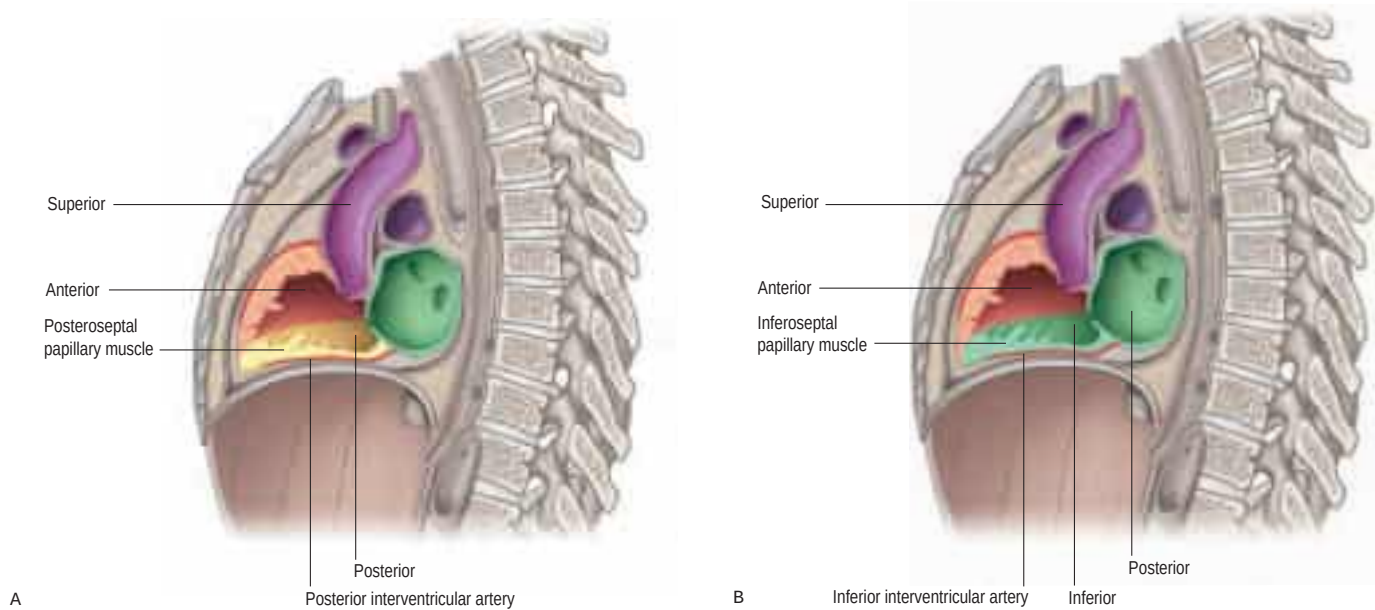


Fig. 57.11 The fundamental differences between describing the heart according to the conventional (A) or the attitudinally correct (B) orientation. (With permission from Anderson R, Spicer D, Hlavacek A, Hill A, Loukas M. 2013. Describing the cardiac components—attitudinally appropriate nomenclature. J Cardiovasc Transl Res. 6:118–123.)

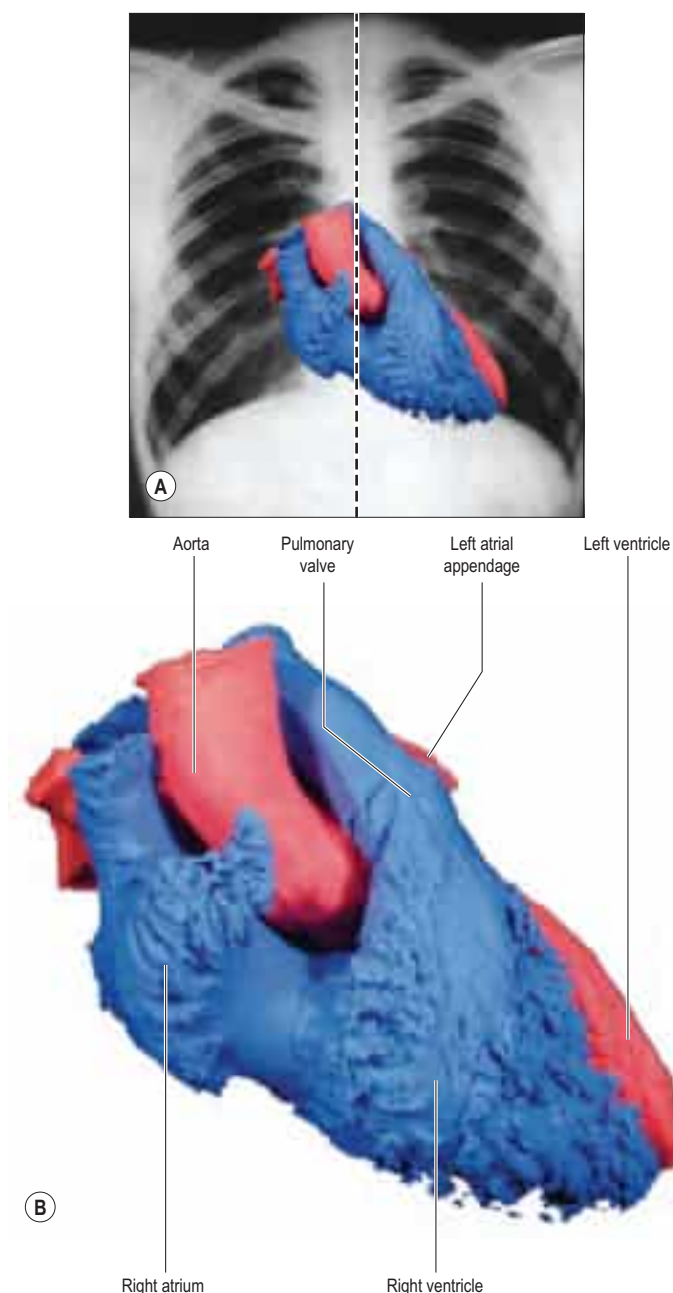


Fig. 57.10 A cast of the so-called right and left chambers of the heart, cast in blue and red, respectively. **A**, The cast superimposed on a chest radiograph from a normal individual. The broken line indicates the midline of the body. The long axis of the heart extends from the right shoulder towards the left hypochondrium. The adult heart is usually positioned within the mediastinum such that one-third is to the right of the midline and two-thirds to the left. **B**, The cast enlarged to show the chambers that form the various borders of the frontal cardiac silhouette. (With permission from: Anderson R, Loukas M. 2009. The importance of attitudinally appropriate description of cardiac anatomy. Clin Anat 22:47–51.)

powerful expulsive contraction that forces blood into the main arterial trunks.

The 'right heart' starts at the right atrium, which receives the superior and inferior venae cavae and the main myocardial venous inflow via the coronary sinus. This systemic venous blood traverses the right atrioventricular orifice, guarded by the tricuspid valve, to enter the inlet component of the right ventricle. Contraction of the ventricle, particularly its apical trabecular component, closes the tricuspid valve and, with increasing pressure, ejects the blood through the muscular right ventricular outflow tract into the pulmonary trunk and then through the relatively low-resistance pulmonary vascular bed. Changes in pressure, time relations and valvular events are described below. Many structural features of the 'right heart', including its overall geometry, myocardial architecture, and the construction and relative strengths of the tricuspid and pulmonary valves, accord with this low resistance, being associated with comparatively low changes in pressure.

The 'left heart' starts at the left atrium, which receives all the pulmonary inflow of oxygenated blood and some coronary venous inflow. It contracts to fill the left ventricle through the left atrioventricular orifice, guarded by its mitral valve. Left ventricular contraction rapidly increases the pressure in the apical trabecular component, closing the mitral valve and opening the aortic valve, enabling the ventricle to eject via the left ventricular outflow tract into the aortic sinuses and the ascending aorta, and thence to the entire systemic arterial tree, including the coronary arteries. This vast vascular bed presents a high peripheral resistance that, with large metabolic demands (especially the sustained requirements of the cerebral tissues), explains the more massive structural organization of the 'left heart'. The ejection phase of the left ventricle is shorter than that of the right but its fluctuations in pressure are very much greater.

The heart is therefore, by necessity, a functionally and structurally markedly asymmetrical pair of combined pumps, with both ventricles designed to deliver the same volume with each contraction into two very different resistive series circulations.

Cardiac size, shape and external features

The heart is a hollow, fibromuscular organ of a somewhat conical or pyramidal form, with a base, apex and a series of surfaces and 'borders'. Enclosed in the pericardium, it occupies the middle mediastinum between the lungs and their pleural coverings (see Fig. 57.3). It is placed obliquely behind the body of the sternum, adjoining costal cartilages and ribs. Approximately one-third of the mass lies to the right of the midline. An average adult heart is 12 cm from base to apex, 8–9 cm at its broadest transverse diameter and 6 cm anteroposteriorly. Its weight varies from 280 to 340 g (average 300 g) in males and from 230 to 280 g (average 250 g) in females. Cardiac weight is 0.45% of body weight in males and 0.40% in females. The difference in weight between males and females appears after the age of 12 years (Kortelainen 1997). Adult weight is achieved between the ages of 17 and 20 years. The heart reaches 50% of its adult dimensions at birth, 75% by 5 years and 90% by 12 years (Nidorf et al 1992).

Grooves on the cardiac surface

The division of the heart into four chambers produces boundaries that are visible externally as grooves (sulci). Some are deep and obvious, and contain prominent structures. Others are less distinct, even barely perceptible, and are sometimes obscured, in part, by the major structures that cross them.

The extensive interatrial groove (of Waterston) separates the two atria. The lateral limits are defined by the borders of the atria. The atrioventricular (coronary) groove separates the atria from the ventricles and contains the main trunks of the coronary arteries. It descends obliquely to the right on the sternocostal surface, separating the right atrium (and its auricle or appendage) from the oblique right margin of the right ventricle and its infundibulum. Superiorly, its left part is obliterated by the crossing of the pulmonary trunk, and posterior to this, the aorta from which the coronary arteries originate. Continuing to the left, the groove curves around the 'obtuse' margin and descends to the right, separating the atrial base from the diaphragmatic surface of the ventricles (see Fig. 57.4). This diaphragmatic part of the atrioventricular groove then curves around the 'acute' margin at its lower right end to become confluent with the sternocostal part. Thus, the groove passes from superior on the left to inferior on the right, with the diaphragmatic part being a little to the left of the sternocostal. A section that includes the atrioventricular groove is at 45° to the sagittal plane and at a greater but variable angle to the transverse and coronal planes. It approximately traverses the lines of attachment of the atrioventricular valves and, even less precisely, those of the aortic and pulmonary valves. A line at right angles to the centre of this plane will descend anteriorly and leftwards to the cardiac apex. Internally, the ventricles are separated by the septum; its mural margins correspond to the anterior and inferior (diaphragmatic) interventricular grooves. The anterior groove, seen on the sternocostal cardiac surface, is near and almost parallel to the left ventricular obtuse margin. On the diaphragmatic surface, the groove is closer to the midpoint of the ventricular mass. The interventricular grooves extend from the atrioventricular groove to the apical notch on the acute margin, the latter a little to the right of the true cardiac apex.

Cardiac base, apex, surfaces and borders

Posterior aspect of the heart The true cardiac base is somewhat quadrilateral, with curved lateral extensions. It faces posteriorly and to

the right, separated from the thoracic vertebrae (fifth to eighth in the recumbent, sixth to ninth in the erect posture) by the pericardium, right pulmonary veins, oesophagus and aorta. It is formed mainly by the left atrium, and only partly by the posterior part of the right atrium (see Fig. 57.4B,D). It extends superiorly to the bifurcation of the pulmonary trunk and inferiorly to the posterior part of the atrioventricular groove, which contains the coronary sinus and coronary arterial branches. It is limited to the right and left by the rounded surfaces of the corresponding atria, separated by the shallow interatrial groove. The point of junction of the atrioventricular, interatrial and posterior (inferior) interventricular grooves is termed the cardiac crux. Two pulmonary veins on each side open into the left atrial part of the base, whereas the superior and the inferior venae cavae open into the upper and lower parts of the right atrial basal region. The area of the left atrium between the openings of right and left pulmonary veins forms the anterior wall of the oblique pericardial sinus (see Fig. 57.3). This description of the anatomical base reflects the usual position of the heart in the thorax. Some confusion is produced by other current usages of the term 'base', often applied to the segment of the atrioventricular and ventriculoarterial junctions seen after dissections through the atrioventricular groove. This area is better termed the base of the ventricles. In clinical practice, auscultation in or near the parasternal parts of the second intercostal spaces is often described as occurring at the 'clinical base', to make the contrast with the 'clinical apex'. Such descriptions, while less than perfect anatomically, will almost certainly persist.

Anatomical apex of the heart This is the apex of the conical left ventricle, which is directed anteroinferiorly and to the left. It is overlapped by the left lung and pleura. The apex is located most commonly behind the fifth left intercostal space, near or a little medial to the mid-clavicular line.

Anterior, sternocostal surface of the heart Facing forwards and upwards, the anterior surface has an acute right and a more gradual left convexity (Fig. 57.12). It consists of an atrial area superior and to the right, and a ventricular part inferior and to the left of the atrioventricular groove. The atrial area is occupied almost entirely by the right

atrium. The left atrium is largely hidden by the ascending aorta and pulmonary trunk. Only a small part of the left atrial appendage projects forwards to the left of the pulmonary trunk. Of the ventricular region, about one-third is made up by the left and two-thirds by the right ventricle. The site of the septum between them is indicated by the anterior interventricular groove. The sternocostal surface is separated by the pericardium from the body of the sternum, the sternocostal muscles and the third to sixth costal cartilages. Because of the bulge of the heart to the left, more of this surface is behind the left costal cartilages than the right. It is also covered by the pleural membranes and the thin anterior edges of the lungs, except for a triangular area at the cardiac incisure of the left lung. The lungs and their pleural coverings are variable in their degree of cardiac overlap.

Inferior, diaphragmatic surface of the heart Largely horizontal, the inferior surface of the heart gently slopes anteroinferiorly towards the apex (see Fig. 57.4B,D). It is formed by the ventricles (chiefly the left) and rests mainly on the central tendon but also, apically, on a small area of the left muscular part of the diaphragm. It is separated from the anatomical base by the atrioventricular groove, and is traversed obliquely by the posterior (inferior) interventricular groove.

Left surface of the heart Facing posterosuperiorly to the left, the left surface consists almost entirely of the obtuse margin of the left ventricle, but a small part of the left atrium and its left atrial appendage contribute superiorly. Convex and widest superiorly, where it is crossed by the atrioventricular groove, it narrows to the cardiac apex. It is separated by the pericardium from the left phrenic neurovascular bundle, and by the left pleura from the deep concavity of the left lung.

Right surface of the heart The right surface is rounded and formed by the right atrial wall. It is separated from the mediastinal aspect of the right lung by the pericardium and pleural coverings. Its convexity merges inferiorly into the short intrathoracic part of the inferior vena cava and above into the superior vena cava. The sulcus terminalis is a prominent landmark separating the systemic atrial venous sinus from

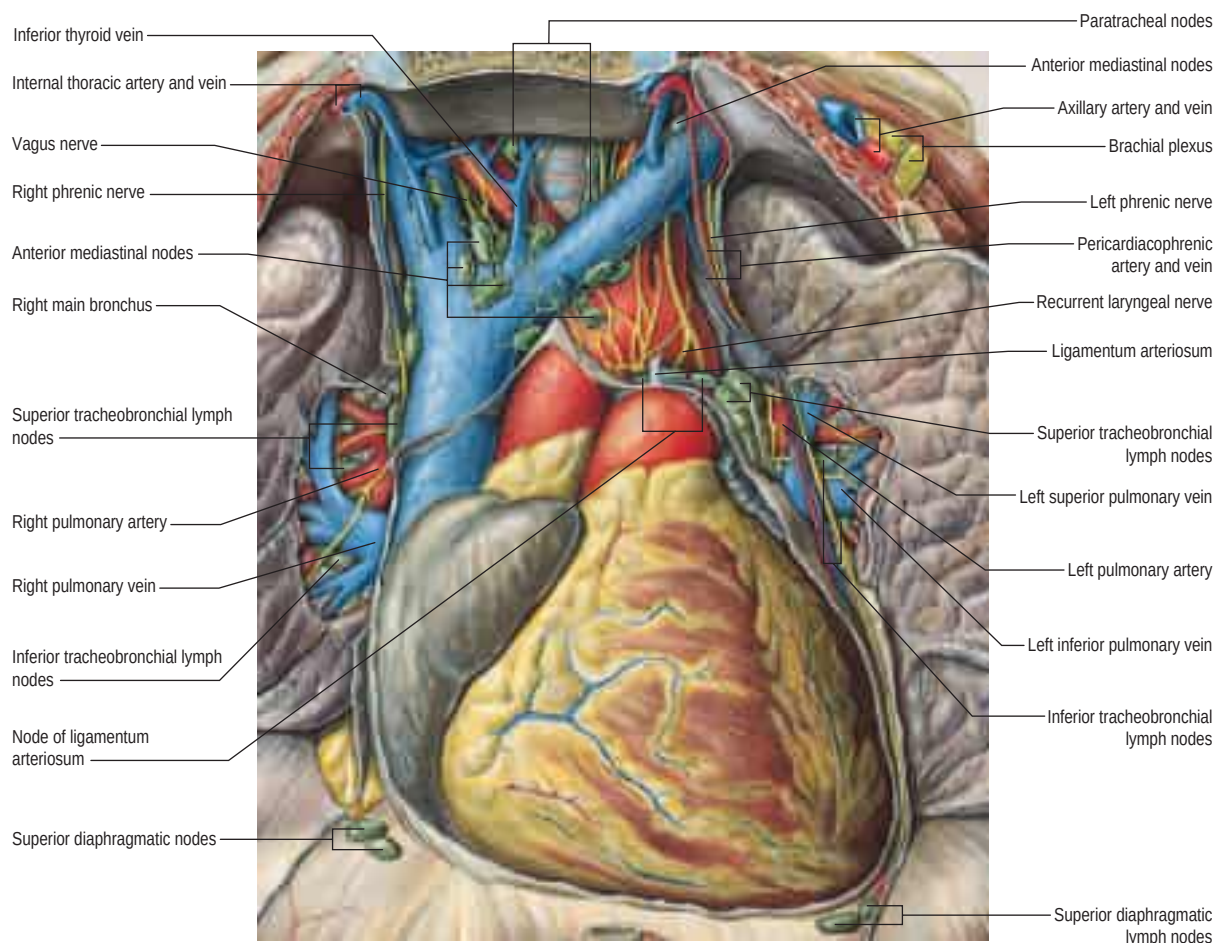


Fig. 57.12 The heart, great vessels and the lungs *in situ*. The manubrium sterni has been retracted cranially, the thymus has been completely removed and the pericardium has been partially removed. The hilum of the lung has been dissected to expose the tracheobronchial lymph nodes. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the appendage; it curves approximately along the junction of the sternocostal and right surfaces, corresponding to the position of the terminal crest.

Upper border of the heart This is atrial (mainly the left atrium). Anterior to it are the ascending aorta and the pulmonary trunk (see Fig. 57.3). At its extremity, the superior vena cava enters the right atrium.

Right border of the heart Corresponding to the right atrium, the profile of the right border is slightly convex to the right as it approaches the vertical.

Inferior border of the heart Also known as the acute margin of the heart, the inferior border is sharp, thin and nearly horizontal. It extends from the lower limit of the right border to the apex and is formed mainly by the right ventricle, with a small contribution from the left ventricle near the apex.

Left border of the heart Also known as the obtuse margin, the left border separates the sternocostal and left surfaces. It is round and mainly formed by the left ventricle but, to a slight extent superiorly, is formed by the left atrial appendage. It descends obliquely, convex to the left, from the appendage to the cardiac apex.

Right atrium

General and external features

The right atrium is both anterior and to the right of the left atrium (see Fig. 57.4), and extends inferior to it. Its walls form the right superior sternocostal surface, the convex right (pulmonary) surface and a little of the right side of the anatomical base. The superior vena cava opens

into its dome, and the inferior vena cava into its inferoposterior part (see Fig. 57.4A,B). An extensive muscular pouch, the appendage, projects anteriorly to overlap the right side of the ascending aorta; it is a broad, triangular, pyramidal structure and has a wide junction with the venous component of the atrium (see Fig. 57.4A,C). The junction between the venous part (sinus venosus) and the atrium proper is marked externally by a shallow groove, the sulcus terminalis, extending between the right sides of the openings of the two venae cavae. The sulcus terminalis corresponds, internally, to the terminal crest (crista terminalis), which is the site of origin of the extensive pectinate muscles that arise serially at right angles from the crest. Posteriorly, the vertical interatrial groove descends to the crux. Anteriorly, the right atrium is related to the anterior part of the mediastinal surface of the right lung, from which it is separated by pleura and pericardium. Laterally, the atrium is also related to the mediastinal surface of the right lung, but anterior to its hilum and separated from it by the pleura, right phrenic nerve and pericardiacophrenic vessels and pericardium. Posteriorly and to the left, the interatrial septum and the surrounding infolded atrial walls separate the atria (the mural infolding is indicated by an extensive interatrial groove). Posteriorly and to the right are the right pulmonary veins. Medially are the ascending aorta and, to a limited extent, the pulmonary trunk and its bifurcation.

Internal features

The interior surface of the right atrium can be divided into three regions: the smooth-walled venous component located posteriorly leads anteriorly to the vestibule of the tricuspid valve and to the appendage (Figs 57.13–57.14). The wall of the vestibule is smooth but its junction with the appendage is ridged all around the atrioventricular junction by pectinate muscles. The smooth-walled systemic venous sinus receives the openings of the venae cavae and coronary sinus; it originates from the sinus venosus of the embryonic heart. The trabeculated appendage is derived from the embryonic atrium proper. The superior and inferior

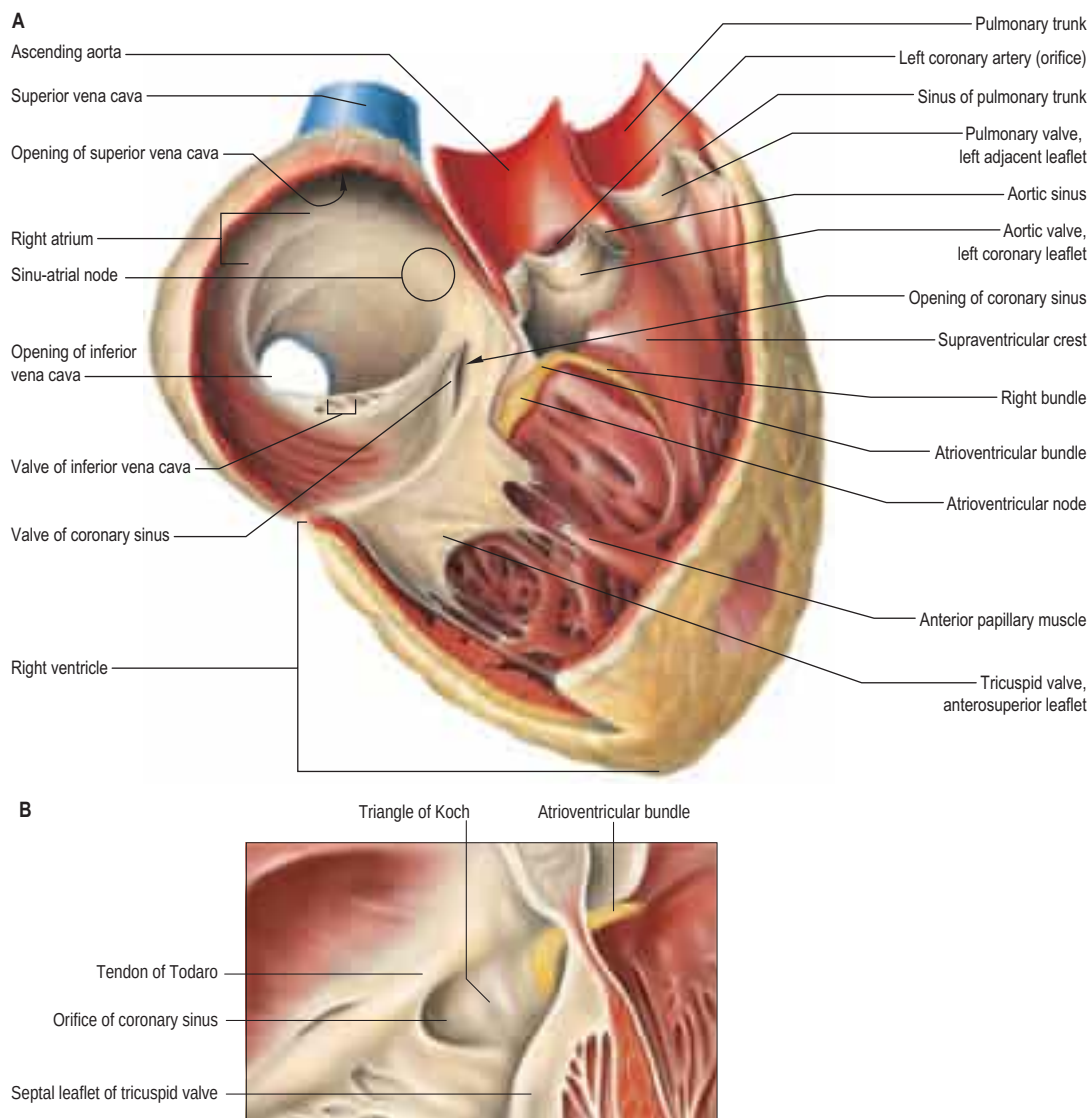


Fig. 57.13 A, The interior of the heart, revealed by incising it along its right, and part of the lower, surfaces. The rest of the heart has been turned over to the left. **B**, The triangle of Koch, defined by the tendon of Todaro, orifice of the coronary sinus and the septal leaflet of the tricuspid valve. (A, With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

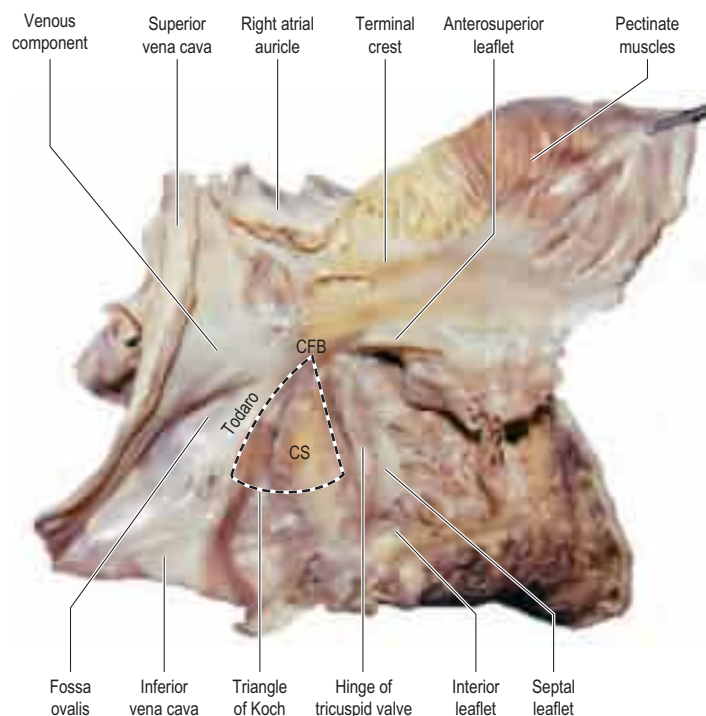


Fig. 57.14 The internal features of the right atrium. The superior and inferior venae cavae have been dissected, exposing the venous component of the right atrium and the fossa ovalis. The dotted line indicates the triangle of Koch, demarcated by the orifice of the coronary sinus (CS) and its valve, the Thebesian valve; the anterior extension of the Eustachian valve, the tendon of Todaro, to the central fibrous body (CFB); and the septal leaflet of the tricuspid valve. (Courtesy of M Loukas MD, PhD.)

venae cavae, together with the coronary sinus, open into the venous component.

The superior vena cava returns blood from head, neck and upper limb through an orifice that faces anteroinferiorly and has no valve; it also receives blood from the chest wall and the oesophagus via the azygos system.

The inferior vena cava is larger than its superior counterpart. It drains blood from all structures below and including the diaphragm into the lowest part of the atrium near the septum. Anterior to its orifice, and found along its lateral or right margin, is the flap-like Eustachian valve, a fold of endocardium that encloses a few muscular fibres and is of varying size (see Fig. 57.13; Fig. 57.15). When traced inferiorly, it forms the Eustachian ridge, which runs into the sinus septum (see below), continuous with the valve of the coronary sinus. The lateral part of the valve becomes continuous with the lower end of the terminal crest. The Eustachian valve is large during fetal life, when it serves to direct richly oxygenated blood (from the placenta) from the right atrium through the foramen ovale of the interatrial septum into the left atrium. The valve varies markedly in size in postnatal life; it is sometimes cribriform or filamentous but often absent.

The coronary sinus opens into the venous atrial component between the orifice of the inferior vena cava, the fossa ovalis and the vestibule of the atrioventricular opening (see Fig. 57.13). It is often guarded by a thin, semicircular Thebesian valve that covers the lower part of the orifice. The upper limb of this valve joins the Eustachian valve and its muscular extension, the Eustachian ridge. It also joins a tendinous structure, the tendon of Todaro, running from this commissure into the sinus septum, which is the septum between the coronary sinus and the fossa ovalis. The tendon of Todaro runs forwards to insert into the central fibrous body and is one of the landmarks of the triangle of Koch (see below). The ostium of the coronary sinus forms a prominent landmark in the right atrium. The sinus itself lies within the left atrioventricular groove (see Fig. 57.4B,D) and is the conduit for return of most of the venous blood from the heart, although some veins drain directly to the atrial chambers. The coronary sinus starts at the confluence of the oblique vein of the left atrium and the great cardiac vein, and receives the middle and small cardiac veins close to its junction with the right atrium. Several small venous ostia, draining the minimal atrial veins, are found scattered around the atrial walls. They return a small fraction of blood from the heart, and are most numerous on the septal aspect. The anterior cardiac veins and, sometimes, the right marginal vein may enter the atrium through larger ostia.

The right atrium and its appendage are separated from the venous sinus by the crista terminalis, a smooth, C-shaped muscular ridge that originates from the upper part of the septal surface and, passing anterior to the orifice of the superior vena cava, skirts its right margin to reach the right side of the orifice of the inferior vena cava. It marks the site of the right venous valve of the embryonic heart, corresponding externally to the terminal groove. The sinu-atrial (sinus) node is located in the superior part of the groove, inferolateral to the orifice of the superior vena cava (see Fig. 57.13). The mean length of the crest is 51 ± 9 mm and its thickness from epicardium to endocardium at the junction of the superior vena cava with the posterior atrial wall is 5.5 ± 1.3 mm. The pectinate muscles (musculi pectinati) are almost parallel muscular ridges that extend anterolaterally from the terminal crest, reaching into the appendage where they form several trabeculations. The largest and most prominent pectinate muscle, forming the bridge of the sulcus terminalis internally, is the taenia sagittalis (second crest or septum spurium). The septal wall presents the fossa ovalis, an oval depression superior and to the left of the orifice of the inferior vena cava. Its floor is the primary atrial septum (septum primum). The rim of the fossa (limbus) is prominent. Often said to represent the edge of the so-called septum secundum, in reality the muscular borders of the fossa are no more than infoldings of the atrial walls, incorporating extracardiac adipose tissue. It is most distinct anterosuperiorly to the fossa, usually deficient inferiorly. Some hearts display an excessive amount of extracardiac adipose tissue within the fold, raising a marked elevation of the superior rim of the fossa (intervenous tubercle of Lower). A small slit is sometimes found at the upper margin of the fossa, ascending beneath the rim to communicate with the left atrium. This represents failure of obliteration of the fetal foramen ovale, which remains patent in up to one-third of all normal hearts. Anteroinferior in the right atrium is the large, oval vestibule leading to the orifice of the tricuspid valve. A triangular zone (of Koch) is defined between the attachment of the septal leaflet of the tricuspid valve, the anteromedial margin of the ostium of the coronary sinus, and the palpable round, collagenous subendocardial tendon of Todaro (see Figs 57.13–57.14). Koch's triangle is a landmark of particular surgical importance, indicating the site of the atrioventricular node and its atrial connections. Anterosuperior to the insertion of the tendon of Todaro, the septal wall is formed by the atrioventricular component of the membranous septum, intervening between the right atrium and subaortic outlet of the left ventricle (Fig. 57.17). The bulging atrial wall anterosuperior to the membranous septum, the aortic mound (torus aorticus), marks the location of the non-coronary aortic sinus with its enclosed valvular leaflet.

Right ventricle

The right ventricle extends from the right atrioventricular (tricuspid) orifice nearly to the cardiac apex. It then ascends to the left to become the infundibulum, or conus arteriosus, reaching the pulmonary orifice and supporting the leaflets of the pulmonary valve. Topographically, the ventricle possesses an inlet component that supports and surrounds the tricuspid valve; a coarsely trabeculated apical component; and a muscular outlet or infundibulum that surrounds the attachments of the pulmonary valve leaflets.

External features

The convex anterosuperior surface of the right ventricle makes up a large part of the sternocostal aspect of the heart, and is separated from the thoracic wall only by the pericardium (see Fig. 57.1). The left pleura and, to a lesser extent, the anterior margin of the left lung are interposed above and to the left. The inferior surface is flat and is related mainly, with the interposition of the pericardium, to the central tendon and a small adjoining muscular part of the diaphragm. The left and posterior wall is the ventricular septum, slightly curved and bulging into the right ventricle so that, in sections across the cardiac axis, the outline of the right ventricle is crescentic. The wall of the right ventricle is relatively thin (3–5 mm), the ratio of the thickness of the two ventricular walls usually being 1:3.

Internal features

The inlet and outlet components of the ventricle, supporting and surrounding the leaflets of the tricuspid and pulmonary valves, respectively, are separated in the roof of the ventricle by the prominent supraventricular crest (crista supraventricularis) (see Fig. 57.13A; Fig. 57.18). The crest is made up predominantly by the inner heart curvature, bounded on the epicardial aspect by the rightward margin of the transverse sinus. The crest is a thick, muscular, highly arched structure, extending obliquely anteriorly and to the right from a septal limb high on the interventricular septal wall to a mural or parietal limb on the

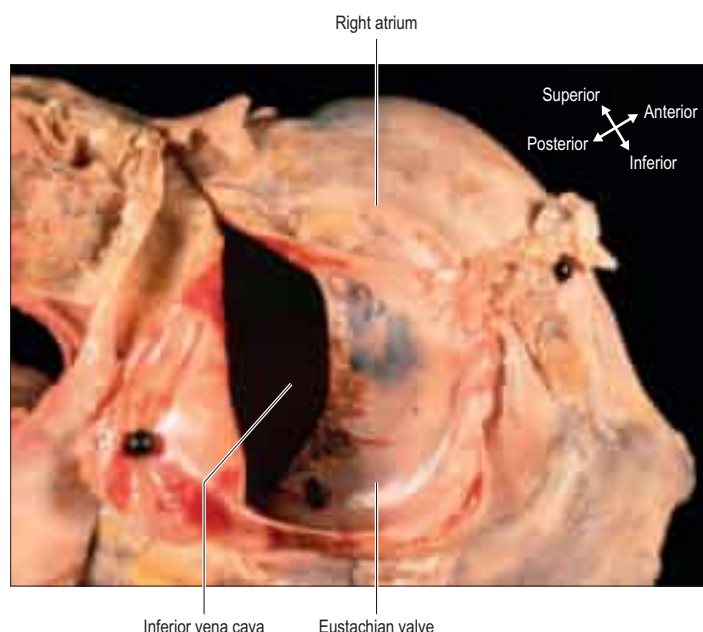


Fig. 57.15 An inferior view of a cadaveric heart, demonstrating an exaggerated Eustachian valve occupying a significant portion of the lumen of the inferior vena cava. (With permission from Loukas M, Sullivan A, Tubbs R, Weinhaus A, Derderian T, Hanna M. 2010. Chiari's network: review of the literature. *Surg Radiol Anat.* 32:895–901.)

Occasionally, a reticulated network, Chiari's network, originates from the Eustachian valve and connects to different parts of the right atrium, including the coronary sinus. It results from incomplete resorption of the right valve of the embryonic sinus venosus (**Fig. 57.16**). Often clinically insignificant, Chiari's network has been associated with the pathogenesis of thromboembolic disease, endocarditis, arrhythmias, cardiac flow obstructions, heart murmurs, infective endocarditis, cardiac tumours and entrapment of catheters on percutaneous intervention.

The prominent sub-Thebesian recess is located posterior to the Eustachian valve and posteroinferior to the ostium of the coronary sinus. The area between the Eustachian and tricuspid valves inferior to the ostium of the coronary sinus (the cavotricuspid isthmus) is the target of catheter-directed ablation procedures, the treatment of choice for atrial flutter ([Sánchez-Quintana et al 2013](#)).

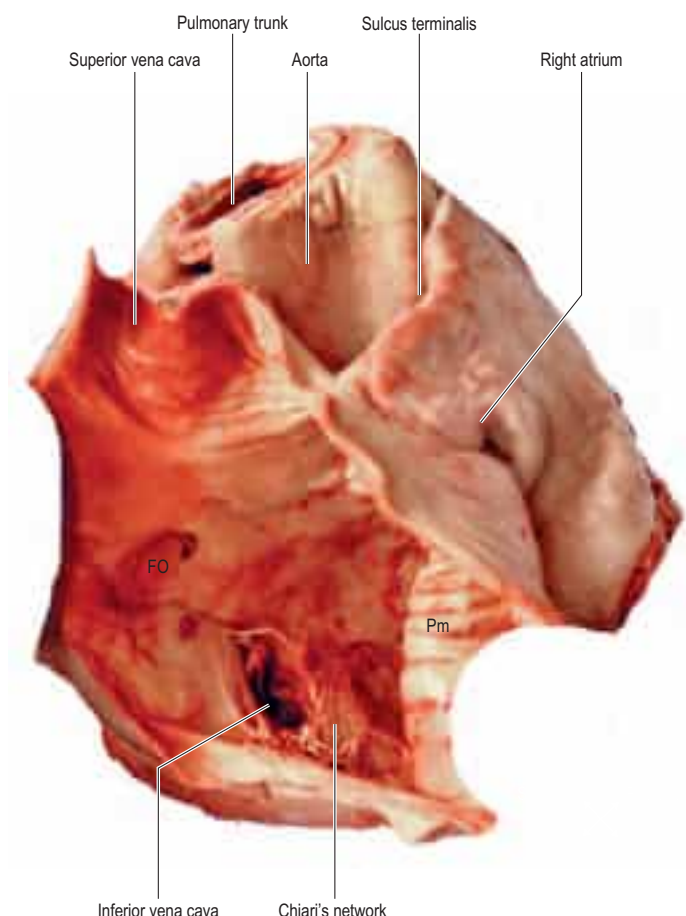


Fig. 57.16 Chiari's network. There are multiple attachment sites around the opening of the inferior vena cava and inferiorly towards the coronary sinus (the reflected part of the right atrium covers the distal attachment to the coronary sinus). Abbreviations: FO, fossa ovalis; Pm, pectinate muscles. (With permission from Loukas M, Sullivan A, Tubbs R, Weinhaus A, Derderian T, Hanna M. 2010. Chiari's network: review of the literature. *Surg Radiol Anat.* 32:895–901.)

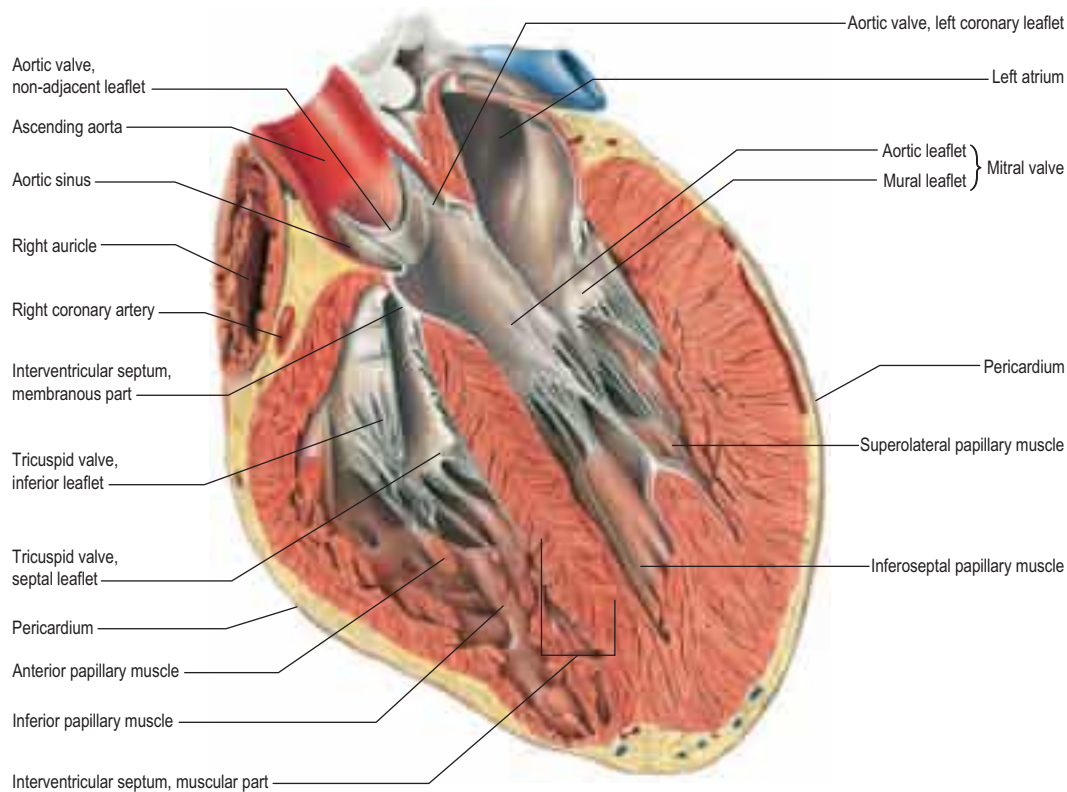


Fig. 57.17 A dissection opening the ventricles, viewed from the front. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

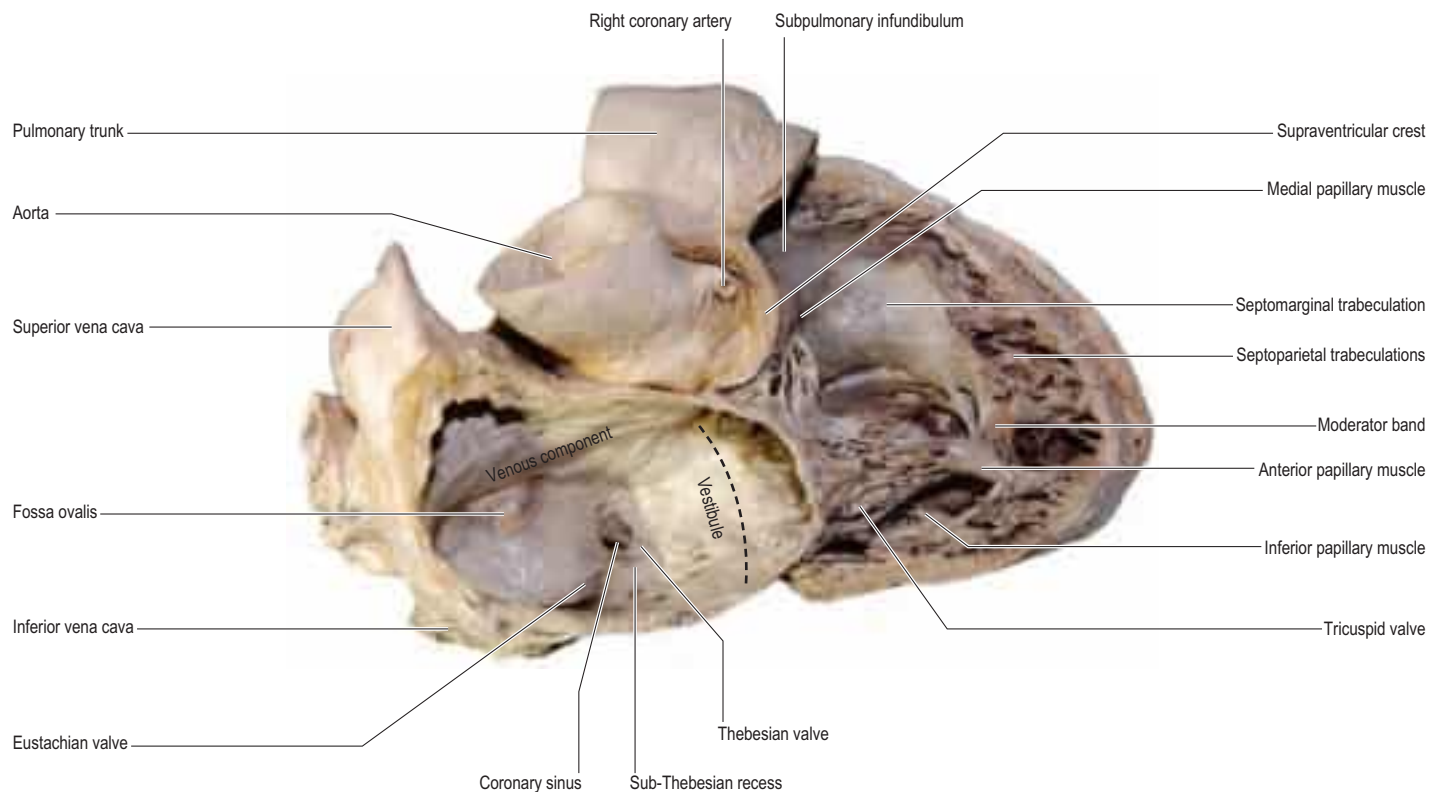


Fig. 57.18 A 'window dissection' of a cadaveric heart, prepared by removing the anterosuperior wall of the right atrium and ventricle, to expose their internal features. The Eustachian valve separates the inferior vena cava from the sub-Thebesian recess. The Thebesian valve guards the entry into the coronary sinus. The smooth circumferential area of atrial wall that surrounds the orifice of the tricuspid valve is the vestibule. Note the location of the supraventricular crest and septomarginal trabeculation. The body of the septomarginal trabeculation continues as an important muscular strand, the moderator band, to the anterior papillary muscle and the parietal wall of the right ventricle. (Specimen courtesy of M Loukas MD, PhD.)

anterolateral right ventricular wall. The posterolateral aspect of the crest provides a principal attachment for the anterosuperior leaflet of the tricuspid valve. The septal limb of the crest may be continuous with, or embraced by, the septal limbs of the septomarginal trabeculation. The inlet and outlet regions extend apically into and from the prominent coarsely trabeculated component of the ventricle. The inlet component itself is also trabeculated, whereas the outlet component (infundibulum) has predominantly smooth walls. The trabeculated appearance is

caused by a myriad of endocardial, lined, irregular muscular ridges and protrusions collectively known as trabeculae carneae. These protrusions and intervening grooves impart great variation in wall thickness; the protrusions vary in extent from mere ridges to trabeculations, fixed at both ends but otherwise free. Other conspicuous protrusions are the papillary muscles, which are inserted at one end on to the ventricular wall and are continuous at the other end with collagenous cords, the chordae tendineae (tendinous cords), inserted on the free edge of the

atrioventricular valves. One protrusion in the right ventricle, the septomarginal trabeculation or septal band, is particularly prominent, reinforcing the septal surface where, at the base, it divides into limbs that embrace the supraventricular crest. Towards the apex, it supports the anterior papillary muscle of the tricuspid valve and, from this point, crosses to the parietal wall of the ventricle as the moderator band (the name reflects an earlier idea that septomarginal trabeculation prevented overdistension of the ventricle). The role of the moderator band as part of the conduction system of the heart involves the right atrioventricular bundle, as conduction cardiomyocytes move towards the apex of the ventricle before entering the anterior papillary muscle. The moderator band may be short/thick, long/thick, short/thin, long/thin: it is occasionally absent. A further series of prominent trabeculations, the septoparietal trabeculations, extend from its anterior surface and run on to the parietal ventricular wall. The smooth-walled outflow tract, or infundibulum, ascends to the left, superior to the septoparietal trabeculations and inferior to the arch of the supraventricular crest to the pulmonary orifice (Spicer and Anderson 2013).

Tricuspid valve

The atrioventricular valvular complex, in both ventricles, consists of the orifice and its associated anulus, the leaflets, the supporting chordae tendineae of various types and the papillary muscles. Harmonious interplay of all of these, together with the myocardial mass, depends on the conduction tissues and mechanical cohesion provided by the cardiac skeleton. All parts change substantially in position, shape, angulation and dimensions during the cardiac cycle (see Fig. 57.20).

Tricuspid valvular orifice

The tricuspid valve orifice is best seen from the atrial aspect. It measures, on average, 11.4 cm in circumference in males and 10.8 cm in females. There is a clear line of transition from the atrial wall or septum to the lines of attachment of the valvular leaflets. Its margins are not precisely in a single plane. It is almost vertical but at 45° to the sagittal plane and slightly inclined to the vertical, such that it 'faces' (on its ventricular aspect) anterolaterally to the left and somewhat inferiorly (Fig. 57.19). Roughly triangular, its margins are described as anterosuperior, inferior and septal, corresponding to the lines of attachment of the valvular

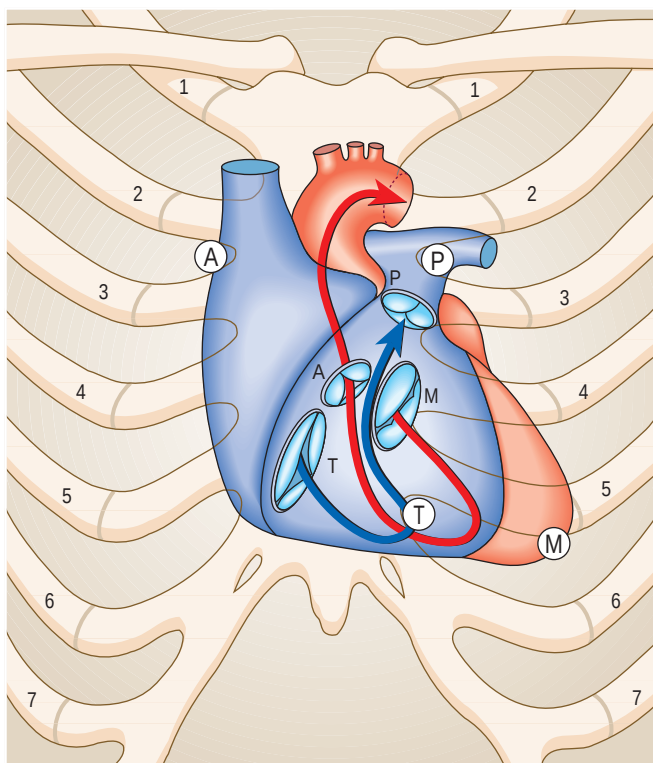


Fig. 57.19 The relation of the sternocostal surface and valves of the heart to the thoracic cage. The right heart is blue, the arrow denoting the inflow and outflow channels of the right ventricle; the left heart is treated similarly in red. The positions, planes and relative sizes of the cardiac valves are shown. The position of the letters A, P, T and M indicate the aortic, pulmonary, tricuspid and mitral auscultation areas of clinical practice, respectively. Note that, for the purpose of illustration, the orifices of the aortic, mitral and tricuspid valves are shown with some separation between them, whereas, in reality, the leaflets of the three valves are in fibrous continuity (see Fig. 57.24).

leaflets. The connective tissues around the orifice of the atrioventricular valves separate the atrial and ventricular myocardial masses completely, except at the point of penetration of the atrioventricular bundle; they vary in density and disposition around the valvular circumference. Extending from the right fibrous trigone component of the central fibrous body is a pair of curved, tapered, subendocardial tendons, or 'prongs' (fila coronaria), which partly encircle the circumference. The latter is completed by more tenuous, deformable fibroblastic sulcal areolar tissue. Although the extent of fibrous tissue varies with sex and age, the tissue within the atrioventricular junction around the tricuspid orifice is always less robust than similar elements found at the attachments of the mitral valve. The topographical 'attachment' of the free valvular leaflets in the tricuspid valve does not wholly correspond to the internal level of attachment of the fibrous core of the valve to the junctional atrioventricular connective tissue. The line of attachment of the leaflet is best appreciated in the heart when examined grossly, this feature being more readily discerned clinically.

Tricuspid valve leaflets

When they are closed, it is usually possible to distinguish the three leaflets in the tricuspid valve on the basis of the zones of apposition between them: hence the name. The leaflets are located anterosuperiorly, septally and inferiorly, corresponding to the marginal sectors of the atrioventricular orifice named in conjunction. The inferior leaflet is often described as being posterior, but when assessed in the attitudinally correct anatomical position, the leaflet is positioned inferiorly (Anderson and Loukas 2009). Each leaflet is a reduplication of endocardium enclosing a collagenous core, continuous marginally and on its ventricular aspect with diverging fascicles of chordae tendineae (see below) and basally confluent with the anular connective tissue. In passing from the free margin to the inserted margin, all leaflets of the atrioventricular valves display rough, clear and basal zones. The rough zone is relatively thick, opaque and uneven on its ventricular aspect where most chordae tendineae are attached; its atrial aspect makes contact with the comparable surface of the adjacent leaflets during full valve closure. The clear zone is smooth and translucent, receives few chordae tendineae and has a thinner, fibrous core. The basal zone, extending 2–3 mm from the circumferential attachment of the leaflets, is thicker from increased connective tissue, vascularized and innervated. It contains the insertions of the atrial myocardium. The anterosuperior leaflet is the largest component of the tricuspid valve, attached chiefly to the atrioventricular junction on the posterolateral aspect of the supraventricular crest, and extending along its septal limb to the membranous septum ending at the anteroseptal commissure. One or more notches often indent its free margin. The attachment of the septal leaflet passes from the inferoseptal commissure on the inferior ventricular wall across the muscular septum, then angling across the membranous septum to the anteroseptal commissure. The septal leaflet defines one of the borders of the triangle of Koch, thereby aiding location of the atrioventricular node at the apex of this triangle, and ensuring avoidance during tricuspid valve surgery (see Figs 57.13, 57.17). The inferior leaflet is wholly mural in attachment and guards the diaphragmatic surface of the atrioventricular junction, its limits being the inferoseptal and anteroinferior commissures. The zone of apposition between the inferior and the anterosuperior leaflets is supported by the septal papillary muscle of the conus.

Opening of the tricuspid valve

Despite its name, the tricuspid valve acts more like a bicuspid valve because its smallest septal leaflet is fixed between the atrial and ventricular septa. The remainder of the tricuspid anulus is muscular. During diastole, the anulus dilates with right ventricular relaxation and the large anterior and posterior leaflets move away from the plane of the anulus into the right ventricle. During systole, the anulus constricts as the right ventricle contracts and the two major leaflets move like sails about a relatively immobile septal leaflet and the septum itself (Fig. 57.20).

Chordae tendineae (tendinous cords)

The chordae tendineae are fibrous collagenous structures that support the leaflets of the atrioventricular valves. Sometimes, false chordae connect the papillary muscles to each other or to the ventricular wall or septum, or pass directly between points on the wall or septum, or both. Their numbers and dimensions vary in the right ventricle; approximately 40% of these false cords contain conduction cardiomyocytes.

The true chordae usually arise from small projections on the tips or margins of the apical third of papillary muscles, although they sometimes arise from the papillary muscle bases or directly from the ventricular walls and septum. They attach to various parts of the ventricular

aspects or the free margins of the leaflets. True chordae are classified into first-, second- and third-order types, according to the distance of the attachment from the margins of the leaflets; the scheme has little functional or morphological merit. Fan-shaped chordae have a short stem from which branches radiate to attach to the margins (or the

ventricular aspect) of the zones of apposition between leaflets and to the ends of adjacent leaflets. Rough-zone chordae arise from a single stem that usually splits into three components that attach to the free margin, the ventricular aspect of the rough zone and to some intermediate point on the leaflet, respectively. Free-edge chordae are single, thread-like and often long, passing from either the apex or the base of a papillary muscle into a marginal attachment, usually near the midpoint of a leaflet or one of its scallops. Deep chordae pass beyond the margins and, branching to various extents, reach the more peripheral rough zone or even the clear zone. Basal chordae are round or ribbon-like, long and slender, or short and muscular; they arise from the smooth or trabeculated ventricular wall and attach to the basal component of a leaflet.

Papillary muscles

The two major papillary muscles in the right ventricle are located in anterior and inferior positions. A third, smaller muscle lies medially, together with several smaller, variable muscles attached to the ventricular septum. The anterior papillary muscle is the largest, its base arising from the right anterolateral ventricular wall inferior to the anteroinferior commissure of the inferior leaflet, also blending with the right end of the septomarginal trabecula. The inferior, papillary muscle, often bifid or trifid, arises from the myocardium inferior to the inferoseptal commissure. The septal (medial) papillary muscle of the conus, the muscle of Lancisi, is almost always present and is the most superior and largest of the small septal papillary muscles. It arises from the posterior septal limb of the septomarginal trabeculation and locates the right bundle branch within the right ventricle. All the major papillary muscles supply chordae to adjacent components of the leaflets they support (see Figs 57.17, 57.18). A feature of the right ventricle is that the septal leaflet is tethered by individual chordae tendineae directly to the ventricular septum; such septal insertions are *never* seen in the left ventricle. When closed, the three leaflets fit snugly together, the pattern of the zones of apposition confirming the trifoliate arrangement of the tricuspid valve.

Pulmonary valve

The pulmonary valve, guarding the outflow from the right ventricle, surmounts the infundibulum and is situated at some distance from the other three cardiac valves (Figs 57.21, 57.23). Its general plane faces superiorly to the left and slightly posteriorly. It has three semilunar leaflets, attached by convex edges partly to the infundibular wall of the right ventricle and partly to the origin of the pulmonary trunk. The line of attachments is curved, rising at the periphery of each leaflet near their zones of apposition (the commissures) and reaching the sinutubular ridge of the pulmonary trunk (Fig. 57.24). Removal of the leaflets reveals that the fibrous semilunar attachments enclose three crescents of infundibular musculature within the pulmonary sinuses, whereas three roughly triangular segments of arterial wall are incorporated within the ventricular outflow tract beneath the apex of each commissural attachment. Thus there is no proper circular 'anulus' supporting the leaflets of the valve, and the fibrous semilunar attachment is an

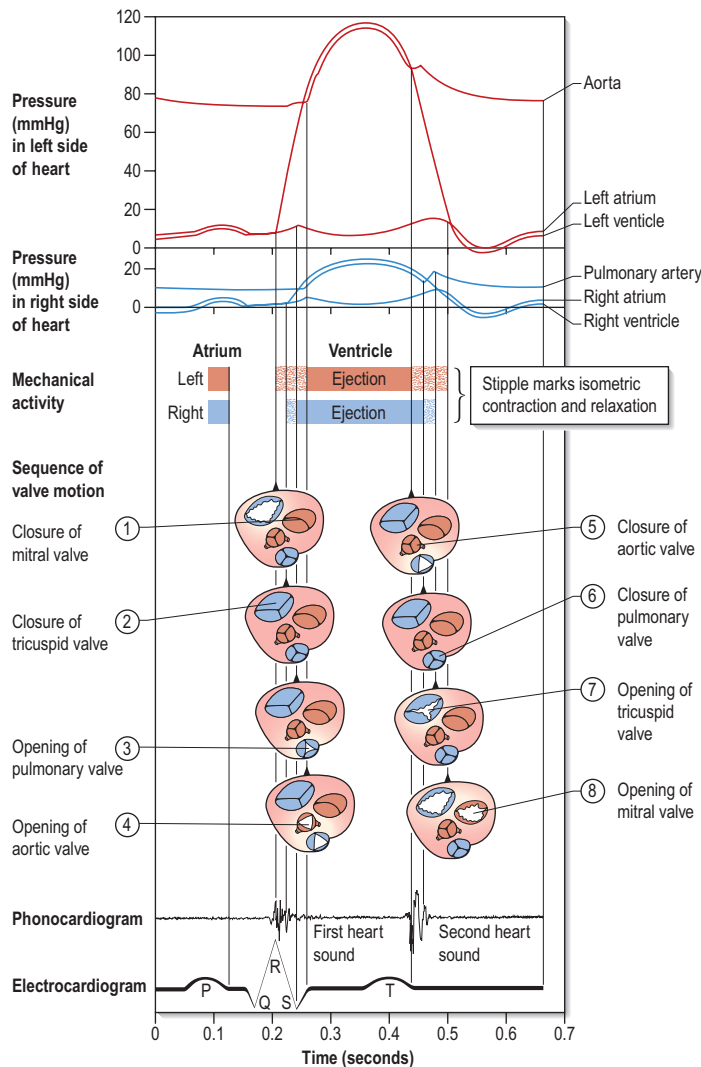


Fig. 57.20 A summary of some of the principal events that occur in the cardiac cycle and that are mentioned at various points throughout this chapter. Systole begins at the onset of the first heart sound (see phonocardiogram) and ends at the onset of the second heart sound, when diastole begins, and this cycle repeats itself.

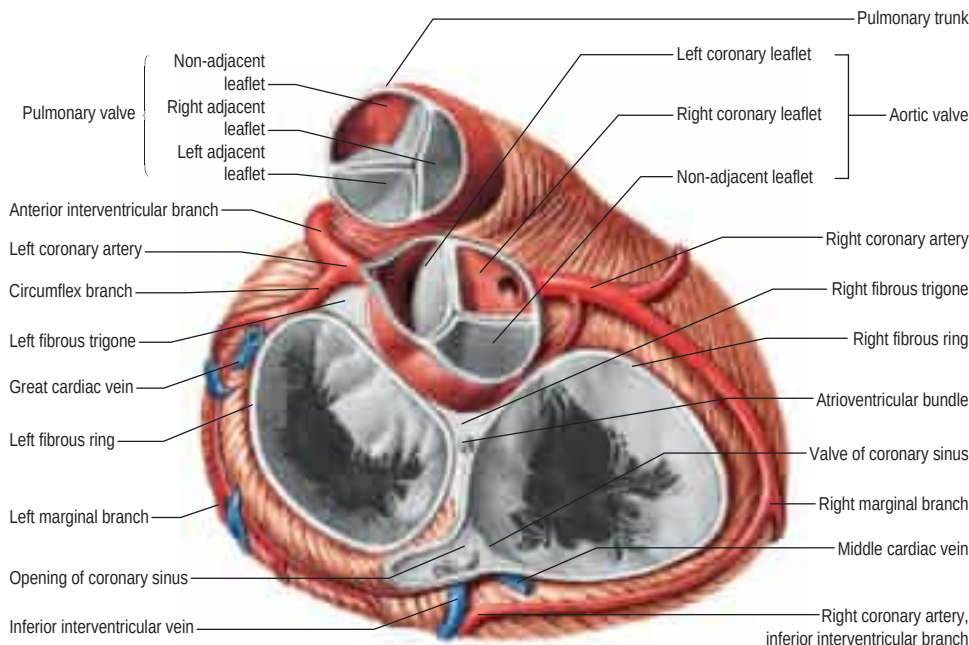


Fig. 57.21 The base of the ventricles, after removal of the atria and the pericardium, exposing the coronary arteries and cardiac veins. Contrast the planes and positions of the aortic and pulmonary valves, and with Figure 57.23. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

essential requisite for snug closure of the nodules and lunules of the leaflets during ventricular diastole (see below).

It is difficult to name the leaflets and corresponding sinuses of the pulmonary valve and trunk precisely according to the coordinates of the body because the valvular orifice is obliquely positioned. *Terminologia Anatomica* (2011) refers to anterior, posterior and septal leaflets on the basis of their fetal position but this changes with development, becoming anterior, right and left, respectively, in the adult. Each leaflet is an endocardial fold with a variably developed intervening substantial fibrous core that traverses both the free edge and the semilunar attached border. The latter is particularly thickened at its deepest central part (nadir) of the base of each leaflet, and therefore never forms a simple complete fibrous ring. The free margin of each leaflet contains a central localized collagenous thickening, the nodule of Arantius. Perforations within the leaflets close to the free margin and near the commissures are frequently present and are of no functional significance. Each semilunar leaflet is contained within one of the three sinuses of the pulmonary trunk.

Opening of the pulmonary valve

During diastole, all three leaflets of the pulmonary valve are tightly apposed. The pulmonary valve is difficult to visualize at echocardiography and usually only the posterior leaflet is visible when the valve is closed; atrial systole may cause a slight posterior movement of the valve leaflets. The pulmonary valve opens passively during ventricular systole and then closes rapidly at the end of systole (see Fig. 57.20).

Left atrium

General, external and internal features

Although smaller in volume than the right, the left atrium has thicker walls (3 mm on average). It possesses a venous component that receives the right and left superior and inferior pulmonary veins, a vestibule and an appendage. Its cavity and walls are formed largely by the proximal parts of the pulmonary veins that are incorporated into the atrium during development. Its extensive body is a remnant of the initial atrial component of the primary heart tube. The left atrium is roughly cuboidal, extending posterior to the right atrium and separated from it by the obliquely positioned septum. The right atrium is therefore anterolateral to the right part of the left atrium. The left part is concealed anteriorly by the initial segments of the pulmonary trunk and aorta: part of the transverse pericardial sinus lies between it and these arterial trunks. Anteroinferiorly, and to the left, it adjoins the base of the left ventricle at the orifice of the mitral valve. Its posterior aspect forms most

of the anatomical base of the heart and is approximately quadrangular, receiving the terminations of (usually) two pulmonary veins from each lung, forming the anterior wall of the oblique pericardial sinus (see Fig. 57.3). This surface ends at the shallow vertical interatrial groove that descends to the cardiac crux.

The left atrial appendage is characteristically longer, narrower and more hooked than the right, and is a finger-like extension with more deeply indented margins. It is constricted at its atrial junction and all its contained pectinate muscles are much smaller than their right counterparts. The left atrium lacks a crista terminalis, and the muscle bundles in the appendage are arranged in a whorl-like fashion rather than being in an array. The tip of the appendage has a variable position lying over the pulmonary trunk and anterior interventricular artery, pointing posteriorly towards the aorta (Fig. 57.22). Its narrow morphology renders the left atrial appendage a potential site for deposition of thrombi.

The four pulmonary veins open into the superior posterolateral surfaces of the left atrium, two on each side (see Figs 57.3, 57.4B). This typical arrangement is present in 20–60% of the population. A common variation includes the presence of a short or long left common venous trunk and multiple pulmonary veins on the right (Fig. 57.25). The right pulmonary veins travel posterior to their respective venae cavae. Their orifices are smooth and oval, the left pair frequently opening via a common channel. Interpulmonary ridges are usually found between ipsilateral orifices; the most prominent is located between the openings of the left atrial appendage and left superior pulmonary vein. The ridges are infoldings of the left atrial wall and contain adipose tissue, atrial arteries and nerve bundles. At the site of the pericardial reflection, the atrial musculature extends into the pulmonary veins, forming myocardial sleeves that are thickest in the inferior wall of the superior pulmonary veins and the superior walls of the inferior pulmonary veins. They lie external to the venous tunica media and internal to the epicardium/adventitia and are often the site of focal electrical activity that initiates atrial fibrillation. Atrial myocardial bridges and crossing strands are often present, connecting the left superior and inferior pulmonary veins.

Several epicardial fat pads on the pulmonary venous component house the superior left, posterolateral, left inferior and posteromedial ganglionated cardiac intrinsic nerve plexuses (typically four). Minimal cardiac veins (venae cordis minimae) return blood directly from the myocardium to the left atrial cavity. The left atrial aspect of the septum has a characteristically rough appearance, bounded by a crescentic, superiorly concave ridge that marks the site of the foramen ovale. The smooth circumferential area of atrial wall that surrounds the orifice of the mitral valve is the vestibule (see Fig. 57.25A, Fig. 57.26). The musculature between the ostium of the inferior pulmonary vein and the

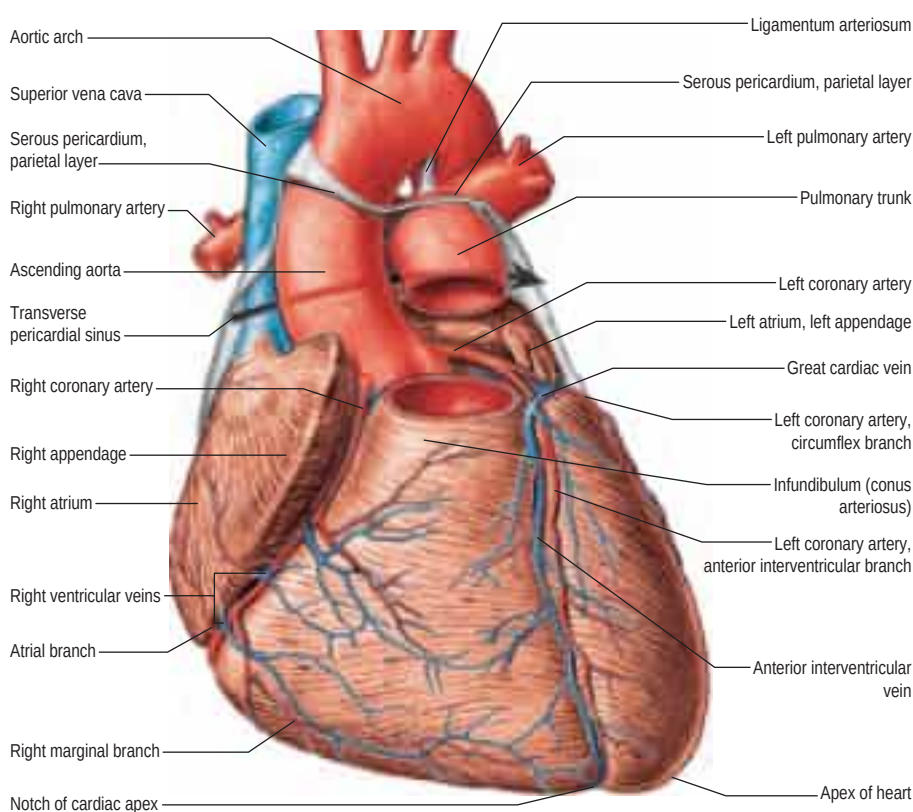


Fig. 57.22 The anterior or sternocostal surface of the heart. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

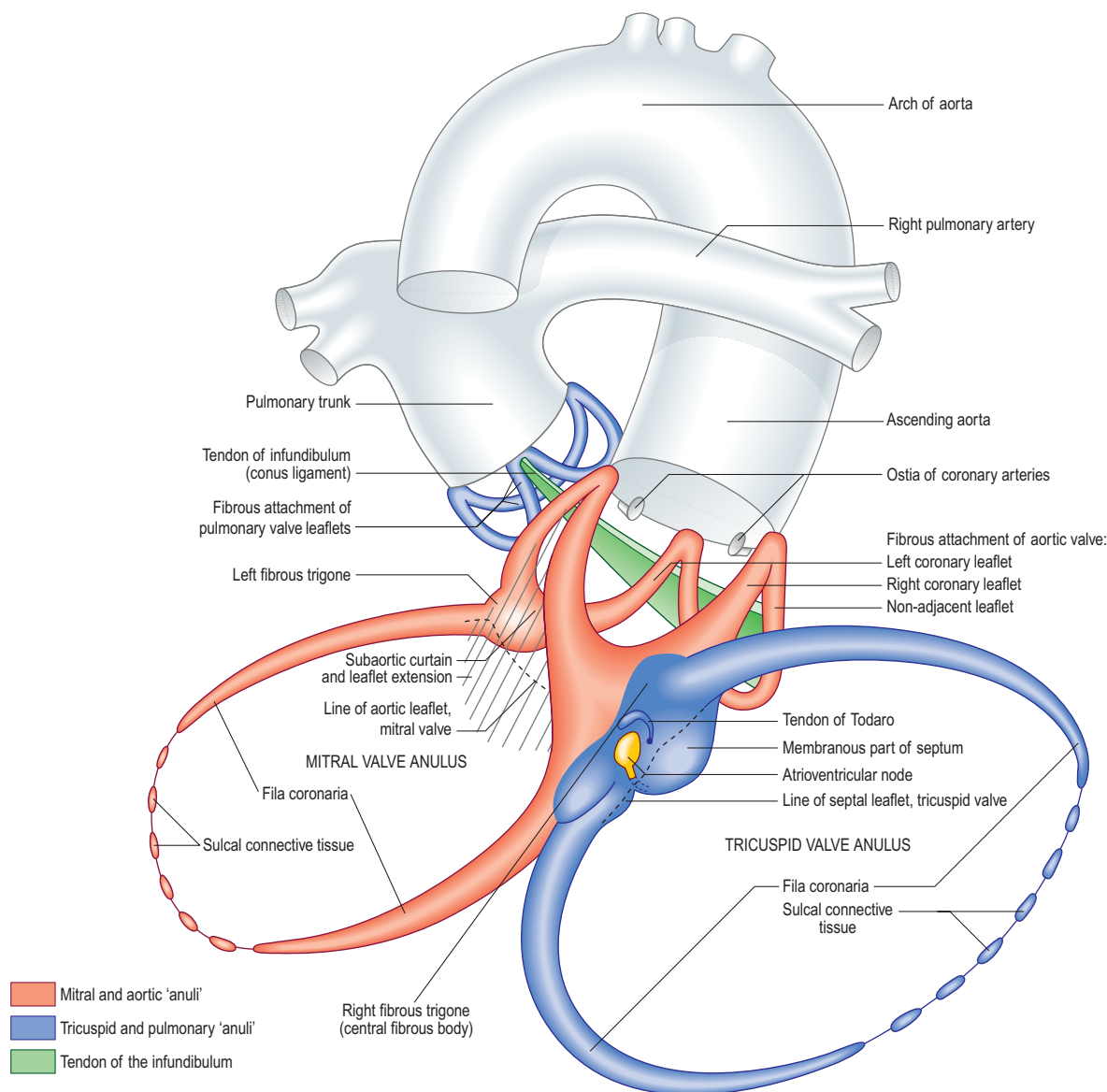


Fig. 57.23 Principal elements of the fibrous skeleton of the heart. For clarity, the view is from the right posteriosuperior aspect. Perspective causes the pulmonary anulus to appear smaller than the aortic anulus, whereas, in fact, the reverse is the case. Consult the text for an extended discussion. (Copyright of The Royal College of Surgeons of England. Reproduced with permission.)

anulus of the mitral valve is the left atrial or mitral isthmus, and is an area where the vestibule of the left atrium directly opposes the wall of the great cardiac vein, coronary sinus and circumflex coronary artery.

Left ventricle

General and external features

The left ventricle is constructed in accordance with its role as a powerful pump for the high-pressured systemic arterial circulation. Various described as half-ellipsoid or cone-shaped, it is longer and narrower than the right ventricle, extending from its base in the plane of the atrioventricular groove to the cardiac apex. Its long axis descends anteriorly and to the left. In transverse section, at right angles to the axis, its cavity is oval or nearly circular, with walls three times thicker (8–12 mm) than those of the right ventricle (Fig. 57.27). It forms part of the sternocostal, left and inferior (diaphragmatic) cardiac surfaces. Except where obscured by the aorta and pulmonary trunk, the base of the ventricular cone is superficially separated from the left atrium by part of the atrioventricular groove with the coronary sinus within in its posterior aspect (see Fig. 57.4B,D). The anterior and posterior (inferior) interventricular grooves indicate the lines of mural attachment of the ventricular septum and the limits of the ventricular territories. The sternocostal surface of the ventricle curves bluntly into its left surface at the obtuse margin.

The shape of the left ventricle changes from elliptical in the neonatal period to the round adult shape later in infancy (Azancot et al 1983).

The effect of obesity on the heart is apparent as early as the second year of life. Obese children aged 2 years have a greater left ventricular mass compared with normal weight controls (de Jonge et al 2011).

Internal features

The left ventricle has an inlet region guarded by the mitral valve (ostium venosum), an outlet region guarded by the aortic valve (ostium arteriosum) and an apical trabecular component. The left atrioventricular orifice admits atrial blood during diastole, the flow being directed towards the cardiac apex (Fig. 57.28). After closure of the mitral leaflets and throughout the ejection phase of systole, blood is expelled from the apex through the aortic orifice. In contrast to the orifices within the right ventricle, those of the left ventricle are in close contact with fibrous continuity between the leaflets of the aortic and mitral valves (the 'subaortic curtain') (Fig. 57.29); the inlet and outlet turn sharply round this fibrous curtain. The anterolateral wall is the muscular ventricular septum, the convexity of which completes the circular outline of the left ventricle (see Fig. 57.27). Towards the aortic orifice, the septum becomes the thin and collagenous interventricular component of the membranous septum, an oval or round area below and confluent with the fibrous triangle separating the right and the non-coronary leaflets of the aortic valve. Between the inferior limits of the free margins of the leaflets of the mitral valve and the ventricular apex, the muscular walls exhibit deeper, finer and more intricate trabeculae carneae than those of the right ventricle, characteristically more developed nearer the apex, and becoming smoother as the superior septal surface is reached.

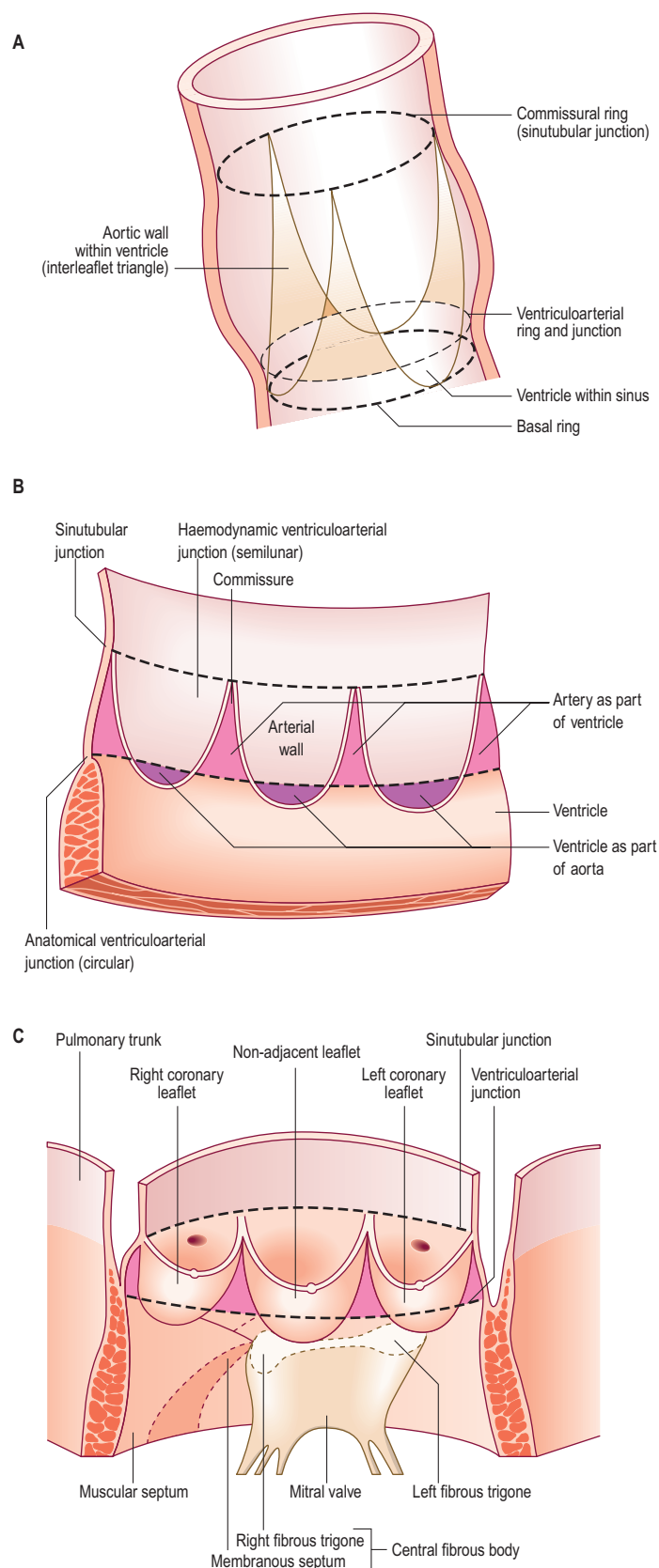


Fig. 57.24 **A**, The structure of the aortic root is best conceptualized in terms of a three-pronged coronet; there are at least three rings within this coronet but none supports the entirety of the attachments of the valvular leaflets (compare with **C**). **B**, The leaflets have been resected at their attachment to the aortic wall. Note the relationship of the leaflet insertions and the ventriculoarterial junction. **C**, The root of the aorta has been cut open and distended, in order to show the insertion of the semilunar leaflets. Note the zone of fibrous continuity between the leaflets of the aortic and mitral valves and their relationship to the fibrous trigones, and the semilunar attachment of the leaflets (compare with **B**). (Redrawn with permission, courtesy of Professor RH Anderson, Institute of Child Health, University College, London.)

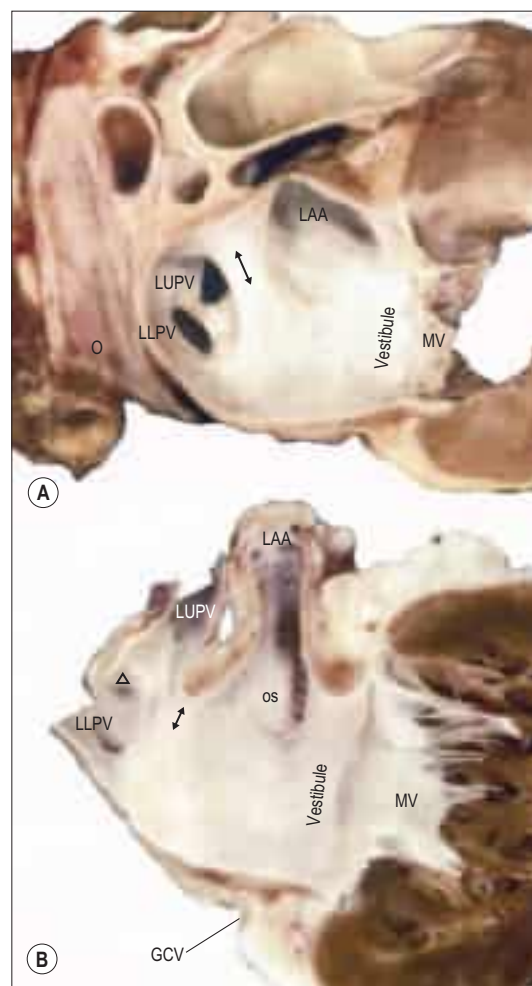


Fig. 57.25 Sagittal sections showing the left side of the left atrium in a cadaveric heart. **A**, The oesophagus (O) passing behind the posterior left atrial wall and a broad left-lateral ridge (double-headed arrow). The left upper (LUPV) and left lower (LLPV) pulmonary veins enter the left atrium via a short common stem. Other abbreviations: LAA, left atrial appendage; MV, mitral valve. **B**, The section passes through the os of the left atrial appendage and the infolding of the ridge. The triangle indicates the carina or interpulmonary ridge between the upper and lower pulmonary veins. The great cardiac vein (GCV) runs underneath the left atrial wall. (With permission from Ho SY, McCarthy KP, Faletra FF. 2011. Anatomy of the left atrium for interventional echocardiography. Eur J Echocardiogr 12:11–15.)

Hypertrophy of heart muscle



Available with the *Gray's Anatomy* e-book

Mitral valve

The general comments already made for the tricuspid valve also apply to the mitral valve. Thus it has an orifice with a supporting anulus, leaflets and a variety of chordae tendineae and papillary muscles.

Mitral valvular orifice

The mitral orifice is a well-defined transitional zone between the atrial wall and the leaflet bases, being smaller than the tricuspid orifice (mean circumference is 9.0 cm in males and 7.2 cm in females). The approximately circular orifice is almost vertical and at 45° to the sagittal plane in diastole, but with a slight anterior tilt. Its ventricular aspect faces anterolaterally to the left and a little inferiorly towards the left ventricular apex. It is almost co-planar with the tricuspid orifice but postero-superior to it, whereas it is posteroinferior and slightly to the left of the aortic orifice. The mitral, tricuspid and aortic orifices are intimately connected at their central fibrous body. When the mitral valve leaflets close, they form a single zone of coaptation, termed the commissure. The anulus of the valve is not a simple fibrous ring but is made up of fibrocollagenous elements of varying consistency, from which the fibrous leaflet cores take origin; the variable consistency is essential to allow the major changes in anular shape and dimensions during the cardiac cycle that are needed for optimal valvular efficiency. The area of

Hypertrophic cardiomyopathy is characterized by myocardial wall thickening, particularly a disproportionate thickening of the interventricular septum in comparison with the posterior wall. Echocardiography accurately assesses the degree of thickening and its effect on systolic function, such as dynamic left ventricular outflow obstruction, systolic anterior motion of the aortic mitral valve leaflet and mid-systolic closure of the aortic valve. There may also be a degree of diastolic dysfunction. Serial short-axis gradient echo MRI allows accurate measurement of wall thickness and is particularly useful in assessing apically confined hypertrophy. A number of histological changes are observed, including cardiomyocytic disarray with replacement fibrosis and collagenous component expansion. Treatment is usually medical, except for refractory cases and those in whom the left ventricular outflow tract obstruction has a gradient of greater than 50 mmHg. Ventricular septal myotomy and myectomy are performed in such cases. Catheter alcohol septal ablation has been introduced as a non-surgical alternative. A number of patients may also require implantation of cardiac defibrillators to prevent sudden cardiac death.

An athlete's heart may physiologically hypertrophy but in a uniform fashion; the left ventricle cavity is usually less than 55 mm in size, and thickness decreases on deconditioning. In contrast, hypertrophic cardiomyopathy reveals asymmetric patterns of left ventricular hypertrophy, often with sharp segmental transitions, left atrial enlargement and bizarre electrocardiographic patterns. Furthermore, there is an autosomal dominant inheritance pattern of abnormalities in genes coding for myocardial proteins associated with hypertrophic cardiomyopathy. Individuals with mutations of the β -MHC (major histocompatibility complex) gene usually develop the classic form of hypertrophy, whereas those with cardiac troponin T gene mutations generally have only mild or clinically undetectable hypertrophy. Rare forms of hypertrophy include localized left ventricular apical hypertrophy as a result of cardiac troponin I mutations, and isolated mid-cavity hypertrophy caused by cardiac actin and MLC (myosin light chain) gene mutations.

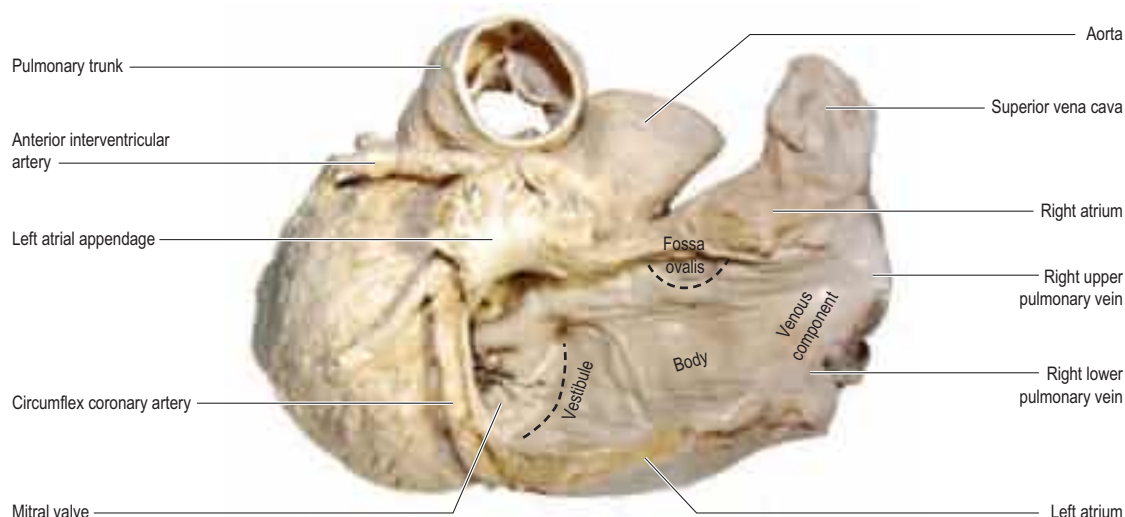


Fig. 57.26 The posterior wall of the left atrium close to the posterior interatrial groove in a cadaveric heart. The smooth-walled venous component of the left atrium is the most extensive component. The septal aspect of the left atrium shows the crescentic line of the free edge of the flap valve against the rim of the fossa ovalis. The orifices of the right superior and inferior pulmonary veins are adjacent to the plane of the septal aspect of the left atrium. (Specimen courtesy of M Loukas MD, PhD.)

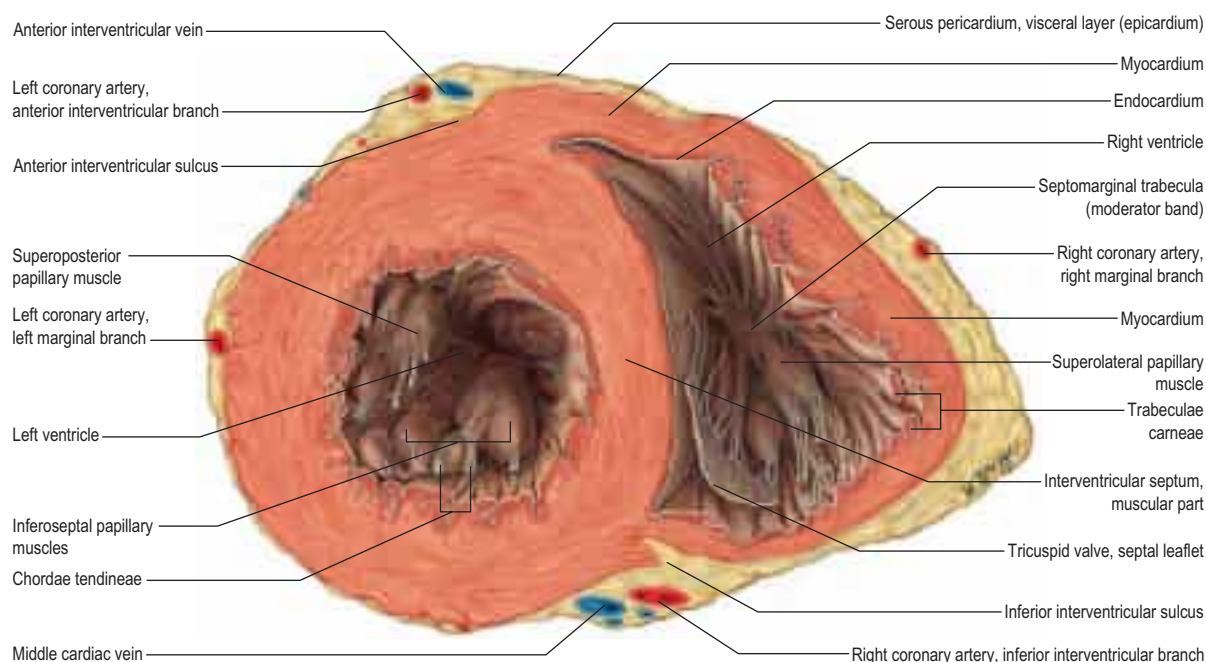


Fig. 57.27 Left and right ventricles: cross-section perpendicular to the axis of the heart, superior aspect. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

the anulus increases linearly with body surface area in children and young adults (Poutanen et al 2006). The anulus is strongest at the internal aspects of the left and right fibrous trigones. Extending from these structures, the anterior and posterior coronary prongs (tapering, fibrous, subendocardial tendons) partly encircle the orifice at the atrio-ventricular junction (see Fig. 57.23). Between the prong tips, the atrial and ventricular myocardial masses are separated by a more tenuous sheet of deformable fibroelastic connective tissue. Spanning anteriorly between the trigones, the fibrous core of the central part of the aortic leaflet of the mitral valve is a continuation of the fibrous subaortic curtain that descends from the adjacent halves of the left and adjacent (non-coronary) valve leaflets (see Fig. 57.29).

Mitral valve leaflets

The mitral valvular leaflets have long been described as paired structures. (The name 'bicuspid valve' is explicit but erroneous because the leaflets are not cuspid, or 'peaked', in form). Small accessory leaflets are almost always found between the two major leaflets and so the mitral valve should be described as a continuous veil that is attached around the entire circumference of the mitral orifice. Its free edge bears several indentations, of which two are sufficiently deep and regular to be nominated as the ends of a solitary and oblique zone of apposition or

commissure. These anteromedial (inferoseptal) and posterolateral (superoposterior) extremities may be regarded as two independent commissures, each positionally named as indicated in brackets. Although simple, the official names for these leaflets – anterior and posterior, respectively – are somewhat misleading because of the obliquity of the valve.

When the valve is laid open, the anterior leaflet (aortic, septal, 'greater' or anteromedial) is seen to guard one-third of the circumference of the orifice and to be semicircular or triangular, with few or no marginal indentations. Its fibrous core (lamina fibrosa) is continuous on the outflow aspect, beyond the margins of the fibrous subaortic curtain, with the right and left fibrous trigones (see Figs 57.17, 57.21, 57.24C). Between the trigones, it is continuous with the fibrous curtain itself and, beyond the trigones, with the roots of the anular fibrous prongs (see Fig. 57.23). The leaflet has a deep crescentic rough zone that receives various chordae tendineae. The ridge limiting the outer margin of the rough zone indicates the maximal extent of surface contact with the mural leaflet in full closure. A clear zone is seen between the rough zone and the valvular anulus; it is devoid of attachments of chordae, although its fibrous core carries extensions from chordae attached in the rough zone. The anterior leaflet has no basal zone and continues into the valvular curtain. Hinging on its anular

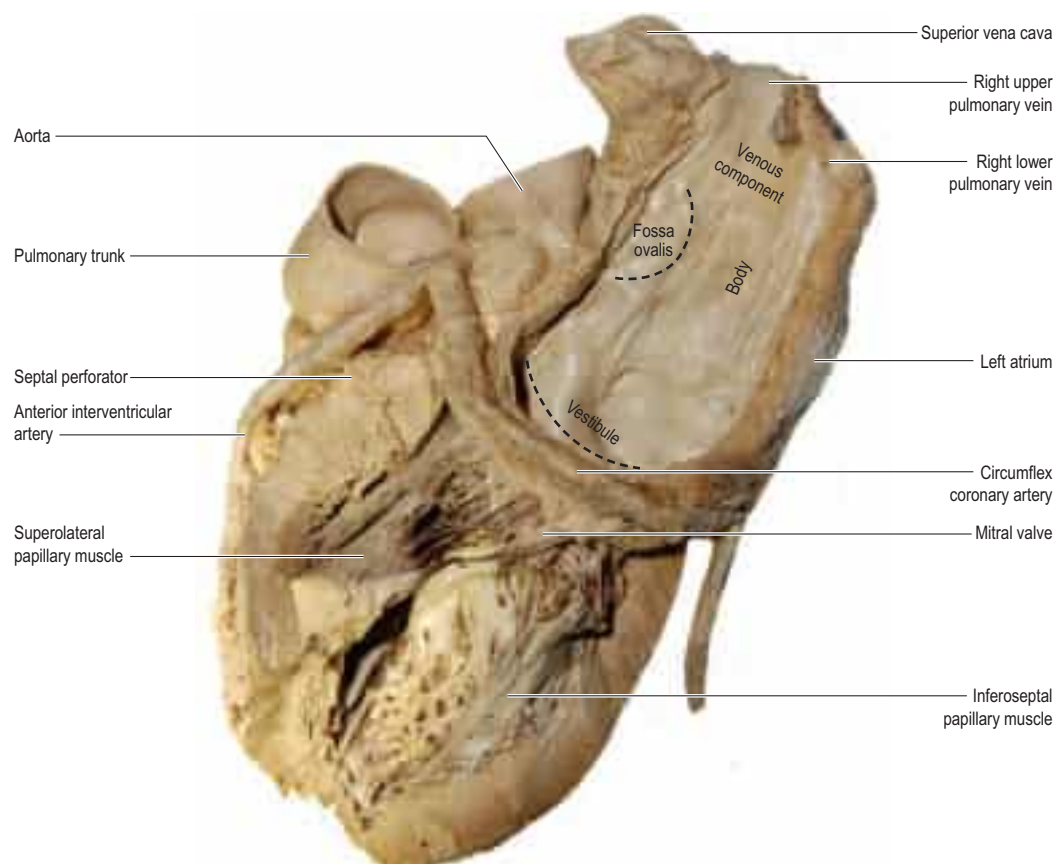


Fig. 57.28 A dissection of the left ventricle in a cadaveric heart, exposing the papillary muscles of the ventricle. Notice the thick wall of the left ventricle and the chordae tendineae of the mitral valve attaching to the papillary muscles. (Specimen courtesy of M Loukas MD, PhD.)

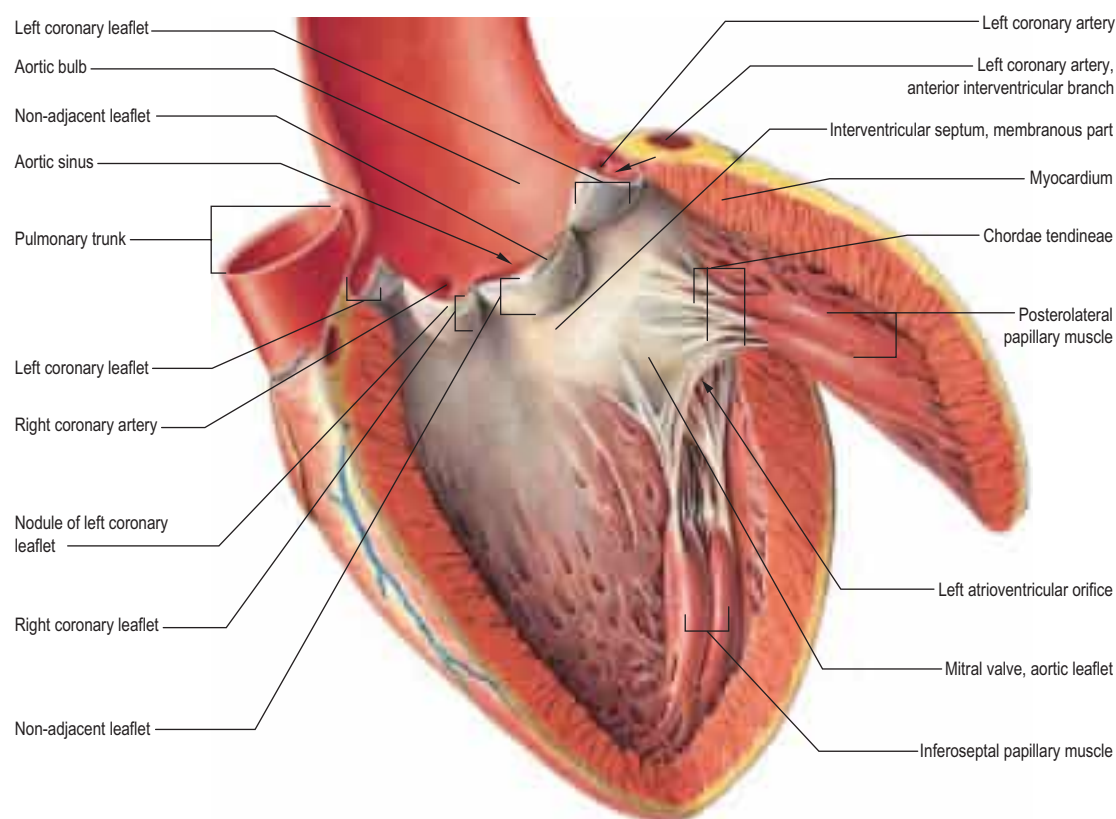


Fig. 57.29 The aortic orifice opened from the front to show the leaflets of the aortic valves, their nodules, lunules, commissures and the triple-scalloped line of their anular attachment. Also shown are the continuity of the subaortic curtain with the mitral aortic leaflet (i.e. 'aortic baffle') and the coronary ostia, and the spatial relationship of the aortic orifice to the pulmonary orifice and to the left ventricle. (With permission from Waschke J, Paulsen F (eds), Sobotta Atlas of Human Anatomy, 15th ed, Elsevier, Urban & Fischer. Copyright 2013.)

attachment, and continuous with the subaortic curtain, it is critically placed between the inlet and the outlet of the ventricle. During passive ventricular filling and atrial systole, its smooth atrial surface is important in directing a smooth flow of blood towards the body and apex of the ventricle. After the onset of ventricular systole and closure of the mitral valve, the ventricular aspect of its clear zone merges into the smooth surface of the subaortic curtain, which, with the remaining fibrous walls of the subvalvular aortic vestibule, forms the smooth boundaries of the ventricular outlet.

The posterior leaflet (mural, ventricular, 'smaller' or posterolateral) usually has two or more minor indentations. Lack of definition of the major intervalvular commissures has led to disagreement and confusion concerning the territorial extent of this leaflet and the possible existence of accessory scallops. Examination of the valve in the closed position reveals that the posterior leaflet may conveniently be regarded as comprising all the valvular tissue posterior to the anterolateral (inferoseptal) and posteromedial (superoposterior) ends of the major zone of apposition with the aortic leaflet. Thus defined, it has a wider attachment to the anulus than does the anterior leaflet, guarding two-thirds of the circumferential attachments. Further indentations usually divide the mural leaflet into a relatively large middle scallop and smaller lateral and septal commissural scallops. Each scallop has a crescentic opaque rough zone, receiving on its ventricular aspect the attachments of the chordae that define the area of valvular apposition in full closure. From the rough zone to within 2–3 mm of its anular attachment, there is a membranous clear zone devoid of chordae. The basal 2–3 mm is thick and vascular, and receives basal chordae. The ratio of rough to clear zone in the anterior leaflet is 0.6 and in the middle scallop of the posterior leaflet is 1.4. Much more of the mural leaflet is in apposition with the aortic leaflet during closure of the mitral valve.

Opening of the mitral valve

At the onset of diastole, opening is passive but rapid, the leaflets parting and projecting into the ventricle as left atrial pressure exceeds left ventricular diastolic pressure. Passive ventricular filling proceeds as atrial blood pours to the apex, directed by the pendant aortic valvular leaflet. The leaflets begin to float passively together, hinging on their anular attachments and partially occluding the ventricular inlet. Atrial systole now occurs, jetting blood apically and causing re-opening of the leaflets. As maximal filling is achieved, the leaflets again float rapidly together. Closure is followed by ventricular systole, which starts in the papillary muscles and continues rapidly as a general contraction of the walls and septum. Coordinated contraction of the papillary muscles increases the tension in the chordae and promotes joining of the corresponding points on opposing leaflets, preventing their eversion. With general mural and septal excitation and contraction, left ventricular pressure increases rapidly (see Fig. 57.20). The leaflets 'balloon' towards

the atrial cavity, and the atrial aspects of the rough zones come into maximal contact. Precise papillary contraction, and increasing tension in the chordae, continue to prevent valvular eversion and maintain valvular competence. The orifices and the leaflets of both atrioventricular valves undergo considerable changes in position, form and area during a cardiac cycle. Both valves move anteriorly and to the left during systole, and reverse their motion in diastole. The mitral valve reduces its orificial (anular) area by as much as 40% in systole; its shape changes from circular to crescentic at the height of systole, the anular attachment of its aortic leaflet being the concavity of the crescent. The attachment of its mural leaflet, although remaining convex, contracts towards the anterior cardiac wall. The smooth left ventricular outflow tract (aortic vestibule) terminates at the aortic valve leaflets. Although stronger in construction, the aortic valve resembles the pulmonary valve (see Figs 57.21, 57.24, 57.29) in possessing three semilunar leaflets, supported within the three aortic sinuses of Valsalva. Although the aortic valve, like the pulmonary valve, is often described as possessing an annulus in continuity with the fibrous skeleton, there is no complete collagenous ring that supports the attachments of the leaflets. As with the pulmonary valve, the anatomy of the aortic valve is dominated by the fibrous semilunar leaflet attachment (see Fig. 57.24C).

Mitral chordae tendineae

Mitral chordae tendineae resemble those supporting the tricuspid valve. False chordae are also irregularly distributed as in the right ventricle. They are single or multiple, thin, fibrous or fibromuscular structures that traverse the cavity of the left ventricle and have no connection with the valvular leaflets (Fig. 57.30). They occur commonly in human left ventricles and often cross the subaortic outflow. Histologically, false chordae sometimes contain extensions from the ventricular conducting tissues; these left ventricular bands are often identified on echocardiography. It has been suggested that false chordae produce premature ventricular contractions and be the possible cause of functional heart murmurs or innocent murmurs in children and young adults.

True mitral valve chordae may be divided into four types: inter-leaflet (commissural), rough zone (including the special strut chordae), 'cleft' and basal chordae. Most true chordae divide into branches from a single stem soon after their origin from the apical third of a papillary muscle, or proceed as single chordae that divide into several branches near their attachment. Basal chordae, in contrast, are solitary structures passing from the ventricular wall to the mural leaflet. There is such marked variation between the arrangement of the chordae that any detailed classification loses much of its clinical significance. Suffice it to say that, in the majority of hearts, the chordae support the entire free edges of the valvular leaflets, together with varying degrees of their ventricular aspects and bases, and there is some evidence to suggest that those valves with unsupported free edges become prone to prolapse in later life.

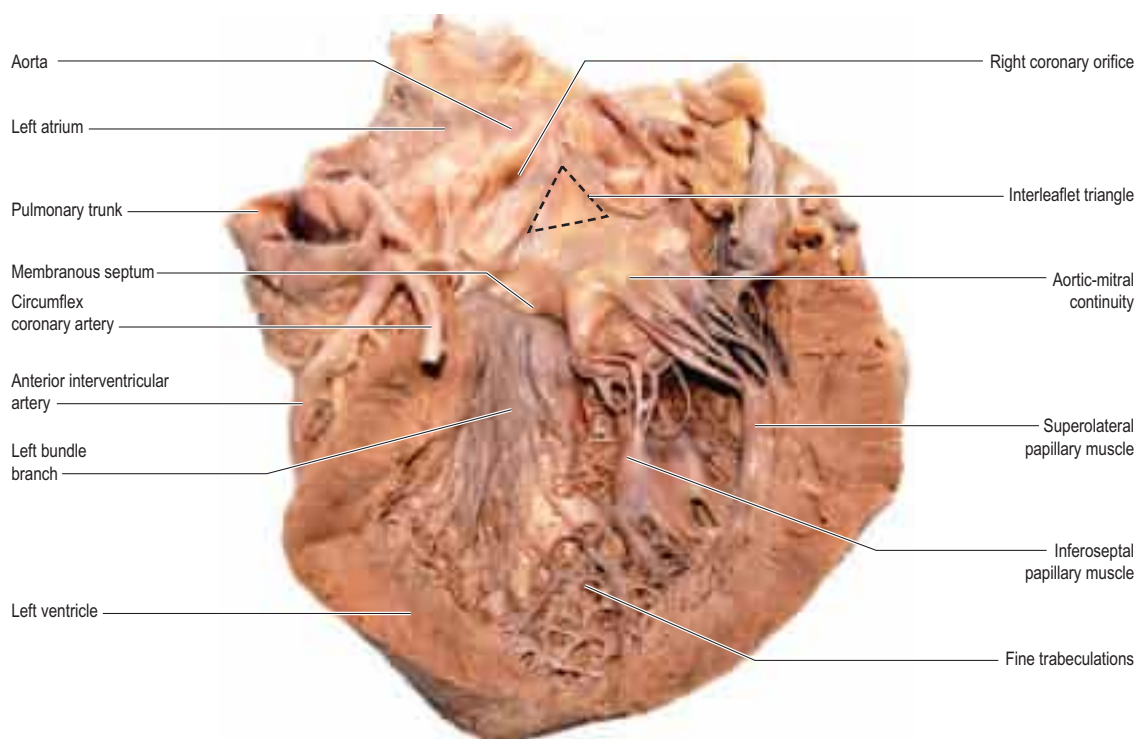


Fig. 57.30 The aorta has been transected at the level of the sinotubular junction in this cadaveric heart in order to reveal the closed leaflets of the aortic valve. Note how the supporting sinuses may be described as left coronary, right coronary and non-adjacent, according to the origin of the coronary arteries. However, this terminology could be ambiguous if the coronary arteries arise in variable locations, whereas the terminology of right, left and non-facing aortic sinus, respectively, is not ambiguous. Note the inter-leaflet triangles, membranous septum, and fibrous continuity between the aorta and mitral valve. (Specimen courtesy of M Loukas MD, PhD.)

Papillary muscles

The two muscles supporting the leaflets of the mitral valve vary in length and breadth, and may be bifid. The anterolateral (superolateral) muscle arises from the sternocostal mural myocardium, and the posteromedial (inferoseptal) from the diaphragmatic region (see Fig. 57.27). Chordae tendineae arise mostly from the tip and apical third of each muscle but sometimes take origin near their base. The chordae from each papillary muscle diverge and are attached to corresponding areas of closure on both valvular leaflets.

Aortic root

The aortic root, the anatomical bridge between the left ventricle and the ascending aorta, consists of the aortic valvular leaflets (supported by the aortic sinuses of Valsalva) and the inter-leaflet triangles interposed between their basal attachments (see Fig. 57.24). As such, it possesses significant length, but because of the semilunar attachment of the leaflets, it has no discrete proximal border. It is limited distally by the sinutubular junction (see Fig. 57.30). The essential feature is the semilunar attachments of the valve leaflets; their hinge lines cross the anatomical ventriculoarterial junction, marking a transition from the myocardium of the left ventricle to the fibroelastic tissue of the valve sinuses (Fig. 57.31). The muscular portion of the aortic root is roughly two-thirds of its widest circumference. Descriptions of the aortic root over the years have been bedevilled by accounts of a valve anulus. Although echocardiographers used to describe this proximal border in terms of an 'anulus', examination of cross-sections of the left ventricular outflow tract has never found any circular anatomical boundary or a distinct boundary of any kind (Loukas et al 2014a). There are at least two rings within the root; neither serves to support the valve leaflets, which are attached in semilunar fashion from the sinutubular junction to a basal ventricular attachment. Two leaflets are supported by muscle, and the third has an exclusively fibrous attachment. The root acts as a bridging structure not only anatomically, separating the myocardial and arterial components of the left ventricular pathway, but also functionally because its proximal and distal components can withstand considerable changes in ventricular and arterial pressures.

Aortic valve leaflets

The aortic valve leaflets are attached in part to the aortic wall and in part to the supporting ventricular structures. The situation is more complicated than in the pulmonary valve because parts of the leaflets also take origin from the fibrous subaortic curtain, and are continuous with the aortic leaflet of the mitral valve (see Fig. 57.29). This area of continuity is thickened at its two ends to form the right and left fibrous trigones (see Fig. 57.21). As with the pulmonary valve, the semilunar attachments incorporate segments of ventricular tissue within the bases of two of the aortic sinuses. The sinuses and leaflets are conveniently named as right, left and non-coronary, according to the origins of the coronary arteries (see Fig. 57.24C). However, the so-called non-coronary leaflet is better termed the non-adjacent leaflet because it rarely gives rise to a coronary artery.

The semilunar attachments incorporate three triangular areas (trigones) of aortic wall within the apex of the left ventricular outflow tract. They are interposed between the bulbous aortic sinuses and separate the cavity of the left ventricle from the pericardial space. Removal of the trigones in an otherwise intact heart is instructive in demonstrating the relationships of the aortic valve, which, justifiably, may be considered as the keystone of the heart. The first triangle, between the non-coronary and left coronary leaflets, has a base continuous inferiorly with the fibrous aortic-mitral curtain. The second triangle, between right and non-coronary leaflets, has the membranous components of the inter-ventricular septum as its base and thus 'faces' the right ventricle, whereas its apex 'points' towards the transverse pericardial space behind the origin of the right coronary artery. The third triangle, between the two coronary leaflets, has its base on the muscular interventricular septum and its apex 'points' to the plane of space found between the aortic wall and the free-standing sleeve of right ventricular infundibular musculature that supports the leaflets of the pulmonary valve. Although the basal attachments of each aortic valvular leaflet are thickened and collagenous at their ventricular origins, the leaflets lack a continuous collagenous circular skeletal support; valvular function depends primarily upon the semilunar attachments of the leaflets.

The leaflets are endocardial folds with a central fibrous core. With the valve half-open, each equals slightly more than a quarter of a sphere, an approximate hemisphere being completed by the corresponding sinus. Each leaflet has a thick basal border, deeply concave on its aortic aspect, and a horizontal free margin that is only slightly thickened,

except at its midpoint, where there is an aggregation of fibrous tissue, the valvular nodule of the semilunar leaflet. The fibrous core that flanks each nodule is tenuous, forming the lunules of translucent and occasionally fenestrated valvular tissue; fenestrations are of no functional significance. The aortic surface of each leaflet is rougher than its ventricular aspect.

Confusingly, three sets of names are used to describe the aortic leaflets. Posterior, right and left refer to their fetal positions before full cardiac rotation has occurred (Ch. 52). Corresponding terms based on their approximate positions in maturity are anterior, left and right posterior. Widespread clinical terminology, which links both leaflets and sinuses to the origins of the coronary arteries, has replaced anterior, left and right leaflets with right and left coronary and non-adjacent (and, usually, non-coronary) leaflets, respectively; these clinical terms are preferable in the normal heart because they are simple and unambiguous.

Aortic sinuses (of Valsalva)

The aortic sinuses are more prominent than those in the pulmonary trunk. The upper limit of each sinus reaches considerably beyond the level of the free border of the leaflet and forms a well-defined circumferential sinutubular ridge on the aortic inner surface, just above the aortic valvular leaflets (see Fig. 57.24C). Coronary arteries usually open near this ridge within the upper part of the sinus but are markedly variable in their origin. The walls of the sinuses are largely collagenous near the attachment of the leaflets but the amount of lamellated elastic tissue increases with distance from the zone of attachment. At the midlevel of each sinus, its wall is about half the thickness of the supravalvular aortic wall and less than one-quarter of the thickness of the sinutubular ridge. At this level, the mean luminal diameter at the commencement of the aortic root is much larger than that of the ascending aorta; these details are functionally significant in the mechanism of valvular motion. A linear relationship between the diameter of the aortic sinus and the square root of body surface area has been described in children (Kaiser et al 2008).

Opening of the aortic valve

During diastole, the closed aortic valve supports an aortic column of blood at high but slowly diminishing pressure (see Fig. 57.20). Each sinus and its leaflet form a hemispherical chamber. The three nodules are apposed and the margins and lunular parts of adjacent leaflets are tightly apposed on their ventricular aspects. From the aortic aspect, the closed valve is triradiate, three pairs of closely compressed lunules radiating from their nodules to their peripheral commissural attachments at the sinutubular junction (see Fig. 57.21). As ventricular systolic pressure increases, it exceeds aortic pressure and the valve is passively opened.

The fibrous wall of the sinuses nearest the aortic vestibule is almost inextensible but, more superiorly, the wall is fibroelastic. Under left ventricular ejection pressure, the radius here increases 16% in systole, as the commissures move apart to form a fully open triangular orifice. The free margins of the leaflets then become almost straight lines between peripheral attachments. However, they do not flatten against the sinus walls, even at maximal systolic pressure, which is probably an important factor in their subsequent closure. During ejection, most blood enters the ascending aorta but some enters the sinuses, forming vortices that help to maintain the triangular 'mid position' of the leaflet during ventricular systole and also probably initiate their approximation at the end of systole. Tight and full closure ensues with the rapid decrease in ventricular pressure in diastole. Commissures narrow, nodules aggregate and the valve reassumes its triradiate form. Experiments indicate that 4% of ejected blood regurgitates through a valve with normal sinuses, whereas 23% regurgitates through a valve without them. The normal structure of the aortic sinuses also promotes non-turbulent flow into the coronary arteries.

Echocardiography

Echocardiography allows a detailed assessment of the functional anatomy of the heart. The gross anatomy of the heart can be evaluated by two-dimensional echocardiography in the parasternal, apical, suprasternal and subcostal positions (Fig. 57.32). The standardized planes used are long-axis, short-axis and four-chamber. The long-axis view is obtained by placing the ultrasound transducer in the left apico-sternal position and provides detailed images of the left ventricle, aorta, left atrium, and mitral and aortic valves (Fig. 57.32C). Angling the beam towards the right also allows assessment of the right atrium, right ventricle and tricuspid valves. Rotating the transducer by 90° in the

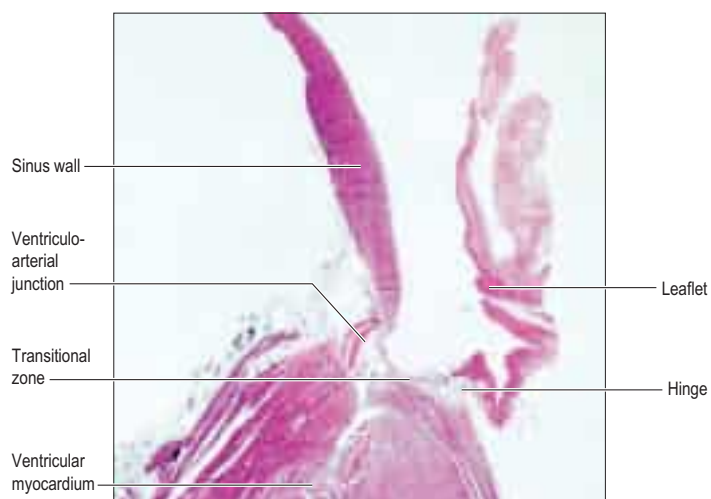


Fig. 57.31 A histological section through one of the coronary aortic sinuses to demonstrate the way in which ventricular muscle supports the transition from the fibroelastic wall of the sinus to the tissues of the leaflet. The transitional area is anchored to the ventricular muscle. The hinge of the leaflet is well below the level of the ventriculoarterial junction. Importantly, there is no ring-like structure in the form of an 'anulus' supporting the hinge of the leaflet within the ventricle. The muscular tissue of the ventricle is stained pink; fibrous tissue, which would show up as a characteristic light colour with this stain, is absent. Haematoxylin and eosin stain. (Courtesy of Diane E Spicer.)

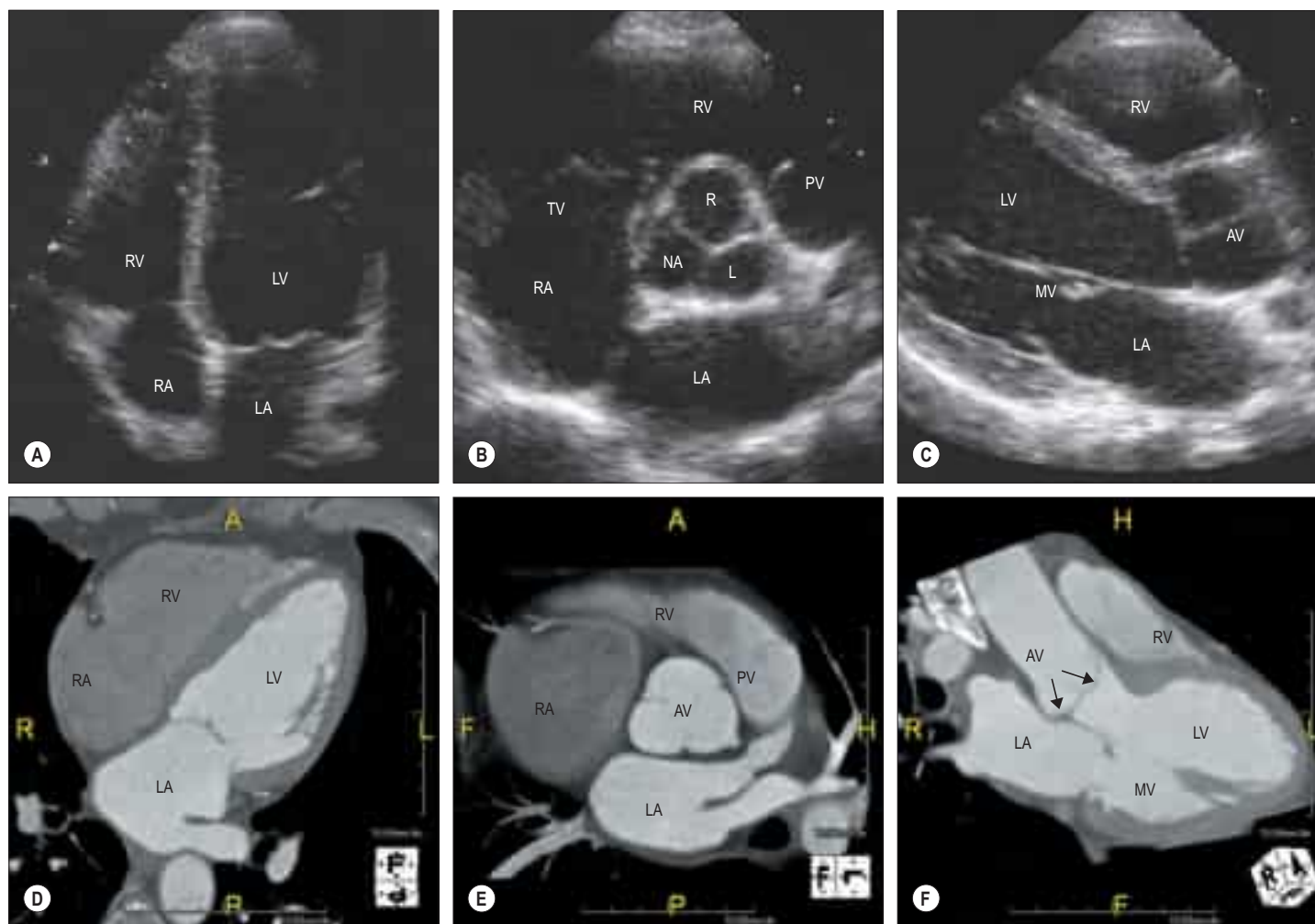


Fig. 57.32 Cardiac anatomy shown by transthoracic echocardiography and CT. **A**, Four-chamber view. **B**, Short-axis view at aortic valve level. Note three aortic leaflets – right, left and non-adjacent – and the central position of the aorta. **C**, Parasternal long-axis view. **D–F**, Corresponding images by CT. Note the different orientation compared with transthoracic echocardiography (looked at from below, foot to head). Abbreviations: A, anterior; AV, aortic valve; F, foot; H, head; L, left; LA, left atrium; LV, left ventricle; MV, mitral valve; NC, non-adjacent; P, posterior; PV, pulmonary valve; R, right; RA, right atrium; RV, right ventricle; TV, tricuspid valve. (Images courtesy of Dr Konstantinos Dimopoulos, Royal Brompton and Chelsea and Westminster Hospitals, London.)

clockwise direction produces the short-axis view, which allows assessment of the left ventricle, papillary muscles, chordae tendineae and mitral valves (Fig. 57.32B). The four-chamber view demonstrates the ventricles, atria, and mitral and tricuspid valves (Fig. 57.32A). Rotation of the transducer allows two-chamber views of the heart and more detailed assessment of the aorta and aortic valves. Cardiac MRI and CT provide similar information on cardiac structure and function (see Figs 57.2, 57.35A,B), together with complementary information on great vessels (Fig. 57.32D–F) and other extracardiac intrathoracic structures.

CONNECTIVE TISSUE AND FIBROUS SKELETON OF THE HEART

From epicardium to endocardium and from the orifices of the great veins to the roots of the arterial trunks, the intercellular spaces between contractile and conduction elements are permeated by connective tissue; the amount, arrangement and texture vary greatly with location. Over much of the heart, a fine layer of areolar tissue is found beneath the mesothelium of the serous (visceral) epicardium that accumulates subepicardial fat, concentrated along the acute margin, the atrioventricular and interventricular grooves, and their side channels. The coronary vessels and their main branches are embedded in this fat; the amount increases with age. The endocardium also lies on a fine areolar tissue rich in elastic fibres. Fibrocellular components of these subepicardial and subendocardial layers blend on their mural aspects with the endomysial and perimysial connective tissue on the myocardium. Each cardiac myocyte is invested by a delicate endomysium composed of fine reticular, collagen and elastin fibres embedded in ground substance. This matrix is lacking only at desmosomal and gap junctional contacts of intercalated discs. Similar arrangements apply to ventricular conduction myocytes and their extensive working myocardial contacts. The

connective tissue matrix itself is interconnected laterally to form bundles, strands or sheets of macroscopic proportions showing a complex geometric pattern. Surrounding and attaching to larger myocardial bundles are stronger perimysial condensations. The overall pattern is described in terms of struts and weaves. Despite its importance, the myocardial matrix cannot be grossly dissected.

Running at the ventricular base is a complex framework of dense collagen with membranous, tendinous and fibroareolar extensions, intimately related to atrioventricular valves and the aortic orifice. The whole is sufficiently distinct to be termed the fibrous skeleton of the heart, but although it is often stated that all four valves are contained within this skeleton, this is *not* the case. The leaflets of the pulmonary valve are supported on a free-standing sleeve of right ventricular infundibulum that can easily be removed from the heart without disturbing either the fibrous skeleton or the left ventricle. Another point of confusion is the idea that the ‘skeleton’ provides the support for the cardiac valves. In reality, it is the overall structure of the atrioventricular junctions that supports the mitral and tricuspid valves, whereas the arterial valves are hinged within the valvular sinuses. The fibrous skeleton is strongest at the junction of the aortic, mitral and tricuspid valves, the so-called central fibrous body (see below) (see Figs 57.21, 57.23). Two pairs of curved, tapering, collagenous prongs, fila coronaria, extend from the central fibrous body. They are stronger on the left, where they pass partially around the mitral and tricuspid orifices. These orifices are almost co-planar and incline to face the cardiac apex. In contrast, the aortic valve faces superiorly, lying anterosuperior and to the right of the mitral orifice. Two of the leaflets of the aortic valve are in fibrous continuity with the aortic leaflet of the mitral valve; this subaortic curtain is also an integral part of the fibrous skeleton (see Figs 57.21, 57.24C). The two ends of the curtain are strengthened as the right and left fibrous trigones, which are the strongest parts of the skeleton. The right trigone, together with the membranous septum, constitutes the central fibrous

body, which is penetrated by the atrioventricular bundle of His (see below). The membranous septum is crossed on its right aspect by the attachment of the tricuspid valve, dividing the septum into atrioventricular and interventricular components.

The aortic root is central within the fibrous skeleton and is often described in terms of an 'anulus' integrated within the fibrous skeleton (Anderson et al 2013b). However, as with the pulmonary valve, the structure of the aortic root corresponds to the triple fibrous semilunar attachments of its leaflets. Within this complex circumferential zone are three crucially important triangular areas that separate, on the ventricular aspect, the aortic bulbous sinuses that house the valvular leaflets. As a whole, three triangles, known as the subaortic spans, can be conceptualized in terms of a three-pointed coronet; their triangular apices correspond to the tips of the valvular commissures and their walls, significantly thinner than those of the sinuses, consist variously of collagen or admixed muscle strands and fibroelastic tissue. They form the subvalvular extensions of the aortic vestibule. The first span is the interval between the non-coronary and left coronary sinuses that is filled with the deformable subaortic curtain. The second span between the non-coronary and right coronary sinuses is continuous with the anterior surface of the membranous septum. The third span, between the two coronary aortic sinuses, is filled with loose fibroelastic tissue and separates the extension of the subaortic root from the wall of the free-standing subpulmonary infundibulum. Previously, this was thought to be the location of the tendon of the infundibulum (conal (conus) ligament) (Loukas et al 2007); research using mouse hearts has shown that the ligament is found at the level of the sinutubular junctions at the site of initial fusion of the distal outflow cushions. The line of proximal fusion is the longitudinal raphe sometimes seen in the muscular subpulmonary infundibulum.

Similar fibrous triangles are found separating the sinuses of the pulmonary trunk but these are significantly less robust. Use of the terms mitral and tricuspid 'anuli' implies that the atrioventricular valves are supported by discrete circular fibrous rings, when, in reality, they are D-shaped structures integrally attached to the fibrous cardiac skeleton. It is therefore inaccurate to consider these valvular orifices as circular. Rather, the straight edge of the 'D'-shaped mitral valve represents a fibrous continuity between the anterior leaflet and the aortic root, and the remainder supports the mural leaflet. The straight edge of the tricuspid valve represents the attachment of the septal leaflet, marking the inferior border of the triangle of Koch, and the remainder supports the anterosuperior and inferior leaflets. The valvular attachments of the atrioventricular valves are not simple, rigid collagenous structures but dynamic, deformable lines that vary greatly at different peripheral points and change considerably with each phase of the cardiac cycle and with increasing age. The tricuspid attachments are even less robust than those of the mitral valve. At several sites, only fibroareolar tissue separates the atrial and ventricular muscular masses.

The fibrous skeleton ensures electrophysiological discontinuity between the atrial and ventricular myocardial masses (except at the site of penetration of the conduction tissue). It also provides direct attachment for the myocardium and for fibrous tissue throughout the heart as a support matrix for a three-dimensional meshwork of cardiomyocytes.

Arrangement of cardiomyocytes The microstructure of cardiac muscle is described in detail in Chapter 6. Ventricular myocytes form a three-dimensional mesh in a supporting fibrous matrix. One population is aligned so that the long axis of the aggregated cells is tangential to the epicardial and endocardial borders, albeit with marked variation in the angulation relative to the ventricular equator. Correlation with measurements taken using force probes shows that these myocytes produce the major unloading of the blood during ventricular systole. A second population is aligned at angles of up to 40° from the epicardium towards the endocardium and produces auxotonic forces during the cardiac cycle. The three-dimensional arrangement of the mesh as demonstrated in MRI of the porcine heart also mediates the realignment of cardiomyocytes that must take place during ventricular contraction and accounts for the extent of systolic mural thickening (Fig. 57.33).

The finding that the number of cardiomyocytes in the left ventricle increases between the first and the twentieth year of life suggests that children may be able to regenerate myocardium (Mollova et al 2013).

CONGENITAL CARDIAC MALFORMATIONS

Congenital malformations of the heart are common and amount to about one-quarter of all developmental anomalies. Their incidence is



Fig. 57.33 A reconstruction of a porcine heart made following diffusion tensor MRI of an autopsied heart. The tracks are created from the first eigenvectors and represent the alignment of the aggregated myocytes. This technique enables the identification of reproducible tracks, or pathways, through the myocardium that connect remote end and epicardial regions by means of simultaneous changes in helical and intrusion angles. (With permission from Anderson R, Smerup M, Sanchez-Quintana D, Loukas M, Lunkenheimer P. 2009. The three-dimensional arrangement of the myocytes in the ventricular walls. Clin Anat 22:64–76.)

estimated at 8 per 1000 live births but they are found in up to 2% of stillbirths. Only a small proportion of the anomalies are directly attributable to genetic or environmental factors and the majority are the result of multifactorial events.

Abnormalities of cardiac position

 Available with the *Gray's Anatomy* e-book

Acyanotic cardiac defects

 Available with the *Gray's Anatomy* e-book

Cyanotic cardiac defects

 Available with the *Gray's Anatomy* e-book

CONDUCTION TISSUES

The cells of cardiac muscle differ from those of skeletal muscle in having the inherent ability to contract and relax spontaneously. This myogenic rhythm is shown by small pieces of cardiac tissue, and even isolated myocytes. Ventricular cells contract and relax at a lower frequency than atrial cells, but in the intact heart, both are synchronized to a more rapid rhythm, generated by pacemaker tissue in the sinus and atrioventricular nodes and conveyed by a system of specialist conduction fibres in the atrioventricular conduction axis and the ventricular conducting pathways (Purkinje cells). For an account of excitation–contraction coupling in cardiac muscle, see page 138.

The anatomical arrangement of these tissues is described in the context of the heart. Here, consideration is restricted to the cells that make up the impulse-generating and conduction system. All are modified cardiac cells. Three types within a continuum of morphology may be distinguished from normal working cells, namely: nodal cells, transitional cells and Purkinje cardiomyocytes.

Overview of the conduction system

Of all the cells in the heart, those of the sinu-atrial node generate the most rapid rhythm, and therefore function as the cardiac pacemaker. The impulse, believed to be generated in the nodal cells, is transmitted

The most severe abnormality of position is an extrathoracic heart (ectopia cordis), where the heart usually projects to the surface through the lower thoracic and upper abdominal wall, remaining covered, in most instances, by the fibrous pericardium, and usually accompanied by herniation of the abdominal contents. A mirror image, i.e. reversal in cardiac shape and position, occurs predominantly in the right hemithorax (dextroposition). The apex of the heart may be directed to the right instead of the left (dextrocardia). This arrangement may be part of 'situs inversus totalis', where the heart, great vessels and abdominal organs all occupy mirror-imaged positions. The heart may also be right-sided in Kartagener's syndrome (a subgroup of ciliary dyskinesias).

More usually, an abnormal location of the heart occurs in cases of isomerism, in which both sides of the thorax, including the main bronchi and the atrial appendages, retain features of either morphological right- or left-sidedness. Isomerism is also commonly associated with the anomalous arrangement of the abdominal organs: right isomerism with absence of the spleen (asplenia) and left isomerism with multiple spleens (polysplenia). Intracardiac anatomy in cases of isomerism is almost universally abnormal, and there is a range of heart defects (usually more simple in left and more complex in right isomerism).

These intrathoracic abnormal arrangements, with or without abdominal abnormalities, often first manifest themselves incidentally following chest radiography. Reversal of normal anatomy has obvious clinical implications, a typical example being a patient with mirror-image arrangements and appendicitis, who presents with an acute left lower quadrant pain.

Most congenital heart abnormalities can be detected during antenatal ultrasound screening (Fig. 57.34). Neonates with severe congenital heart disease may present with tachypnoea, difficulty in feeding, cyanosis and/or cardiovascular collapse, and there may be audible murmurs on auscultation. The vast majority of abnormalities may be detected by postnatal echocardiography; in a very few cases, cardiac catheterization and direct measurements of pressure, oxygen saturations and angiography may be required.

Acyanotic cardiac defects are the result of either left-to-right cardiac shunting through heart defects (intra- or extracardiac) or of obstruction. Left-to-right shunting leads to an increased workload and stress on the heart and the lungs as a consequence of increased pulmonary blood flow and increased pulmonary venous return. Depending on the location of the shunt and the magnitude of left-to-right shunting, patients are at risk of developing pulmonary arterial hypertension unless timely heart surgery is undertaken. Examples of defects leading to left-to-right shunting are simple septal defects such as atrial or ventricular septal defects or patent ductus arteriosus, or more complex atrioventricular septal defects, and/or a combination of any of these defects with abnormal atrioventricular or ventriculoarterial connections (e.g. a double-outlet right ventricle, where more than 50% of both the aorta and the pulmonary artery originate from the right ventricle) (Ch. 52). Obstructive lesions may involve the atrioventricular or arterial valves (mitral, aortic or pulmonary valve stenosis), or narrow vessels (e.g. coarctation of the aorta or interrupted aortic arch). Depending on the level and severity of the obstruction, patients may have a range of

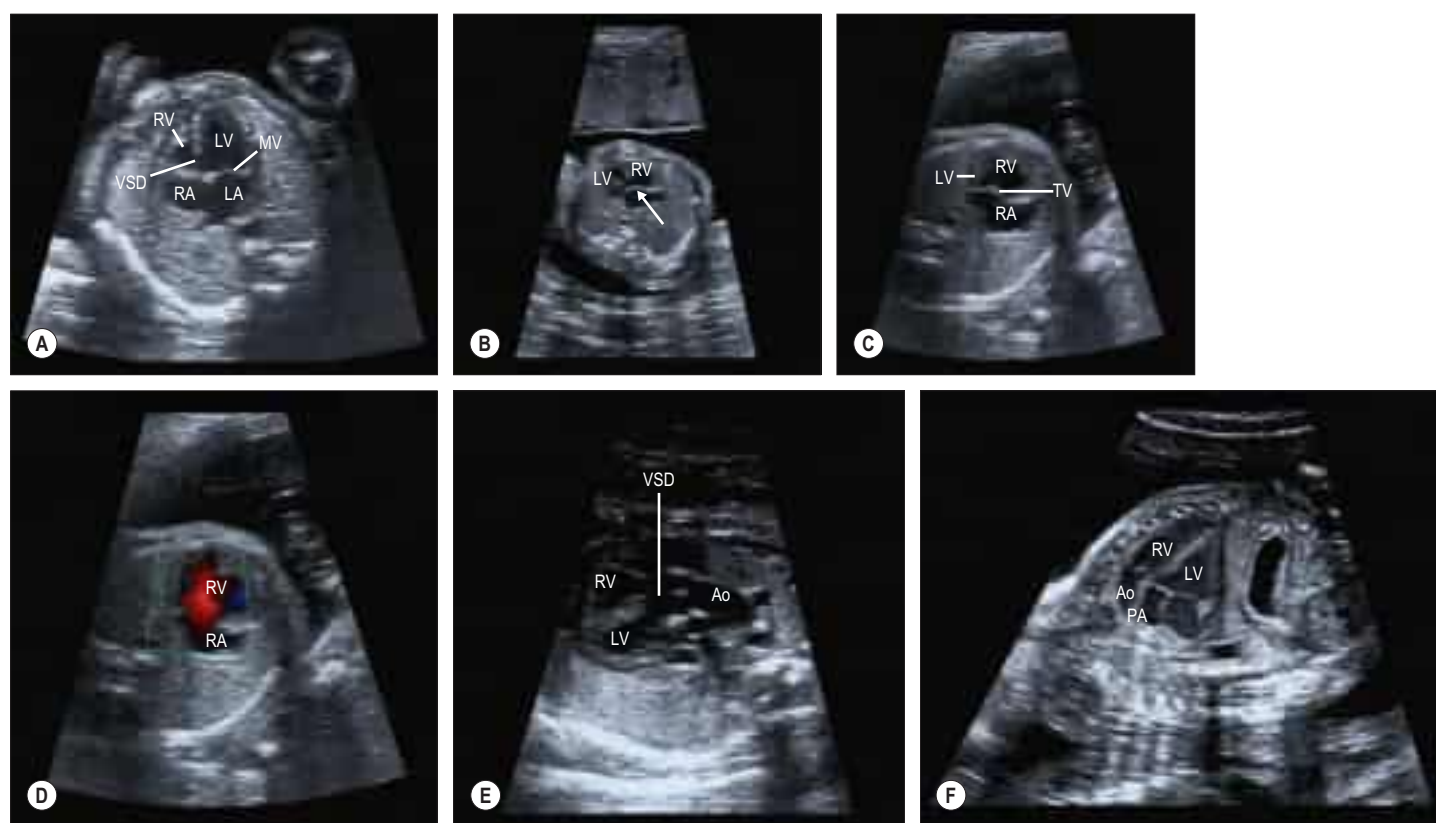


Fig. 57.34 **A**, Tricuspid atresia at 21 weeks' gestation. A four-chamber view showing a large left ventricle and a hypoplastic right ventricle. The mitral valve is seen; no tricuspid valve is identified. There is an associated ventricular septal defect. **B**, An atrioventricular septal defect at 20 weeks' gestation. A common atrioventricular valve (arrow) is associated with a defect in the atrioventricular septum. **C**, Hypoplastic left heart syndrome at 32 weeks' gestation. This four-chamber view shows a large right atrium, right ventricle and tricuspid valve. There is a hypoplastic left ventricle; no mitral valve is identified. **D**, Hypoplastic left heart syndrome at 32 weeks' gestation. This four-chamber view shows colour flow from right atrium to right ventricle across the tricuspid valve; no mitral valve is identified and there is no flow in the left side of the heart. **E**, Tetralogy of Fallot at 33 weeks' gestation. The aorta arises astride a ventricular septal defect. **F**, Transposition of the great arteries at 21 weeks' gestation. The aorta and pulmonary artery arise in parallel orientation. The aorta arises from the right ventricle and the pulmonary artery from the left ventricle. Abbreviations: Ao, aorta; LA, left atrium; LV, left ventricle; MV, mitral valve; PA, pulmonary artery; RA, right atrium; RV, right ventricle; TV, tricuspid valve; VSD, ventricular septal defect. (Courtesy of Gurleen Sharland, Evelina London Children's Hospital.)

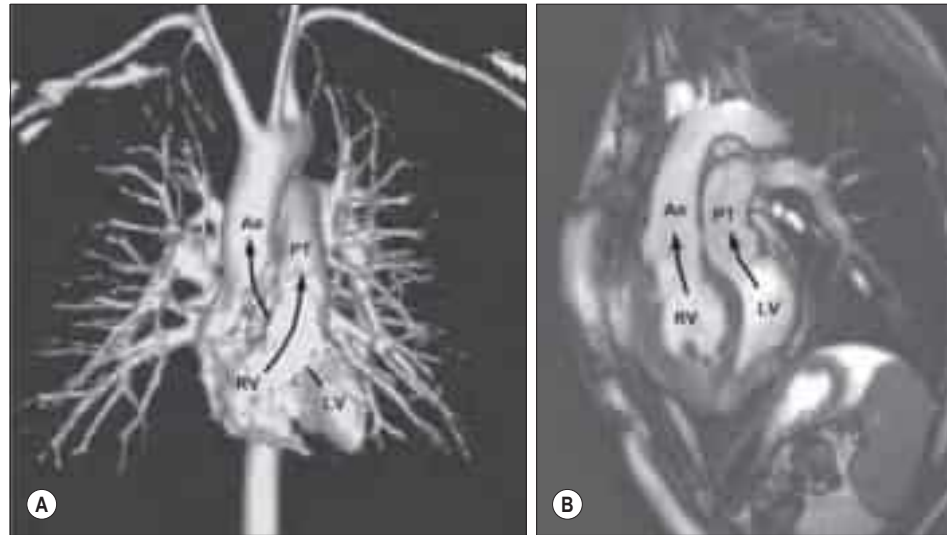


Fig. 57.35 **A**, Cardiac magnetic resonance angiography in a patient with a normal heart. Note the criss-cross relationship of the great arteries; the aorta arises from the left ventricle and runs to the right and posteriorly to the pulmonary trunk (at its origin) and towards the head. The pulmonary artery arises from the right and anterior right ventricle, crosses over and then runs to the left of the aorta and towards the back (where it bifurcates into the right and left pulmonary arteries at the underside of the aortic arch). **B**, A cardiac magnetic resonance angiogram from a patient with transposition of the great arteries. Note the parallel or side-by-side relationship of the great vessels and loss of their normal criss-cross relationship. The aorta runs anteriorly and comes off the right ventricle. The hypertrophied right ventricle supports the systemic circulation, and the ventricular septum bows from right to left. Abbreviations: Ao, aorta; LV, left ventricle; PT, pulmonary trunk; RV, right ventricle. (Courtesy of Dr Philip Kilner, Royal Brompton Hospital, London.)

clinical presentations from an asymptomatic heart murmur (common) to cardiovascular collapse (uncommon). Treatment in all cases is directed towards normalizing heart workload, systemic and pulmonary blood flow, and cardiac output, and may be surgical and/or catheter-based.

Cyanotic cardiac defects may be attributable to a right-to-left intra- or extracardiac shunt or to a severe reduction in pulmonary blood flow. They may be caused by simple lesions such as severe pulmonary stenosis with an atrial septal defect and right-to-left shunting; moderate lesions, including Fallot's tetralogy, in which there is a ventricular septal defect, right ventricular outflow obstruction, right ventricular hypertrophy and an overriding aorta; or more complex lesions, including trans-

position of the great arteries (Fig. 57.35; p. 924) and tricuspid and pulmonary atresia (often in the setting of hypoplastic right ventricle and functionally univentricular physiology). Neonates with severe cyanotic congenital heart defects may be dependent on the patency of the ductus arteriosus; early diagnosis, treatment with intravenous prostaglandin infusion and timely transfer to a tertiary centre for more definitive therapy is key to survival and good long-term outcomes.

Major advances in the diagnosis and management of infants with congenital cardiac malformations in recent years have resulted in an increased number of adolescents and adults with palliated or repaired, but not cured, congenital heart disease. This is an area that poses multiple challenges to both cardiovascular and other medical disciplines.

over preferentially conducting pathways to right and left atria, and to the atrioventricular node. At the atrioventricular node, the impulse is delayed by 40 ms, allowing the atria to eject their contents fully before commencement of ventricular contraction, and also placing an upper limit on the frequency of signals that can be transmitted to the ventricles. The transitional cells interpose between the atrial cardiomyocytes and the nodal cells. There is no transition in humans between the node and the penetrating bundle, and the cells are virtually identical. Here, they become continuous with the more distinctively appearing Purkinje cells. Conduction of the impulse is rapid in the bundle and its branches (2–3 m/sec, as opposed to 0.6 m/sec in normal myocardium). The cardiac impulse therefore arrives at the apex of the heart before spreading through the ventricular walls, producing a properly coordinated ventricular ejection. The human heart beats ceaselessly at 70 cycles every minute for many decades, maintaining perfusion of pulmonary and systemic tissues. The rate and stroke volume fluctuate in response to prevailing physiological demands. The principal events in a cardiac cycle are summarized in [Figure 57.20](#), including the electrical events recorded in the electrocardiogram; the mechanical sequences of diastole, atrial systole, volumetric contraction, ejection and isovolumetric relaxation in ventricular systole; the acoustic phenomena recorded in the phonocardiogram; the pressure profiles of right and left hearts and arterial trunks; and the sequences of valvular events.

Cardiac efficiency depends on precise timing of the operation in interdependent structures. Passive diastolic filling of the atria and ventricles is followed by atrial systole, stimulated by discharge from the sinu-atrial node, which completes ventricular filling. Excitation and contraction of the atria must be synchronous and finish before ventricular contraction; this is effected by a delay in the conduction of excitation from atria to ventricles. Thereafter, ventricular contraction proceeds in a precise manner. The atrioventricular valves are closed by ventricular systole, which must occur prior to closure; ventricular activation must therefore precede valvular closure, which spreads from the ventricular apices towards the outflow tracts and orifices.

Cardiac contraction originates unequivocally in specialized cardiomyocytes, but neural influences are important in adapting the intrinsic cardiac rhythm to functional demands from the entire body. All cardiomyocytes are excitable. They display autonomous rhythmic depolarization and repolarization of their cell membranes, conduction of waves of excitation via gap junctions to adjacent cardiomyocytes, and excitation–contraction coupling to their actomyosin complexes. These properties are developed to different degrees in different sites and in different types of cardiomyocyte ([Ch. 6](#)). The rate of depolarization and repolarization is slowest in the ventricular myocardium, intermediate in the atrial muscle, and fastest in the myocytes of the sinu-atrial node. The latter override those generating slower rhythms and, in the normal heart, are the locus for the rhythmic initiation of cardiac cycles. Conversely, conduction velocity is slow in nodal myocytes, intermediate in general ‘working’ cardiac myocytes, and fastest in the myocytes of the ventricular conduction system.

The nodes and networks of the so-called specialized myocardial cells constitute the cardiac conduction system ([Figs 57.36–57.37](#)). The components of this system are the sinus and atrioventricular nodes, the atrioventricular bundle with its left and right bundle branches, and the subendocardial plexus of ventricular conduction cells (Purkinje cells). The main pacemaker rhythm of the heart (sinus rhythm) is generated within the system but is influenced by nerves. It is transmitted specifically from atria to ventricles via the atrioventricular node and bundle. The spread of excitation is very rapid but not instantaneous. Different parts of the ventricles are excited at slightly different times, with important functional consequences. Failure of the conduction system will not block cardiac contraction but the system will become poorly coordinated. The rhythm will be slower because it originates from a spontaneous (myogenic) activity in the working cardiac myocytes or in a subsidiary pacemaker in a more distal part of the diseased or disrupted conduction system. There are no specialized internodal and interatrial conduction pathways; the excitation emanating from the sinu-atrial node spreads to the atrial musculature and to the atrioventricular node through ordinary atrial working myocardium. The geometric arrangement of fibres along well-organized atrial muscle bundles, e.g. the terminal crest, the rims of the fossa ovalis and Bachmann’s bundle, ensures that conduction is marginally more rapid than elsewhere within the atrium.

Sinu-atrial node

In the majority of hearts, the sinu-atrial node is a crescent-shaped structure, between 8 and 25 mm long, located at the embryonic junction between the venous part (sinus venosus) and the atrium proper derived

parts of the right atrium (see [Figs 57.13A, 57.37](#)). In hearts over 65 years of age, a layer of connective or fatty tissue between the subendocardium and the body of the node may sometimes render it visible to the naked eye. Extending on the right from the crest of the right appendage, the node typically courses posteroinferiorly into the upper part of the terminal groove; in about 1 in 10 hearts it extends in horseshoe fashion across the crest of the appendage. Nodal tissue does not occupy the full thickness of the right atrial wall from epicardium to endocardium but sits as a wedge of specialized subepicardial tissue within the terminal groove. Its location is consistently marked by a large central artery, which originates from either the proximal part of the right coronary or circumflex arteries in equal proportion. Nodal cells are grouped circumferentially around the sinu-atrial nodal artery, packed within a dense matrix of connective tissue as interlacing strands of myocytes. They are smaller, paler and more empty-looking than working atrial myocardial fibres, being 5–10 µm in their greatest diameter, with a large central nucleus. Their pale appearance is attributable to the sparsity of organelles; myofibrils are few and irregularly arranged, and there is no proper sarcotubular system with little glycogen.

The sinu-atrial node is described as having a head, body and tail, from which 5–8 short digitations of nodal tissue radiate towards the superior vena cava, subepicardium and crista terminalis, penetrating the working atrial myocardium ([Figs 57.38–57.39](#)). At the nodal periphery, nodal cells intermingle with slender, fusiform, transitional ‘linking’ cells that are part of a heterogeneous cellular group intermediate in appearance between nodal cells and normal working atrial myocytes. (For further reading on the use of molecular markers to map the extent of nodal tissue, see [Monfredi et al \(2010\)](#).) There are no autonomic ganglion cells within the node, although many border it anteriorly and posteriorly. Nerve fibres are present but do not appear to contact the nodal myocytes.

Paranodal area

A paranodal area within the terminal crest, between the cells of the sinu-atrial node and the working atrial cardiomyocytes, and possessing properties of both nodal and atrial tissues, has been identified ([Molenaar et al 2011](#)). Its precise function remains to be determined, but computer simulations suggest that it is an integral part of normal atrial activation and may play a role in pacemaking.

Internodal atrial myocardium

Specialized pathways have been alleged to connect the sinus and atrioventricular nodes, but there is no evidence that tracts insulated from the working myocardium, as occurs in the ventricular conduction pathways, exist within the atrial walls. The impulse generated by the sinu-atrial node is conducted more rapidly via the long axis of the atrial muscle bundles than it is transversely. The main pathways for conduction towards the atrioventricular node are the terminal crest and the margins of the oval fossa. Conduction to the left atrium is preferentially through Bachmann’s bundle; additional pathways occur through the margins of the oval fossa and through the muscular connections between the walls of the coronary sinus and left atrium. During development, the walls of the systemic venous sinus are exclusively composed of primary myocardium. Although pathways within these walls may be identified in prenatal stages, primary myocardium is slowly conducting, which means that these pathways do not represent areas of preferentially rapid conduction.

Atrioventricular node

The atrioventricular node is an oblique, half-oval atrial structure, located within the atrial component of the muscular atrioventricular septum and separated from the ventricular musculature by the insulating tissues of the atrioventricular groove (see [Fig. 57.37](#)). Its anatomical landmarks are the boundaries of the triangle of Koch, i.e. the attachment of the septal leaflet of the tricuspid valve inferiorly, the ostium of the coronary sinus basally and the tendon of Todaro superiorly (see [Fig. 57.13](#)). The atrial aspect of the node is convex and overlain by atrial myocardium. Its left margin is concave and abuts on to the superior aspect of the central fibrous body. The basal end projects into the atrial muscle and the anteroinferior end enters the central fibrous body to become the penetrating atrioventricular bundle.

The node consists of two zones, compact and transitional, pervaded by an irregular collagenous reticulum that enmeshes the myocytes ([Fig. 57.40](#)). The compact zone consists of frequently stratified, interlocking nodal cells. Cells from the right and left atrial walls and the atrial septum feed directly into the compact zone. The transitional zone envelops the compact portion of the node and consists of elongated ‘transitional cells’, intermediate in morphology and function between compact nodal cells and working atrial cardiomyocytes.

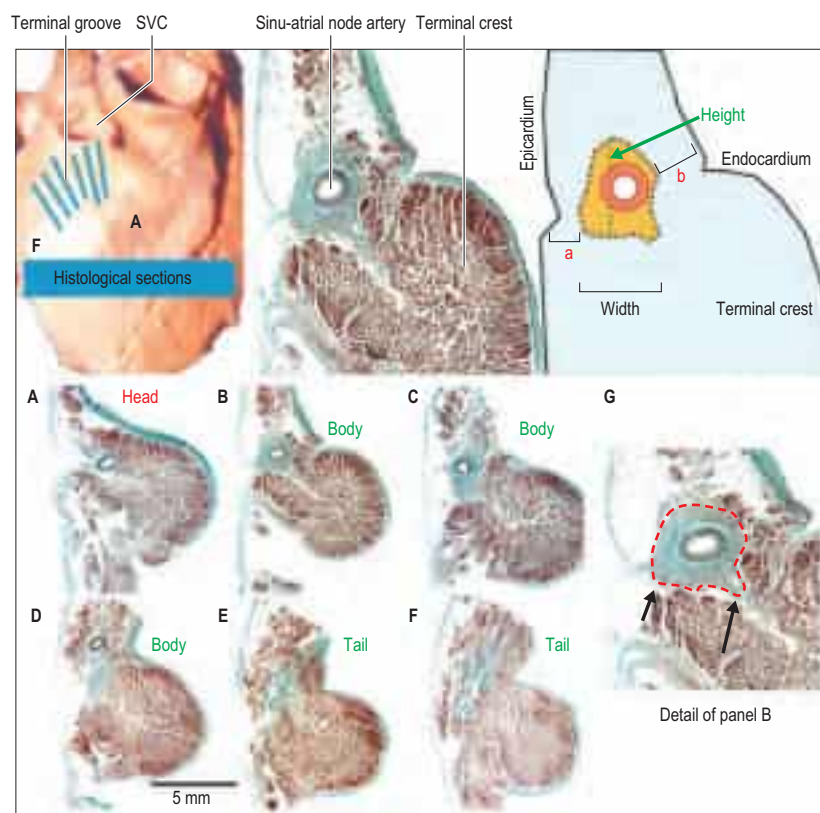


Fig. 57.38 Measurements of the width and height of the sinu-atrial node and the distances of nodal tissue to epicardium (a) and endocardium (b), measured in cadaveric tissue. (Top) A gross cadaveric specimen together with an example of a histological section and a diagram to indicate where measurements were made on each histological section. (Bottom) Histological sections (Masson trichrome stain) taken through levels A–F as indicated on the gross specimen. (G) The red dotted line delineates the nodal boundaries. Note the irregular contour of the node and the extensions towards the neighbouring myocardium (arrows). Abbreviations: SVC, superior vena cava. (With permission from Sánchez-Quintana D, Cabrera JA, Farré J, Climent V, Anderson RH, Ho SY. 2005. Sinu-atrial node revisited in the era of electroanatomical mapping and catheter ablation. 2005 91:189–94.)

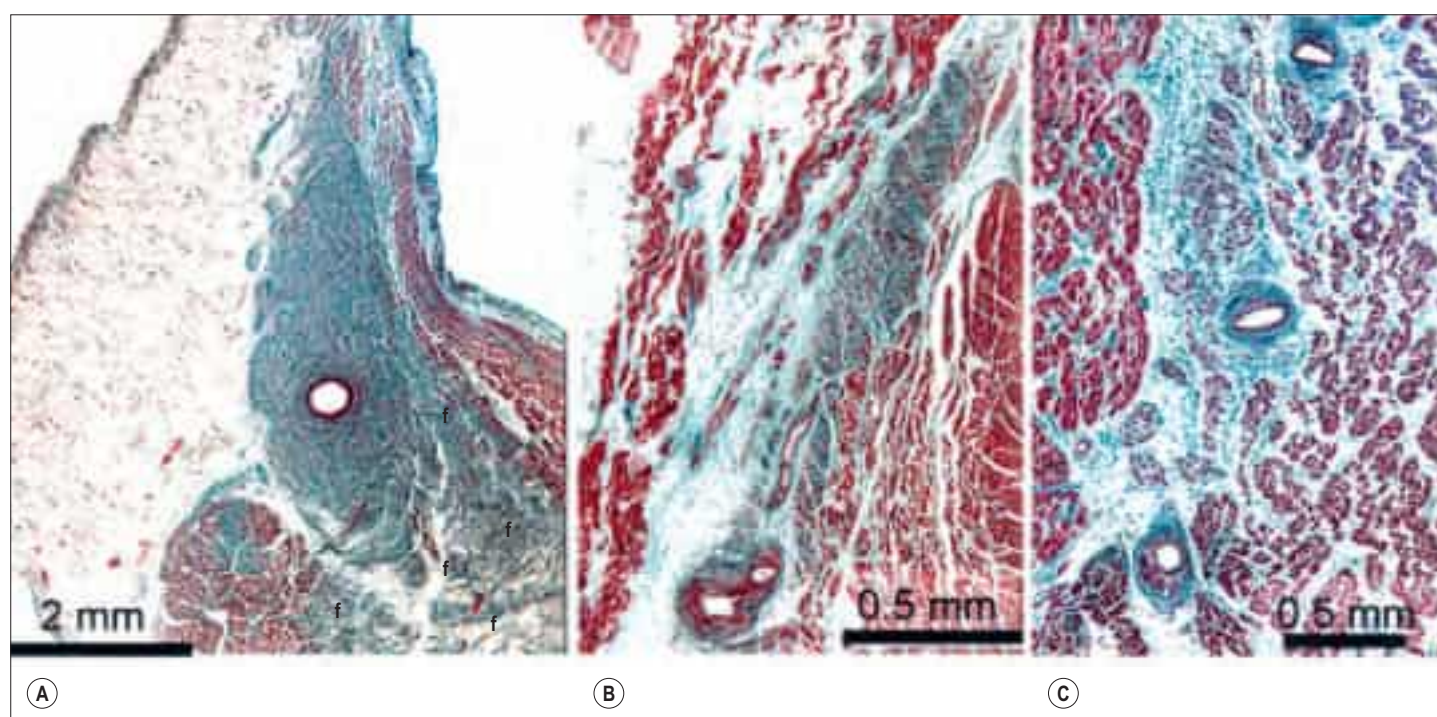


Fig. 57.39 Histological sections of the nodal body. **A**, Fibro-fatty tissue (f) between the caudal aspect of the nodal body and the subendocardium. **B–C**, Fragmentation of the nodal tail into clusters. Masson trichrome stain. (With permission from Sánchez-Quintana D, Cabrera JA, Farré J, Climent V, Anderson RH, Ho SY. 2005. Sinu-atrial node revisited in the era of electroanatomical mapping and catheter ablation. 2005 91:189–94.)

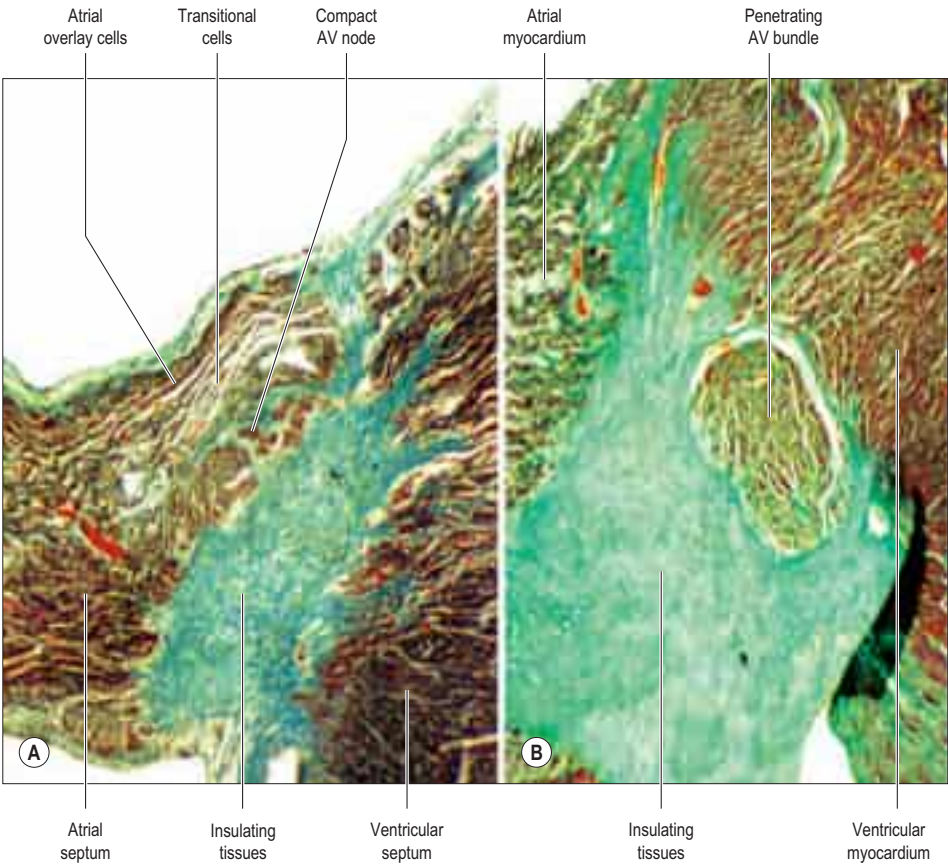


Fig. 57.40 Histological sections showing the features of **(A)** the atrioventricular (AV) node and **(B)** the penetrating atrioventricular bundle. Trichrome method, fibrous tissue stained green. (With permission from [Anderson RH](#), [Boyett MR](#), [Dobrzynski H](#), [Moorman AF](#). 2013. The anatomy of the conduction system: implications for the clinical cardiologist. *J Cardiovasc Transl Res.* 6:187–196.)

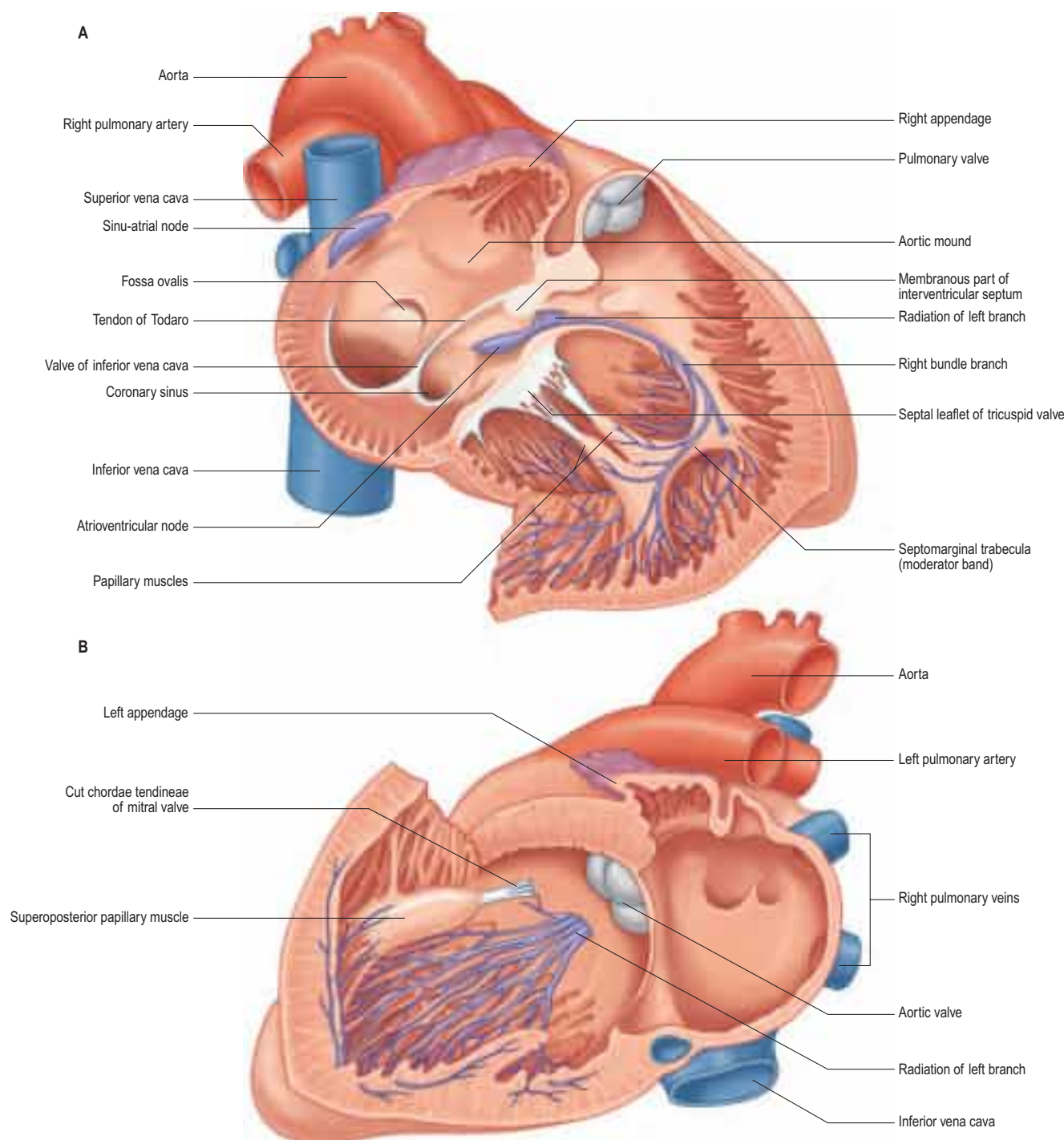


Fig. 57.36 The conduction tissue of the heart. **A**, Right aspect. **B**, Left aspect. The elements of the conduction system are shown in purple. Conduction tissue accompanies fine trabeculae carneae and false chordae. In reality, the radiation of the left bundle branch is directly related to the leaflets of the aortic valve.

Extensions of nodal cardiomyocytes run in the direction of the coronary sinus along the tricuspid anulus (the putative 'slow pathway'); in the anterior portion of the triangle of Koch near the compact portion of the atrioventricular node (the putative 'fast pathway'); and in the direction of the mitral anulus (the left atrial extension) (Mani and Pavri 2014). In both sinus and atrioventricular nodes, the intercellular contacts between nodal cells, and between nodal and transitional cells, are much less specialized than the intercalated discs between normal cardiac cells (p. 137). A sparsity of gap junctions is consistent with the absence from these areas of connexin-43 (a major protein component of mammalian gap junctions), and probably accounts for the observed difficulty in exciting these cells from adjacent cells. The atrioventricular delay may owe much to this relative non-excitability, which appears to disturb the spread of potential, delaying propagation; the narrow diameter of the transitional cells may also contribute to conduction delay.

The arterial supply to the atrioventricular node is from a characteristic vessel that originates from the dominant coronary artery at the cardiac crux. Autonomic ganglia are present between the node and the coronary sinus.

Atrioventricular bundle

The atrioventricular bundle (of His) is the direct continuation of the atrioventricular node. It becomes oval, quadrangular or triangular in transverse sectional profile as it enters the central fibrous body. It traverses the fibrous body and branches on the crest of the muscular interventricular septum; the branching tract is sandwiched between the muscular and the membranous components of the septum.

The right branch of the bundle (*crus dextrum*) is a narrow, discrete, rounded group of fascicles that courses at first within the myocardium and then subendocardially towards the ventricular apex, entering the septomarginal trabecula to reach the anterior papillary muscle. Its branches to the ventricular walls are few in its septal course. At the origin of the anterior papillary muscle, it divides profusely into fine subendocardial fascicles that diverge, first embracing the papillary muscle, then recurring subendocardially for distribution to the remaining ventricular walls.

The left branch (*crus sinistrum*) arises as numerous fine, intermingling fascicles that leave the left margin of the branching bundle through

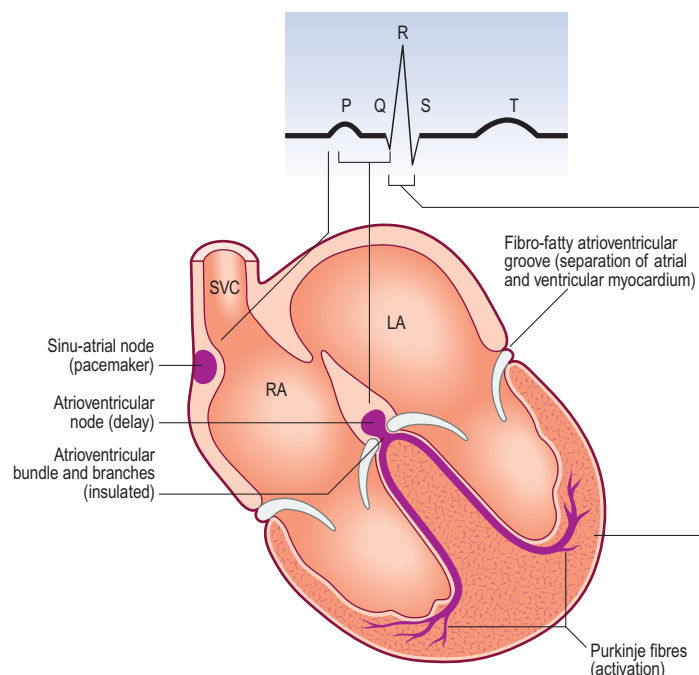


Fig. 57.37 The basic structure of the conduction system and its relationship with the electrocardiogram. Note the foramen ovale allowing for communication between the right atrium (RA) and the left atrium (LA). Other abbreviations: SCV, superior vena cava. (Redrawn by courtesy of Professor RH Anderson, Institute of Child Health, University College, London.)

most of its course along the crest of the muscular ventricular septum (see Fig. 57.36). These fascicles form a flattened sheet down the smooth left ventricular septal surface that diverges apically and subendocardially across the left aspect of the ventricular septum, trifurcating into anterior, septal and posterior divisions. Fine branches leave the sheets, forming subendocardial networks, which first surround the papillary muscles and then curve back subendocardially to be distributed to all parts of the ventricle. The principal branches of the bundle are insulated from the surrounding myocardium by sheaths of connective tissue.

Functional contacts between ventricular conduction and working myocytes become numerous only in the subendocardial terminal ramifications. Hence, papillary muscles contract first, followed by a wave of excitation and ensuing contraction that travels from the apex of the ventricle to the arterial outflow tract. Because the Purkinje network is subendocardial, muscular excitation proceeds from endocardium to epicardium.

In the developing heart, the bundle responsible for atrioventricular conduction is a much more extensive structure. Immunohistochemical analysis has revealed that the precursor of the system is a ring of cells that surrounds the inlet and outlet components of the developing ventricular loop (Ch. 52).

Cardiac pacing



Available with the *Gray's Anatomy* e-book

Cardiac conduction studies



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Congenital conduction abnormalities



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VASCULAR SUPPLY AND LYMPHATIC DRAINAGE



Coronary arteries

The right coronary artery arises from the anterior ('right coronary') sinus and the left coronary artery from the left posterior ('left coronary') sinus of the ascending aorta (see Figs 57.21–57.22; Fig. 57.41); ostia levels vary. The two arteries, as indicated by their generic name, form an

oblique inverted crown within the atrioventricular groove. They also form a variable and often insignificant anastomosis (in the non-pathological state) via marginal and interventricular (descending) loops that intersect at the cardiac apex. The main arteries and major branches are usually subepicardial, but those in the atrioventricular and interventricular grooves are often deeply sited, and occasionally hidden by overlapping myocardium or embedded in it (myocardial bridging). The term 'dominant' is used to refer to the coronary artery giving off the posterior interventricular (posterior descending or inferior interventricular) branch, which supplies the posterior (inferior) part of the ventricular septum and often part of the posterolateral (inferolateral) wall of the left ventricle. The right artery is the dominant artery in 60% of hearts. Anastomoses between right and left coronary arteries are abundant in the fetus but are much reduced by the end of the first year of life. Anastomoses providing collateral circulation may become prominent in conditions of chronic hypoxia and in coronary artery disease. An additional collateral circulation is provided by small branches from mediastinal, pericardial and bronchial vessels.

Coronary arterial original calibre, based on arterial casts or angiogram measurements, ranges between 1.5 and 5.5 mm. The calibre of the left origin exceeds the right in 60% of hearts, the right being larger in 17%, and both vessels being of approximately equal calibre in 23%. The external diameter of the left coronary artery increases from 2.1 mm at the age of 1 year to 3.3 mm at the age of 15 years (Pesonen et al 1991) and the diameters of the coronary arteries may increase up to the 30th year. A significant association has been found between the diameters of the right and left coronary arteries at birth and at 1 and 6 months of age and birth weight, height and body surface area (Karagol et al 2012).

Right coronary artery

The right coronary artery arises from the anterior ('right coronary') aortic sinus; its ostium is usually below the sinutubular junction. The artery is usually single but as many as four right ostia have been observed, reflecting the independent origins of the conal, sinu-atrial node and ventricular arteries (Figs 57.42–57.43). The right coronary artery passes at first anteriorly and slightly to the right between the right atrial appendage and pulmonary trunk, where the sinus usually bulges. On reaching the atrioventricular groove, it descends almost vertically to the right (acute) cardiac border, curving around it into the posterior (inferior) part of the groove, where the latter approaches its junction with both interatrial and interventricular grooves, the appropriately termed cardiac crux. The artery ends a little to the left of the crux, often by anastomosing with the circumflex branch of the left coronary artery. The right coronary artery may end near the right cardiac border, between this and the crux, or more often it reaches the left border, replacing the more distal part of the circumflex artery. Its branches supply the right and variable parts of the left chambers and atrioventricular septum. Usually, the first branch is the right conal artery (conus artery, artery of the conus, arteria coni arteriosi; Loukas et al 2014b). (This vessel arises independently from the anterior aortic sinus in approximately one-third of hearts and is therefore sometimes termed the 'third coronary artery'; a similarly named vessel arises from the left coronary circulation and so this title is inappropriate.) The right conal artery ramifies antero-inferiorly over the pulmonary conus and over the superior aspect of the right ventricle, sometimes anastomosing with a similar branch from the left interventricular (anterior descending) artery to form the anulus of Vieussens, a tenuous anastomosis around the right ventricular outflow tract (Figs 57.44–57.45) (Loukas et al 2009).

The first segment of the right coronary artery (between its origin and the right cardiac margin) gives off anterior atrial and ventricular branches that diverge widely, approaching a right angle in the case of the ventricular arteries, an arrangement that is in marked contrast to the more acute origins of the left coronary ventricular branches. The anterior ventricular branches, usually two or three, ramify towards the cardiac apex, which they rarely reach (unless the right marginal artery is included within this group of branches, as it is by some authors). The right marginal artery is greater in calibre than the other anterior ventricular arteries and long enough to reach the apex in most hearts. When it is very large, there may be only one other remaining anterior ventricular branch, or even no other branches. Up to three small posterior (inferior) ventricular branches, most often two, arise from the second segment of the right coronary artery between the right border and crux to supply the right ventricular diaphragmatic aspect. Their size is inversely proportional to that of the right marginal artery, which usually extends to the cardiac diaphragmatic surface. On approaching the crux, the right coronary artery normally produces up to three posterior (inferior) interventricular branches (occasionally none). The posterior (inferior) interventricular artery itself, lying within the interventricular groove, may therefore be flanked, either to the right or to the left, or on

This ring is in continuity with the atrioventricular bundle, itself part of an interventricular ring of specialized cardiomyocytes. The segment of the interventricular ring that encircles the developing left ventricular outflow tract subsequently breaks down. The segment that persists within the ventricles extends along the septum beyond the branching atrioventricular bundle and forms a dead-end tract. The dorsal component of the interventricular ring is part of the definitive atrioventricular node, continuous with the atrioventricular bundle. The rightward component of the atrioventricular ring system, emerging inferiorly from the atrioventricular node, is part of the slow nodal pathway. The cardiomyocytes that formed the ring initially possessed a primary phenotype, i.e. they would have conducted slowly.

Wolff–Parkinson–White syndrome is caused by abnormal small strands of otherwise unremarkable ventricular myocardium that run through the fibroareolar tissue of the atrioventricular groove, connecting the atrial and ventricular myocardial masses at some point around the atrioventricular junctions.

Temporary pacing wires are usually inserted by cannulation of either the internal jugular or subclavian veins; the latter approach carries a slightly greater risk of a pneumothorax because of the proximity of the pleural cavity. Other potential risks are brachial plexus injury if the entry site is too posterior, and thoracic duct injury if the left subclavian vein is cannulated. The most common location for permanent pacemaker devices is subcutaneously on the anterior chest wall. Access to the chambers and endocardium of the right heart is gained via the cephalic vein within the deltopectoral groove.

Intracardiac electrocardiography and electrophysiology are used to assess cardiac conduction and rhythm abnormalities. A catheter is inserted via the femoral, subclavian or internal jugular veins using a guidewire technique. Fluoroscopy, echocardiography and, more recently, cardiac magnetic resonance are used to guide accurate placement of the catheters to the appropriate position. The sites of study are the high right atrium (for assessing the atrioventricular bundle and right bundle branch) and the coronary sinus (for evaluating atrioventricular junctional arrhythmias and accessory pathways). The multipolar electrodes provide detailed electro-anatomical mapping of the sequence of excitation from the atria, atrioventricular junction and ventricles. The origin of supraventricular arrhythmias, ventricular tachycardias, accessory conduction pathways and re-entrant pathways can be identified and used to guide radiofrequency ablation.

In the past, there was an anatomical basis for the majority of presenting congenital conduction abnormalities: they were the product of either accessory pathways or conduction tissue dysgenesis at any point from the atrioventricular node to the atrioventricular bundle. Today, conduction abnormalities are increasingly likely to relate to long-standing haemodynamic problems and/or the effects of previous surgery for patients with congenital heart defects. This reflects the fact that, although surgery for most of these defects has been available for the past four decades, surgery itself has not been curative; more often than not, patients develop conduction abnormalities and arrhythmia from surgical scars that cause haemodynamic problems such as chamber dilation and/or hypertrophy. Occasionally, conduction abnormalities are caused by tumours such as multifocal Purkinje cell tumours, or benign congenital polycystic tumours of the atrioventricular node.

Congenital abnormalities of the coronary arteries are found rarely in children. The two most common abnormalities are coronary arteriovenous fistula and anomalous left coronary artery arising from the pulmonary artery (Uysal et al 2014). Other congenital abnormalities include ectopic origin of left circumflex artery from right coronary artery, single coronary artery arising from right sinus of Valsalva, and ectopic right coronary artery arising from the left sinus of Valsalva (Clemente et al 2010).

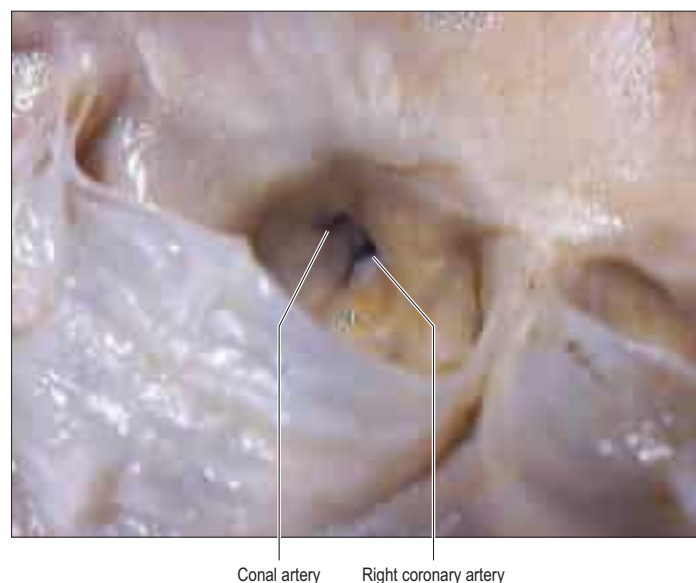


Fig. 57.43 The right coronary sinus, revealing a coronary orifice for the right coronary artery and a coronary orifice for the conal artery. (With permission from Loukas M, Sharma A, Blaak C, Sorenson E, Mian A. 2013. The clinical anatomy of the coronary arteries. *J Cardiovasc Transl Res.* 6:197–207.)

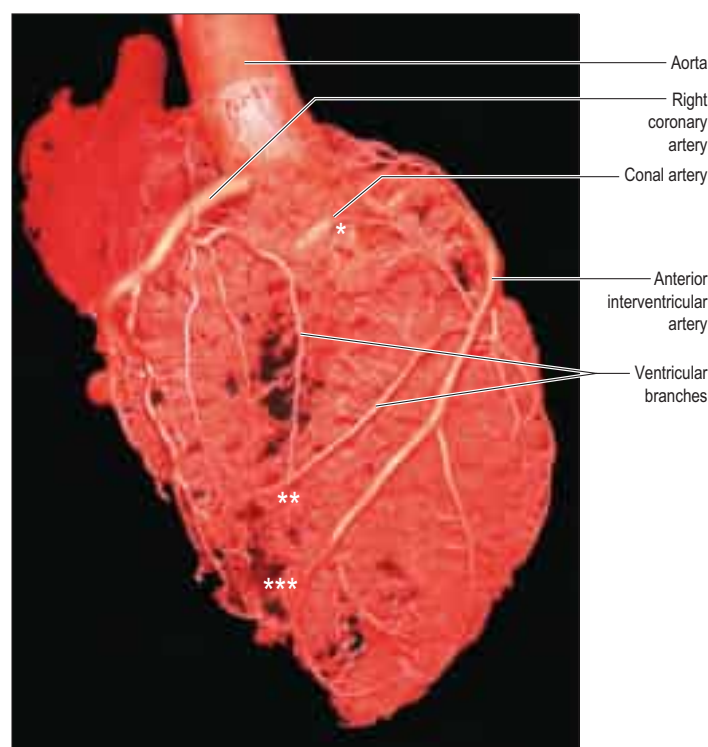


Fig. 57.44 A variation of the anulus of Vieussens (*). This corrosion cast specimen shows the right conal artery, a branch of the right coronary artery, anastomosing with a proximal ventricular branch of the anterior interventricular artery. A second anastomosis (**) occurs between the ventricular branches of the right coronary and anterior interventricular arteries. Close to the apex, a third anastomosis (***) appears between the distal part of the anterior interventricular artery and branches of the acute marginal artery. (With permission from Loukas M, Bilinsky S, Bilinsky E, Matusz P, Anderson RH. 2009. The clinical anatomy of the coronary collateral circulation. *Clin Anat.* 22:146–160.)

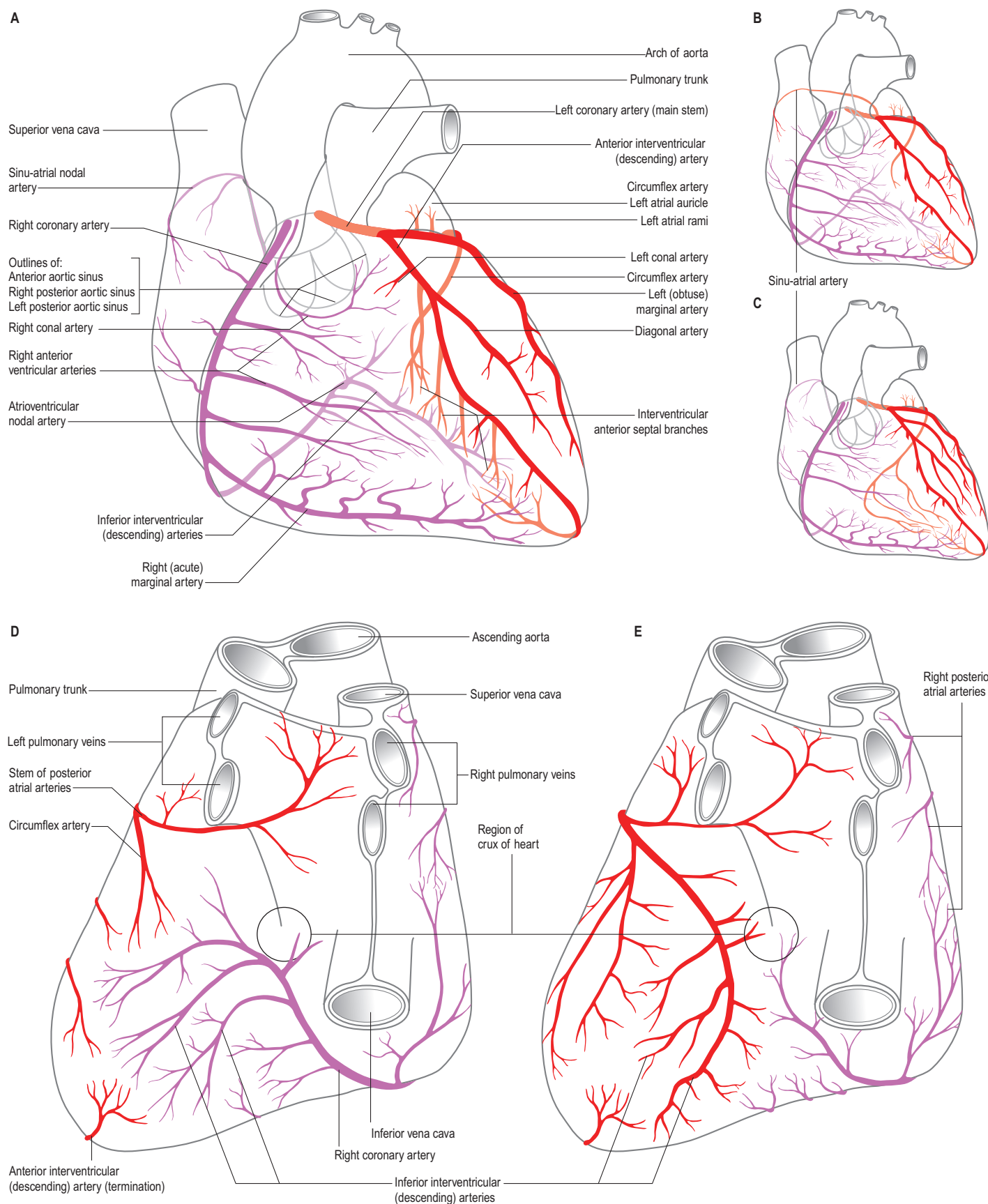


Fig. 57.41 Anterior views of the coronary arterial system, with the principal variations. The right coronary arterial tree is shown in purple, the left in red. In both cases, posterior distribution is shown in a paler shade. **A**, The most common arrangement. **B**, A common variation in the origin of the sinu-atrial nodal artery. **C**, An example of left 'dominance' by the left coronary artery, also showing an uncommon origin of the sinu-atrial nodal artery. Posteroinferior views of the coronary arterial system. The right coronary arterial tree is shown in purple, the left in red. **D**, An example of the more normal distribution in right 'dominance'. **E**, A less common form of left 'dominance'. In these 'posterior' views, the diaphragmatic (inferior) surface of the ventricular part of the heart has been artificially displaced and foreshortening ignored, to clarify the details of the so-called posterior (inferior) distribution of the coronary arteries.

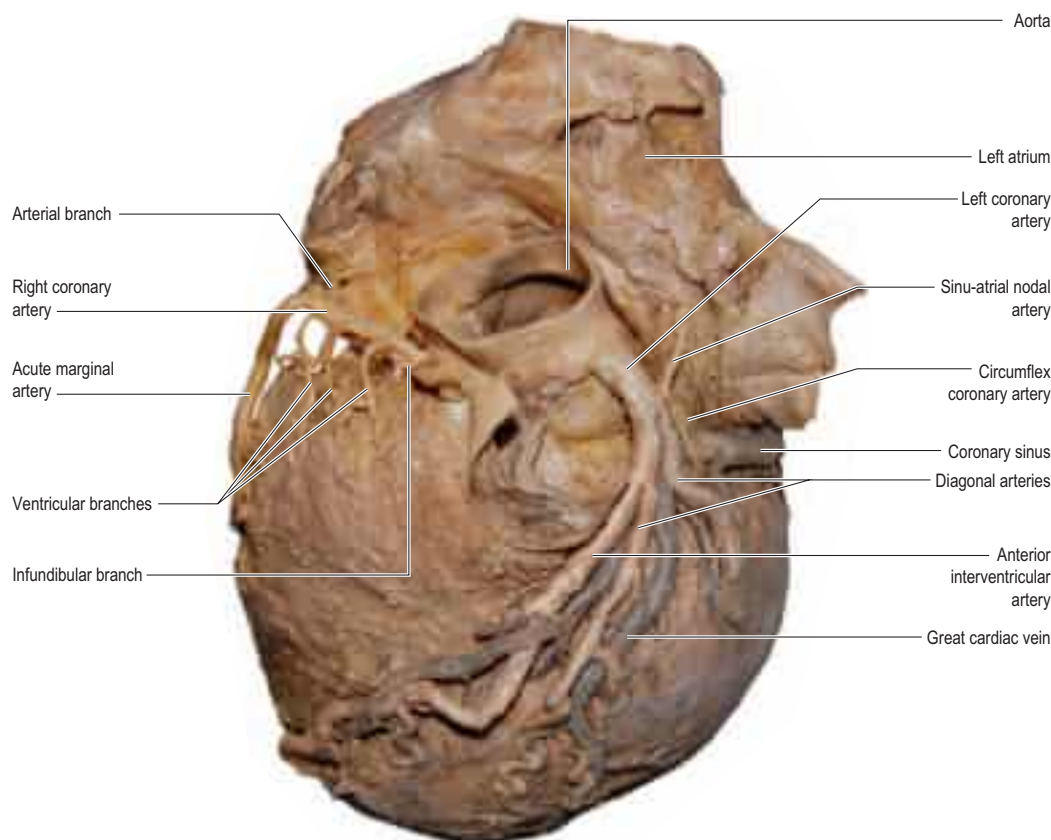


Fig. 57.42 A dissection of a cadaveric heart in which both coronary arteries have been exposed to reveal their branches to the right and left ventricles, respectively. Notice how the great cardiac vein is intertwined around the anterior interventricular artery. (Specimen courtesy of M Loukas MD, PhD.)

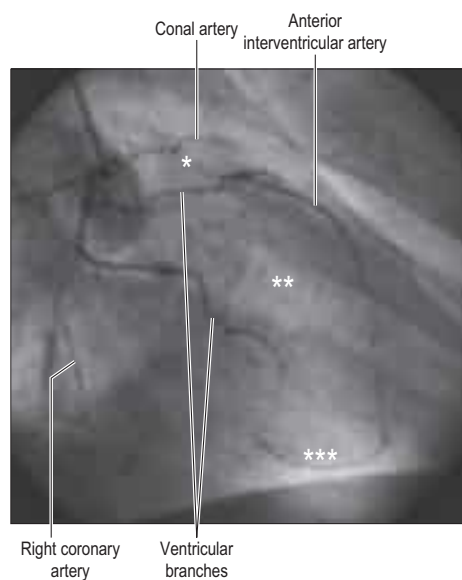


Fig. 57.45 A coronary angiogram demonstrating three collateral arterial anastomoses. At the area of the subpulmonary infundibulum, the right conal artery, a branch of the right coronary artery, anastomoses with the left conal artery, a branch of the anterior interventricular artery, to form the anulus (arterial circle or vascular ring) of Vieussens (*). Ventricular branches of the right coronary artery anastomose with the proximal (**) and distal portions (***) of the anterior interventricular artery, forming two collateral pathways. As a result, there is contrast medium in the anterior interventricular artery. (With permission from Loukas M, Bilinsky S, Bilinsky E, Matusz P, Anderson RH. 2009. The clinical anatomy of the coronary collateral circulation. Clin Anat. 22:146–160.)

both sides, by these parallel branches. When these flanking vessels exist, branches of the posterior (inferior) interventricular artery are small and sparse. The posterior (inferior) interventricular artery is occasionally replaced by a left coronary branch. Although the atrial branches of the right coronary artery are sometimes described as part of anterior, lateral (right or marginal) and posterior groups, they are usually small single vessels, 1 mm in diameter (Figs 57.46–57.48). The right anterior and

lateral branches – occasionally double, and very rarely triple – mainly supply the right atrium. The posterior branch is usually single and supplies both atria.

The artery of the sinu-atrial node is an atrial branch, distributed largely to right atrial myocardium. Its origin is variable: most commonly, it arises from the anterior atrial branch of the right coronary artery, less often from its right lateral part, and least often from its posterior atrioventricular part. This ‘nodal’ artery thus usually passes posteriorly in the groove between the right atrial appendage and aorta. It may originate from the circumflex branch of the left coronary artery. Whatever its origin, it usually branches around the base of the superior vena cava, typically as an arterial loop from which small branches supply the right atrium. A large branch, the ramus cristae terminalis, traverses the sinu-atrial node; it would seem more appropriate to name this branch the ‘nodal artery’, on the grounds that the vessel that currently bears this name actually supplies the atria and serves as the ‘main atrial branch’.

Septal perforating branches of the right coronary artery are relatively short, and leave the posterior (inferior) interventricular branch to supply the posterior interventricular septum. They are numerous but do not usually reach the septal apex. The largest posterior septal perforating artery, usually the first, commonly arises from the inverted loop said to characterize the right coronary artery at the crux; it almost always supplies the atrioventricular node. Small, recurrent atrioventricular branches originate from the ventricular branches of the right coronary artery as they cross the atrioventricular groove, and supply adjacent atrial myocardium.

Left coronary artery

The left coronary artery is usually larger in calibre than the right. It supplies a greater volume of myocardium, including almost all of the left ventricle and atrium, and most of the interventricular septum (see Fig. 57.42; Figs 57.49–57.51). In hearts with ‘right dominance’, the right coronary artery supplies a variable posterior region of the left ventricle (see Fig. 57.41A–C).

The left coronary artery arises from the left posterior (left coronary) aortic sinus; the ostium sometimes lies inferior to the margin of the leaflets and may be double, leading into major initial branches, usually the circumflex and anterior interventricular (descending) arteries. Its initial portion, between its ostium and its first branches, varies in length from a few millimetres to a few centimetres. The artery lies between the pulmonary trunk and the left atrial appendage, emerging into the atrioventricular groove, where it turns left. This part is loosely embedded in subepicardial fat and usually has no branches, but may give off a small

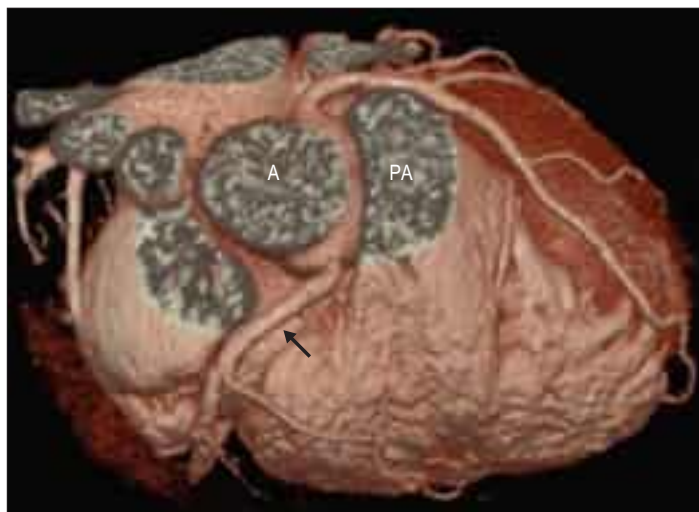


Fig. 57.46 Normal ECG-gated CT anatomy of the right coronary artery and its branches. This oblique volume-rendered image of the superior aspect of the heart shows the right coronary artery (arrow) arising from the right sinus of Valsalva and coursing in the right atrioventricular groove towards the posterior interventricular septum. Abbreviations: A, aorta; PA, pulmonary artery. The conal artery and the sinu-atrial nodal artery were too small to be seen in this case. (With permission from Kim SY, Seo JB, Do KH, et al 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)

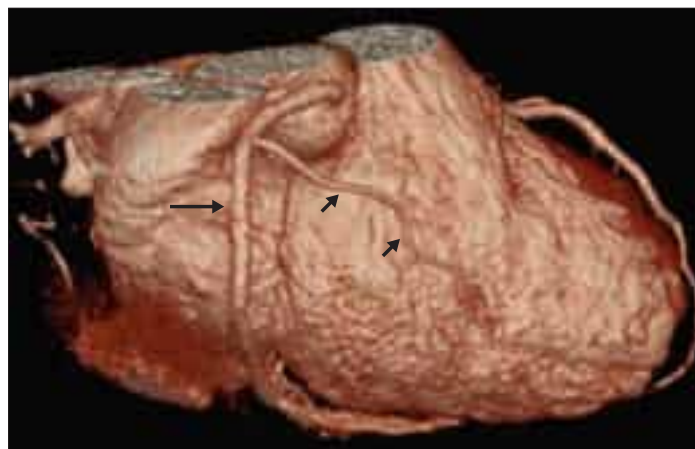


Fig. 57.47 Normal ECG-gated multidetector row CT anatomy of the right coronary artery and its branches. This lateral oblique volume-rendered image shows the caudal course of the proximal right coronary artery (long arrow), which gives off an acute marginal branch (short arrows) to the right ventricle. (With permission from Kim SY, Seo JB, Do KH, et al. 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)

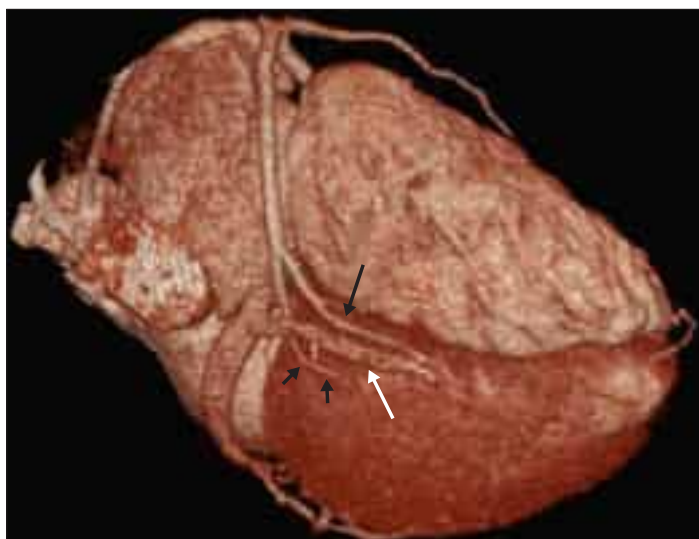


Fig. 57.48 Normal ECG-gated multidetector row CT anatomy of the right coronary artery and its branches. This posterior oblique volume-rendered image shows that the distal right coronary artery divides into the posterior (inferior) interventricular artery (long black arrow) and posterior left ventricular branches (short black arrows). The posterior (inferior) interventricular artery courses in the posterior (inferior) interventricular groove, parallel to the middle cardiac vein (white arrow). (With permission from Kim SY, Seo JB, Do KH, et al. 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)

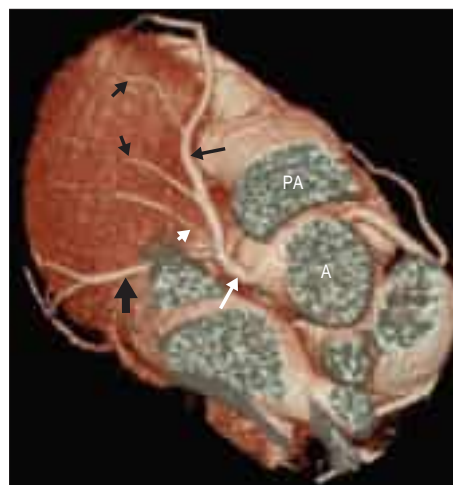


Fig. 57.49 Normal ECG-gated multidetector row CT anatomy of the left coronary artery and its branches. This oblique volume-rendered image of the top of the heart shows the left coronary artery (long white arrow) arising from the left sinus of Valsalva and trifurcating into the left anterior interventricular artery (thick black arrow), the left circumflex artery (thin black arrow), and the ramus intermedius (short white arrow), which takes a course similar to that of the usual first diagonal branch. The left anterior interventricular artery then gives rise to diagonal branches (short black arrows) to the anterior free wall of the left ventricle. Abbreviations: A, aorta; PA, pulmonary artery. (With permission from Kim SY, Seo JB, Do KH, et al. 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)

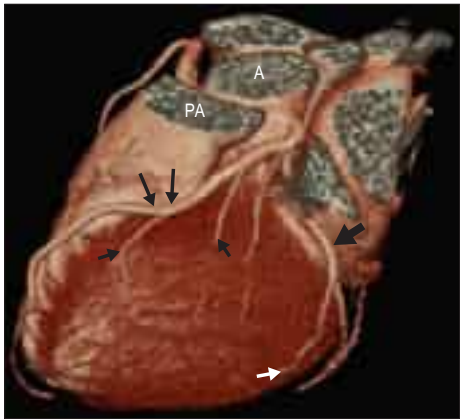


Fig. 57.50 An anterior oblique volume-rendered image showing the left anterior ventricular artery (thin black arrows) coursing along the anterior interventricular groove, and the left circumflex artery (thick black arrow) coursing in the left atrioventricular groove. Obtuse marginal branches (short white arrow) and diagonal branches (short black arrows) are also shown. Abbreviations: A, aorta; PA, pulmonary artery. (With permission from Kim SY, Seo JB, Do KH, et al. 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)



Fig. 57.51 A posterior oblique volume-rendered image showing the left anterior ventricular artery (long white arrows) coursing along the anterior interventricular groove, and the left circumflex artery (thick black arrow) coursing in the left atrioventricular groove. Obtuse marginal branches (short white arrows) and a diagonal branch (short black arrow) are also shown. (With permission from Kim SY, Seo JB, Do KH, et al 2006. Coronary artery anomalies: classification and ECG-gated multi-detector row CT findings with angiographic correlation. Radiographics. 26:317–33.)

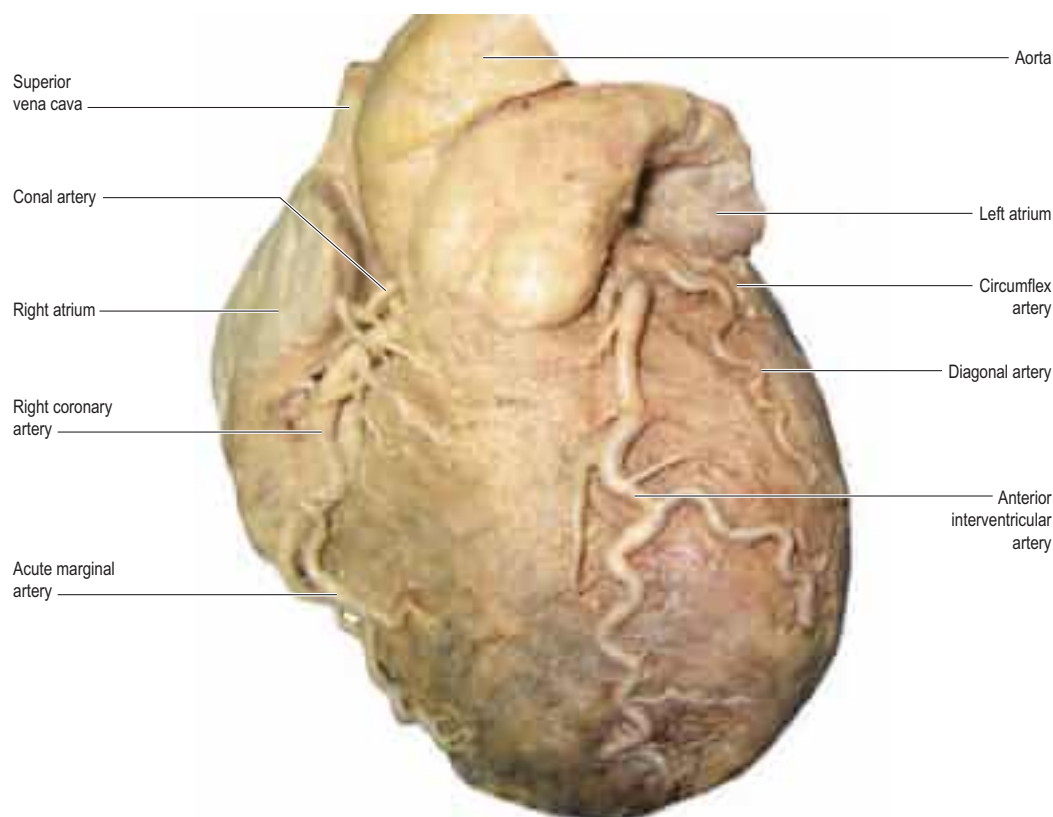


Fig. 57.52 A dissection of a cadaveric heart revealing the course of the left coronary artery and its branches. (With permission from Loukas M, Sharma A, Blaak C, Sorenson E, Mian A. 2013. The clinical anatomy of the coronary arteries. *J Cardiovasc Transl Res.* 6:197–207.)

atrial ramus and, rarely, the sinu-atrial nodal artery. Reaching the atrio-ventricular groove, the left coronary artery divides into its main branches: namely, the circumflex and anterior interventricular arteries.

The anterior interventricular artery is commonly described as the continuation of the left coronary artery. It descends obliquely forwards and to the left in the interventricular groove (**Fig. 57.52**), sometimes deeply embedded in or crossed by bridges of myocardial tissue (myocardial bridges), and by the great cardiac vein and its tributaries.

Almost invariably, the anterior interventricular artery reaches the apex, where it terminates in one-third of hearts. More frequently, it turns round the apex into the posterior interventricular groove and passes one-third to one-half of the way along its length, meeting the terminal twigs of the posterior (inferior) interventricular branches of the right coronary artery (see above).

The anterior interventricular artery gives off right and left anterior ventricular and anterior septal branches, and a variable number of corresponding posterior branches. Anterior right ventricular branches are small and rarely number more than one or two; the right ventricle is supplied almost entirely by the right coronary artery. Up to nine large left anterior ventricular arteries branch at acute angles from the anterior interventricular artery, crossing the anterior aspect of the left ventricle diagonally, with the largest reaching the rounded (obtuse) left cardiac border. One often dominates, sometimes arising separately from the left coronary trunk, which then ends by trifurcation. This left diagonal artery, reported to exist in at least 33–50% of hearts, may be doubled (20%). A small left conal artery frequently leaves the anterior interventricular artery near its origin, and anastomoses on the conus with its counterpart from the right coronary artery and with the vasa vasorum of the pulmonary artery and aorta. The anterior septal perforating branches leave the anterior interventricular artery almost perpendicularly, and pass posteroinferiorly within the septum, usually supplying its ventral two-thirds. The first septal perforator artery usually supplies the atrioventricular bundle at the point of its division. Small posterior septal branches from the same source supply the posterior third of the septum for a variable distance from the cardiac apex (**Loukas et al 2009**).

The circumflex artery, comparable to the anterior interventricular artery in calibre, curves left in the atrioventricular groove, and continues round the left cardiac border into the posterior part of the groove, terminating left of the crux in most hearts, although sometimes continuing as the posterior (inferior) interventricular artery. Proximally, the left atrial appendage usually overlaps it. A large ventricular branch, the left marginal artery, normally arises perpendicularly from the circumflex artery and ramifies over the rounded 'obtuse' margin, supplying much of the adjacent left ventricle, typically to its apex. Smaller anterior and

posterior branches of the circumflex artery also supply the left ventricle. Anterior ventricular branches (from one to five, commonly two or three) course parallel to the diagonal artery when it is present, and replace it when it is absent. Posterior ventricular branches are smaller and fewer because the left ventricle is partly supplied by the posterior (inferior) interventricular artery. When this artery is small or absent, it is accompanied or replaced by often doubled or tripled interventricular continuations of the circumflex artery.

The artery to the sinu-atrial node is often derived from the anterior circumflex segment (less often from the circum-marginal segment). It passes over and supplies the left atrium, encircles the superior vena cava (like a right coronary nodal branch) and sends a large branch through the node; despite its name, it is predominantly atrial in distribution. The artery to the atrioventricular node, sometimes the terminal branch of the circumflex artery (20%), arises near the crux. When this occurs, the circumflex artery usually gives off the posterior interventricular artery, an example of so-called 'left dominance'.

Coronary arterial distribution

Details of coronary arterial distribution require integration into a concept of total cardiac supply. Most commonly, the right coronary artery supplies all of the right ventricle (except a small region to the right of the anterior interventricular groove); a variable part of the diaphragmatic aspect of the left ventricle; the posteroinferior third of the interventricular septum; the right atrium and part of the left atrium; and the conduction system as far as the proximal parts of the right and left crura. Left coronary distribution is reciprocal and includes most of the left ventricle; a narrow strip of right ventricle; the anterior two-thirds of the interventricular septum; and most of the left atrium. As noted previously (see **Fig. 57.41**), variations in the coronary arterial system mainly affect the diaphragmatic aspect of the ventricles and reflect the relative 'dominance' of coronary arterial supply. The term is misleading because the left artery almost always supplies a greater volume of tissue than the right. In 'right dominance', the posterior (inferior) interventricular artery is derived from the right coronary artery; in 'left dominance', it is derived from the left coronary artery. In the so-called 'balanced' pattern, branches of both arteries run in or near the posterior (inferior) interventricular groove.

Less is known about variations in atrial supply because the small vessels involved are not easily preserved in the corrosion casts that are used for analysis. In more than 50% of individuals, the right atrium is supplied only by the right coronary artery, and in the remainder the supply is dual. More than 62% of left atria are supplied mainly by the left coronary artery, 27% by the right coronary artery (in each group a small accessory supply from the other coronary artery exists), and 11%

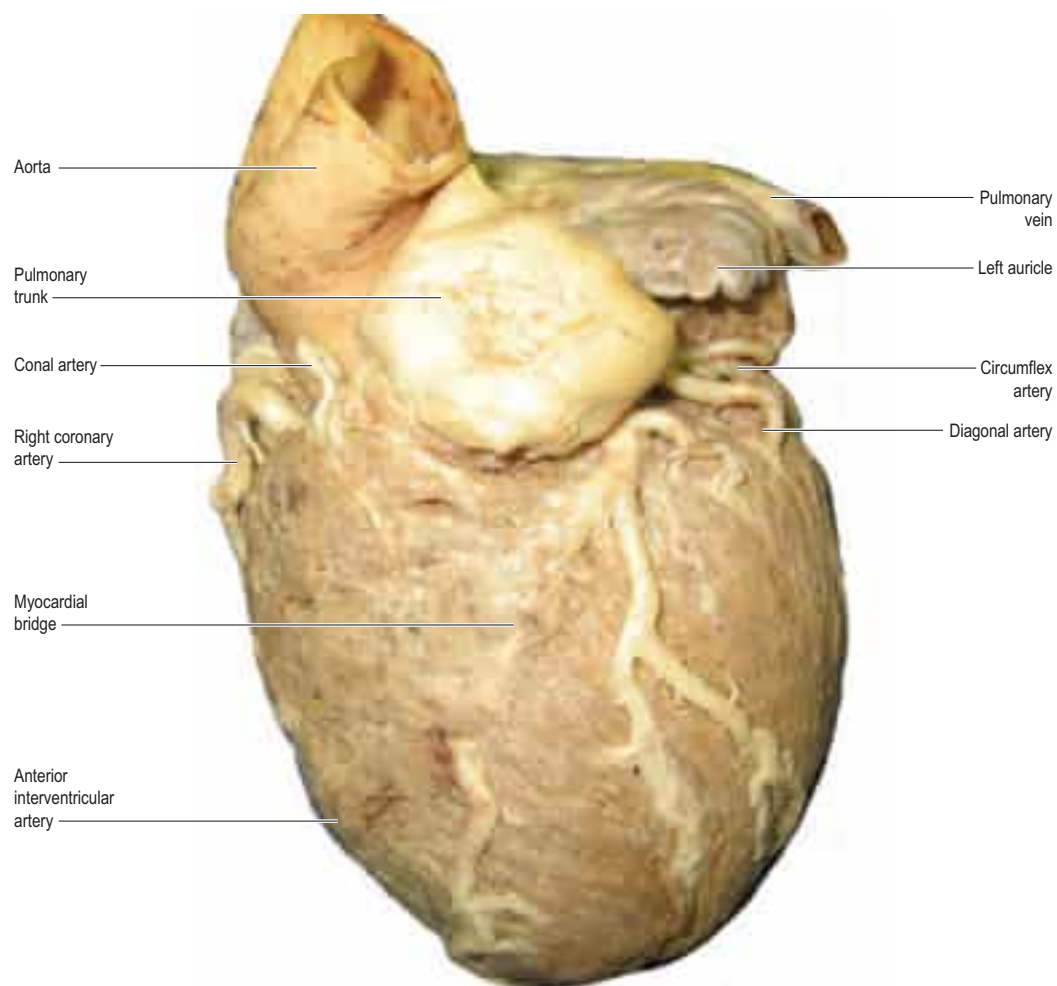


Fig. 57.53 A dissection of a cadaveric heart in which a large part of the anterior interventricular artery is covered by a myocardial bridge.

Myocardial bridges are reported to have a frequency varying from 0.5 to 40% when identified clinically and from 15 to 85% when found at autopsy ([Fig. 57.53](#)). The wide variation in frequency indicates that many bridges may be asymptomatic during life. The major clinical conditions produced by a myocardial bridge are cardiac ischaemia, atherosclerosis and sudden cardiac death. The incidence of atherosclerosis is increased when the right coronary artery is bridged. Although a relationship between myocardial bridges and sudden cardiac death has not been established, autopsy series have shown histological evidence of otherwise unexplained ischaemia in individuals with myocardial bridges; many died during exercise and had no other risk factors for coronary arterial disease.

are supplied almost equally by both arteries. Arterial supply to the sinu-atrial and atrioventricular nodes also varies: the sinu-atrial node is supplied more often by the right coronary artery; fewer than 10% of sinu-atrial nodes receive a bilateral supply. The atrioventricular node is usually supplied by the right coronary artery.

Coronary anastomoses

The cardiac collateral circulation represents a native system for coronary arterial bypass. The first few centimetres of the arterial main stems are devoid of anastomotic branches, but further distally, collateral channels are abundant, exhibit variable calibres and occupy numerous locations, allowing for bidirectional flow between most native arteries. Approximately 30 different sites of collateral extramural vessels have been described, the most frequent being at the apex; the anterior aspect of the right ventricle; the posterior aspect of the left ventricle; the crux; the interatrial and interventricular grooves; and between the sinu-atrial nodal and other atrial vessels (see Figs 57.44, 57.45). Anastomoses between branches of the coronary arteries, both subepicardial and myocardial, and between these arteries and extracardiac vessels, are of prime medical importance. Clinical studies suggest that anastomoses cannot rapidly provide collateral routes sufficient to circumvent sudden coronary obstruction (see Figs 57.44, 57.45, 57.54). Nevertheless, it has long been established that anastomoses do occur, particularly between fine subepicardial branches, and they may increase during individual life by mechanisms of angiogenesis and arteriogenesis. The anulus of Vieussens is a collateral vessel that crosses the subpulmonary infundibulum, providing an anastomosis between the conal branch of the right coronary artery and the anterior interventricular artery. The artery to the sinu-atrial node commonly provides a communication between the proximal parts of the coronary arteries. The apical collateral artery joins the interventricular arteries. The frequent enlargement of the first septal branch of the anterior interventricular artery, Kugel's anastomotic artery (arteria anastomotica auricularis magna), has been described as running between the proximal parts of the coronary arteries along the antero-superior margin of the oval fossa to anastomose with the distal part of the right coronary artery.

Extracardiac anastomoses

Extracardiac anastomoses connect various coronary branches with other thoracic vessels, most commonly involving the bronchial and internal thoracic arteries (Fig. 57.54). To a much lesser degree, anastomoses between coronary arteries and pericardiacophrenic branches of the internal thoracic, anterior mediastinal, intercostal and oesophageal arteries also exist. The posterior pericardium also receives a direct supply from the bronchial arteries; extracardiac coronary anastomoses involving bronchial arteries are typically found at the pericardial reflections, such as the points of entry of the venae cavae. The most common anastomoses are with the circumflex branch of the left coronary artery via the posterior pericardial reflections and reflect the close proximity of the bronchial arteries within the pulmonary hila. In pathological conditions, notably those resulting in pericardial adhesions, it is also possible for extracardiac anastomoses to develop through trans-pericardial vascularization.

Extracardiac communications also exist with coronary atrial branches, especially the sinu-atrial nodal artery. The effectiveness of any of these connections as collateral routes in coronary occlusion is not well quantified. Coronary arteriovenous anastomoses and numerous connections between the coronary circulation and cardiac cavities, producing so-called 'myocardial sinusoids' and 'arterioluminal vessels', have been reported, but their importance in coronary disease remains uncertain.

Coronary angiography

 Available with the *Gray's Anatomy* e-book

Coronary revascularization

 Available with the *Gray's Anatomy* e-book

Coronary artery fistula

A coronary artery fistula is an abnormal connection that directly links one or more coronary arteries to a heart chamber or to major thoracic vessels without an interposed capillary bed.

 Coronary artery fistulae are rare (Fig. 57.58); those that arise from a coronary artery and then terminate in a chamber of the heart are known as coronary cameral fistulae, while those terminating into a vein are coronary arteriovenous fistulae. Fistulae may be congenital or may

develop later in life. Congenital fistulae are more common and account for 50% of paediatric coronary vascular aberrations; they are believed to be derived from Thebesian vessels. Acquired coronary artery fistulae are most commonly iatrogenic in aetiology but may also occur after traumatic injury; these most commonly are of the coronary cameral type, from the right coronary artery into the right side of the heart.

Cardiac veins

The cardiac venous system is divided into two major parts. The greater system consists of large vessels that lie within the subepicardial myocardium and drain most of the outer myocardium. These veins extend over the myocardial surface and do not adhere to topographical borders, although the larger collecting vessels lie within the interventricular grooves before draining into the coronary sinus. The intercommunicating parts of this system are the coronary sinus and its tributaries, the anterior cardiac venous system, and the ventricular septal and atrial veins. The coronary sinus and its tributaries return blood to the right atrium from the entire heart (including its septa), except for the anterior region of the right ventricle and small, variable parts of both atria and the left ventricle. The anterior cardiac veins drain an anterior region of the right ventricle, expanding to include a region around the right cardiac border when the right marginal vein joins this group.

The smaller system functions primarily to return venous blood from the inner myocardial walls into the right atrium and ventricle and, to a lesser extent, into the left atrium and sometimes the left ventricle. It contains the smallest cardiac veins (Thebesian veins) that drain the subendocardial myocardium either directly, via connecting intramural arteries and veins, or indirectly, via subendocardial sinusoidal spaces.

Variation in cardiac veins Attempts to categorize variations in cardiac venous circulation into 'types' have not produced any accepted pattern. The coronary sinus may receive all the cardiac veins other than the Thebesian veins, including the anterior cardiac veins, which may be reduced by diversion into the small cardiac vein and then to the coronary sinus.

Coronary sinus

Most cardiac veins drain into the wide coronary sinus, 2 or 3 cm long, lying in the posterior atrioventricular groove between the left atrium and ventricle (see Fig. 57.4B; Fig. 57.59). The sinus opens into the right atrium between the opening of the inferior vena cava and the right atrioventricular orifice. The opening may be guarded by an endocardial fold (semilunar valve of the coronary sinus, Thebesian valve; see Fig. 57.13A); the fold may be absent or may cover the ostium of the sinus completely. The tributaries of the coronary sinus are the great, small and middle cardiac veins, the posterior vein of the left ventricle and the oblique vein (of Marshall) of the left atrium; all except the oblique vein have orificial valves. Isolated absence of the coronary sinus has been reported, with coronary venous drainage into the pulmonary trunk (Ogawa et al 2013).

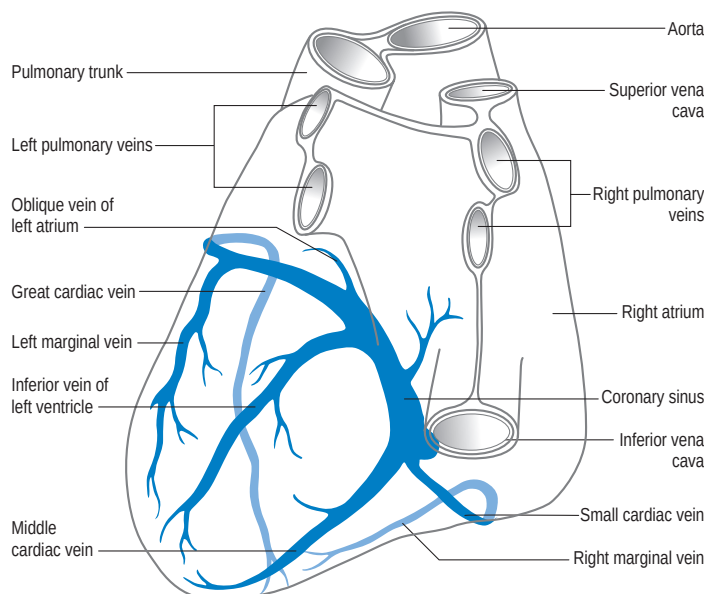


Fig. 57.59 The principal veins of the heart.

Table 57.1 Patterns of cardiac collateral arteries

Grade	Description
0	Total absence of any collateral arteries
1	Poorly developed collateral arteries with no prominent distal channel visualized by angiography
2	Presence of moderate collateral arteries providing faint but delayed opacification of a prominent distal channel
3	Good collateral arteries giving clear opacification of a prominent distal channel
4	Excellent collateralization giving full and brisk opacification of prominent distal vessels

Clinical imaging (catheter angiography and colour Doppler ultrasound) has led to the development of a grading system to describe the overall pattern of cardiac collateral arteries (Table 57.1).

Coronary angiography may be performed by introducing a catheter through the femoral, radial or brachial arteries. The femoral artery is punctured with a needle 3 cm below the inguinal ligament while the leg is held adducted and slightly externally rotated. The exact position is guided by palpation of the femoral arterial pulse, and the needle is inserted at an angle of 45°. After arterial puncture, a fine guidewire is inserted through the needle and fed into the artery. The catheter is then inserted over the guidewire and manipulated via the iliac artery into the aorta, up the aortic arch and into the ascending aorta. The brachial or radial artery may be used for percutaneous access to the circulation. Once the catheter is located in the ascending aorta, a variety of guidewires may be used to enter the coronary vessels for selective arteriography and interventions. Angiography is performed with standard high-osmolality contrast medium with cineangiography. In selected patients, new-generation, low-osmolality contrast medium may also be used. All the coronary arteries are catheterized and evaluated in a variety of views to obtain a full evaluation of their anatomy and to determine the location and degree of any stenoses (Figs 57.55–57.56). The ostium of the left coronary artery arises from the left aortic sinus and is best viewed in the direct frontal and left anterior oblique projections. The right anterior oblique view is useful in demonstrating the diagonal branches and anterior interventricular (descending) coronary artery. The right coronary artery originates from the right sinus of Valsalva and is usually visualized in the right anterior oblique view. Pressure and oxygen saturations can be measured via the catheter; changes in pressure across valves allow the degree of stenosis to be measured. Coronary blood flow and relative flow reserve can also be calculated. Significant stenosis may be treated initially by balloon angioplasty followed by stent insertion. The balloon exerts pressure against the plaque in the arterial wall, fracturing and splitting the plaque. The splinting effect of the plaque and elastic recoil are reduced, resulting in an increase in the arterial lumen. Stent insertion reduces the re-stenosis rate.

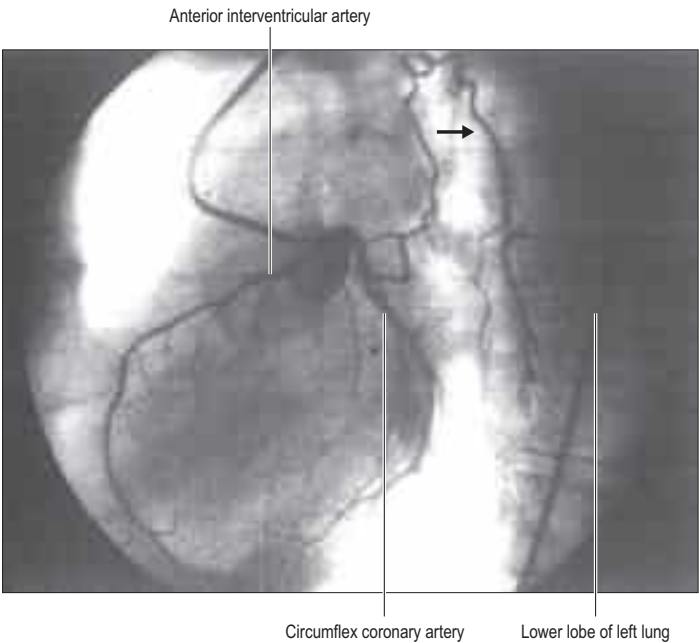


Fig. 57.54 A left coronary angiogram in left lateral projection, showing a large, tortuous vessel (arrow) arising from the proximal circumflex branch of the left coronary artery and anastomosing with the bronchial artery, which goes on to supply the lower lobe of the left lung. (With permission from Stefan L, Assayag P, Aubry P, et al, Coronary to bronchial artery anastomosis with bronchial steal syndrome demonstrated by thallium-201 myocardial tomoscintigraphy. Eur Heart J, 1990, 11, 275–279.)

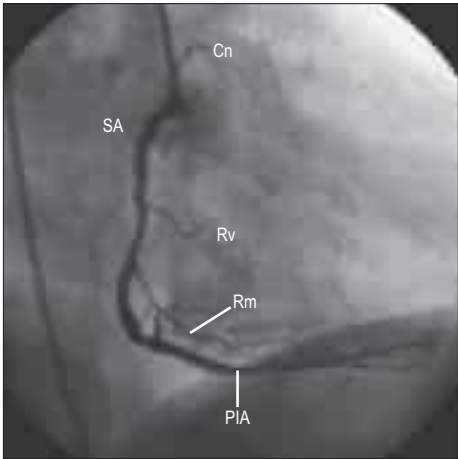


Fig. 57.55 A right coronary angiogram showing several branches of the right coronary artery. Abbreviations: Cn, conal artery; PIA, posterior (inferior) interventricular artery; Rm, right marginal artery; Rv, right ventricular branches; SA, artery to the sinu-atrial node. (Courtesy of M Loukas MD, PhD.)

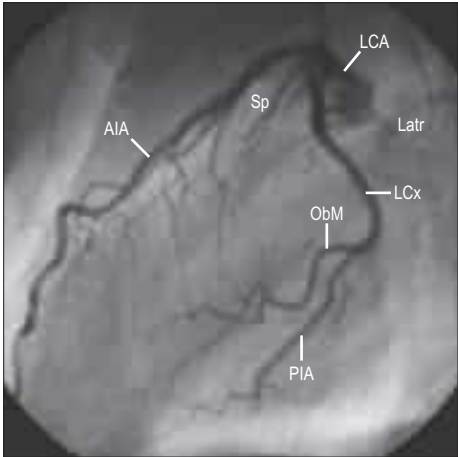


Fig. 57.56 A left coronary angiogram showing several branches of the left coronary artery (LCA). Other abbreviations: AIA, anterior interventricular artery; Latr, left atrial branch; LCx, left circumflex artery; ObM, obtuse marginal; PIA, posterior (inferior) interventricular artery; Sp, septal perforator. (Courtesy of M Loukas MD, PhD.)

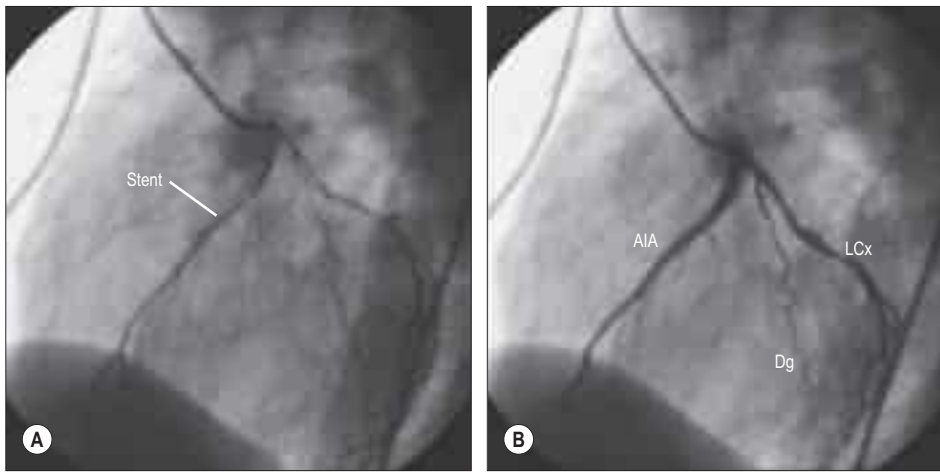


Fig. 57.57 **A**, A left coronary angiogram showing a stent placement within the anterior interventricular artery. **B**, After stent placement (and once the contrast medium has filled the coronary arterial tree), the anterior interventricular artery shows no evidence of stenosis. Abbreviations: AIA, anterior interventricular artery; Dg, diagonal branch; LCx, left circumflex artery. (Courtesy of M Loukas MD, PhD.)

Atherosclerosis causing more than 60% stenosis of the terminal diameter of the coronary arteries is likely to cause significant reduction in myocardial perfusion. Patients with high-grade lesions, left main stem coronary artery or triple-vessel disease with impaired left ventricular function are usually considered for coronary artery bypass grafting. The common grafts that are used are the internal thoracic (mammmary) and radial arteries. The left internal thoracic artery and radial artery grafts have a greater patency rate than saphenous vein grafts. Approximately 15% of saphenous vein grafts occlude in 1 year and, from then on, at an annual rate of 1–2% in the first 6 years and 4% thereafter; between 40% and 50% of saphenous vein grafts have occluded by 10 years, whereas only about 10% of left internal thoracic or radial artery grafts have occluded over this time. The common surgical approach is via a midline sternotomy. If the internal thoracic artery is used as a donor graft, it is divided distally (maintaining its proximal origin from the subclavian artery) and anastomosed to the coronary artery distal to the stenosis. If radial artery grafts are used, they must be anastomosed both proximal and distal to the coronary artery, to bridge the site of the stenosis. In selected cases, minimally invasive direct coronary artery bypass grafting is performed, but the approach is dependent on the vessel being grafted. The anterior approach is via mini-thoracotomy over the fourth intercostal space underneath the nipple for grafting the mid-left anterior interventricular (descending) and diagonal branches. The anterolateral approach is via an incision in the third intercostal space from the mid-clavicular to anterior axillary lines and is used for grafting early marginal branches of the circumflex system. The lateral approach allows grafting of the circumflex vessels via a lateral thoracotomy measuring only 10 cm in size through the fifth or sixth intercostal spaces. Extrathoracic approaches that are occasionally used include the subxiphoid approach for the distal right coronary artery and posterior interventricular (descending) artery. Port access surgery allows for full revascularization with cardiopulmonary bypass but obviates the need for midline sternotomy. However, the vast majority of patients are treated with stent placement (Fig. 57.57).

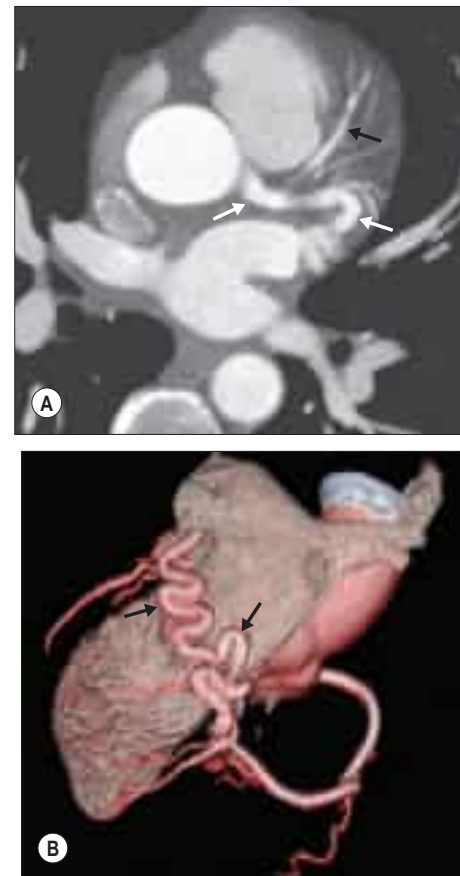


Fig. 57.58 A coronary artery fistula in a 54-year-old woman with palpitations. **A**, A CT image showing a tortuous left circumflex artery (white arrows) that is dilated in comparison with the anterior interventricular artery (black arrow). **B**, A volume-rendered CT image shows the markedly tortuous left circumflex artery (arrows). (With permission from Shriki JE, Shinbane JS, Rashid MA, Hindoyan A, Withey JG, DeFrance A, Cunningham M, Oliveira GR, Warren BH, Wilcox A. 2012. Identifying, characterizing, and classifying congenital anomalies of the coronary arteries. *Radiographics* 32:453–468.)

Great cardiac vein The great cardiac vein begins at the cardiac apex, and ascends in the anterior interventricular groove to the atrioventricular groove, which it follows, passing to the left and posteriorly to enter the coronary sinus at its origin (see Fig. 57.59). It receives tributaries from the left atrium and both ventricles, including the large left marginal vein that ascends the left aspect (obtuse border) of the heart. It usually courses superior to the arterial branches. The valve of Vieussens normally guards the orifice of the great cardiac vein at its junction with the oblique vein; smaller diminutive valves may occur.

Small cardiac vein The small cardiac vein lies in the posterior atrioventricular groove between the right atrium and ventricle, receiving blood from their posterior parts and opening into the atrial end of the coronary sinus (see Fig. 57.59). The right marginal vein passes right, along the inferior cardiac margin (acute border), and sometimes joins the small cardiac vein in the atrioventricular groove, typically opening directly into the right atrium.

Middle cardiac vein The middle cardiac vein begins at the cardiac apex and runs posteriorly in the inferior interventricular groove to end in the coronary sinus near its atrial end (see Fig. 57.59). The vein is also described as meeting the great cardiac vein at the apex, so forming, together with the coronary sinus, a full venous circle.

Inferior vein of the left ventricle The inferior vein of the left ventricle (previously named as the posterior vein of the left ventricle) lies on the diaphragmatic surface of the left ventricle, a little to the left of the middle cardiac vein, and usually opens into the middle of the coronary sinus, sometimes into the great cardiac vein (see Fig. 57.59). Rarely, the inferior vein of the left ventricle is absent, in which case the left marginal vein drains most of the left ventricular wall.

Oblique vein of the left atrium The diminutive oblique vein of the left atrium descends obliquely on the posterior aspect of the left atrium to join the coronary sinus (see Fig. 57.59). It is continuous above with the ligament of the left vena cava; both structures are remnants of the left common cardinal vein.

Left marginal vein

The left (obtuse) marginal vein courses over the left oblique marginal surface of the heart, draining much of the left ventricular myocardium. It runs superficial to the marginal branch of the left coronary artery and usually drains into the great cardiac vein, although may sometimes drain directly into the coronary sinus.

Anterior cardiac veins

The anterior cardiac veins drain the anterior part of the right ventricle. Usually two or three, sometimes even five, they ascend in subepicardial tissue to cross the right part of the atrioventricular groove, passing deep or superficial to the right coronary artery. They end in the right atrium, near the atrioventricular groove, separately or in variable combinations. A subendocardial collecting channel, into which all may open, has been described.

Right marginal vein

The right marginal vein courses along the inferior (acute) cardiac margin, draining adjacent parts of the right ventricle, and usually opens separately into the right atrium, although it may join the anterior cardiac veins or, less often, the coronary sinus. It is often grouped with the small cardiac veins but it is larger in calibre, comparable to the anterior cardiac veins or even wider.

Small cardiac veins

The existence of small cardiac (venae cordis minimae; Thebesian) veins, opening into all cardiac cavities, has been confirmed but they are difficult to demonstrate. Their numbers and size are highly variable: vessels up to 2 mm in diameter open into the right atrium and ones as small as 0.5 mm in diameter open into the right ventricle, all with valveless orifices. Numerous small cardiac veins have been identified in the right atrium and ventricle but they are rarely found in the left side. Four types of Thebesian veins have been described: venoluminal veins drain directly into the cardiac chambers; venosinusoidal veins drain into subendocardial sinusoids (which, in turn, drain into the cardiac chambers); arteriololuminal veins connect small arteries and arterioles directly with the cardiac chambers; and arteriosinusoidal veins connect thin arteries or arterioles with subendocardial sinusoidal spaces.

Cardiac venous anastomoses

Widespread anastomoses occur at all levels of the cardiac venous circulation, on a scale exceeding that of the arteries and amounting to a

veritable venous plexus. Not only are adjacent veins often connected, but connections also exist between tributaries of the coronary sinus and those of the anterior cardiac veins. Abundant anastomoses occur at the apex and its anterior and posterior aspects. Like the coronary arteries, cardiac veins connect with extracardiac vessels, particularly the vasa vasorum of the large vessels that are continuous with the heart.

Lymphatic drainage of the heart

Cardiac lymphatic vessels form subendocardial, myocardial and subepicardial plexuses. Efferents from the subepicardial plexuses form the left and right cardiac collecting trunks; two or three left-sided trunks ascend the anterior interventricular groove, receiving vessels from both ventricles. On reaching the atrioventricular groove, they are joined by a large vessel from the diaphragmatic surface of the left ventricle, which first ascends in the posterior interventricular groove and then turns left along the atrioventricular groove. The vessel formed by this union ascends between the pulmonary artery and the left atrium, and usually ends in an inferior tracheobronchial node. The right trunk receives afferents from the right atrium and the right border and diaphragmatic surface of the right ventricle. It ascends in the atrioventricular groove, near the right coronary artery, and then anterior to the ascending aorta, and ends in a brachiocephalic node, usually on the left (Fig. 57.60). (For further reading, see Loukas et al (2011).)

INNERVATION

Initiation of the cardiac cycle is myogenic, originating in the sinu-atrial node. It is harmonized in rate, force and output by autonomic nerves that operate on the nodal tissues and their prolongations, on coronary vessels and on the working atrial and ventricular musculature. All the cardiac branches of the vagus (parasympathetic) and all the sympathetic branches (other than the cardiac branch of the superior cervical sympathetic ganglion) contain both afferent and efferent fibres; the cardiac branch of the superior cervical sympathetic ganglion is entirely efferent. Sympathetic fibres accelerate the heart and dilate the coronary arteries when stimulated, whereas vagal fibres slow the heart and cause coronary arterial constriction.

Preganglionic cardiac sympathetic axons arise from neurones in the intermediolateral column of the upper four or five thoracic spinal segments. Some synapse in the corresponding upper thoracic sympathetic ganglia, while others ascend to synapse in the cervical ganglia; postganglionic fibres from these ganglia form the sympathetic cardiac nerves. Preganglionic cardiac parasympathetic axons arise from neurones either in the dorsal vagal nucleus or near the nucleus ambiguus, and run in vagal cardiac branches to synapse in the cardiac plexuses and atrial walls. In humans (like most mammals), intrinsic cardiac neurones are limited to the atria and interatrial septum, and are most numerous in the subepicardial connective tissue near the sinus and atrioventricular nodes. The intrinsic ganglia are thought not to be simple nicotinic relays, but may act as sites for the integration of extrinsic nervous inputs and form complex circuits for the local neuronal control of the heart, and perhaps even local reflexes.

Cardiac plexus

Nearing the heart, the autonomic nerves form a mixed cardiac plexus, usually described in terms of a superficial component inferior to the aortic arch, lying between it and the pulmonary trunk, and a deep part between the aortic arch and tracheal bifurcation. The cardiac plexus is also described by regional names for its coronary, pulmonary, atrial and aortic extensions (Fig. 57.61). Ganglion cells are largely confined to the atrial tissues, with a preponderance adjacent to the sinu-atrial node, but some may also be found within the heart along the branches of the plexuses. Their axons are considered to be largely, if not exclusively, postganglionic parasympathetic. Cholinergic and adrenergic fibres, arising in or passing through the cardiac plexus, are distributed most profusely to the sinus and atrioventricular nodes; the supply to the atrial and ventricular myocardium is much less dense. Adrenergic fibres supply the coronary arteries and cardiac veins. Rich plexuses of nerves containing cholinesterase, adrenergic transmitters and other peptides, e.g. neuropeptide Y, are found in the subendocardial regions of all chambers and within the valvular leaflets.

Superficial (ventral) part of the cardiac plexus The superficial (ventral) part of the cardiac plexus lies inferior to the aortic arch and anterior to the right pulmonary artery. It is formed by the cardiac branch

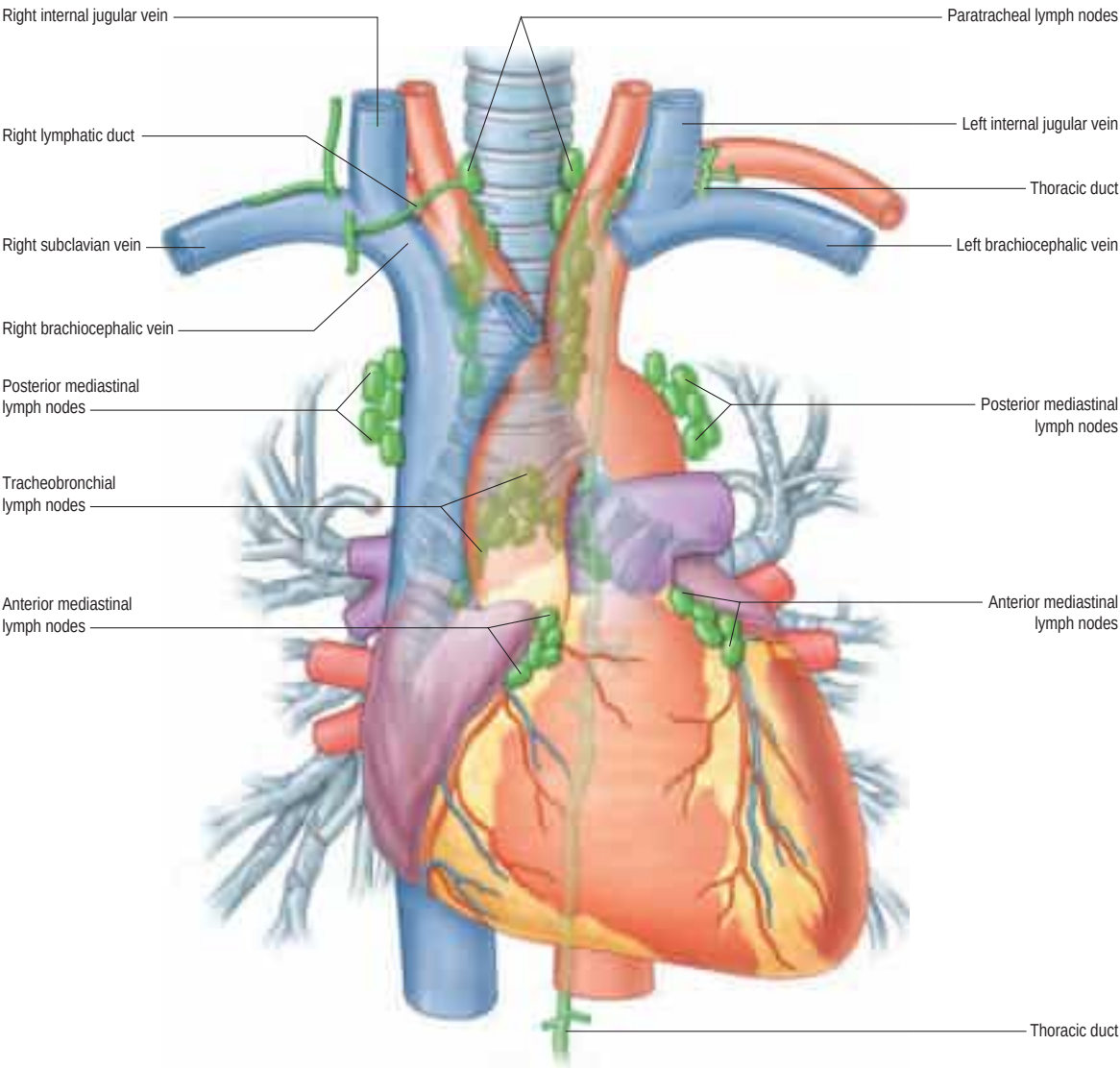


Fig. 57.60 The cardiac lymphatic system. (With permission from Loukas, M, Shah, S, Bhusnurmath S, et al; A general outline of the cardiac lymphatic system. In: The Cardiac Lymphatic System, Karunamuni, G (ed). Springer. 2013.)

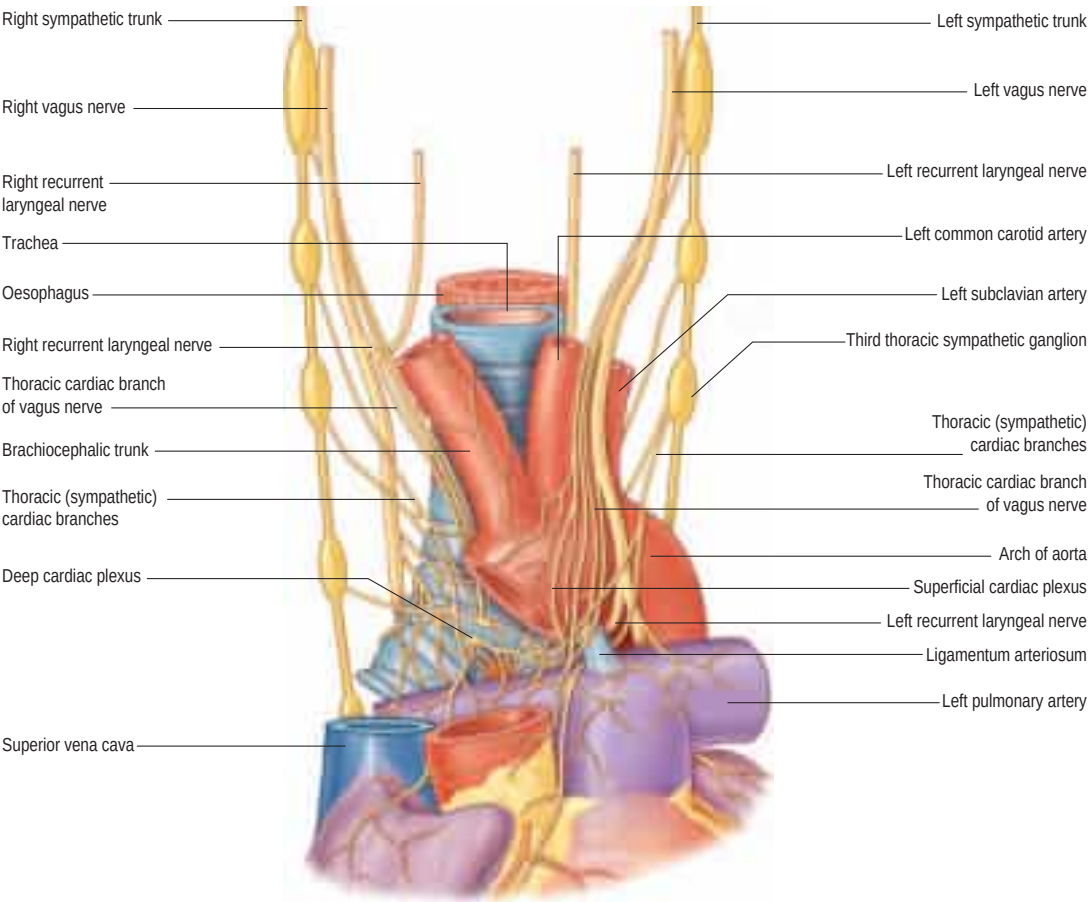


Fig. 57.61 The cardiac plexus: its source from the cervical parts of the vagus nerves and sympathetic trunks and its extensions, the pulmonary, atrial and coronary plexuses. Note the numerous junctions between sympathetic and parasympathetic (vagal) branches that form the plexus.

of the left superior cervical sympathetic ganglion and the lower of the two cervical cardiac branches of the left vagus. A small cardiac ganglion is usually present in this plexus immediately below the aortic arch, to the right of the ligamentum arteriosum. This part of the cardiac plexus connects with the deep part, the right coronary plexus and the left anterior pulmonary plexus.

Deep (dorsal) part of the cardiac plexus The deep (dorsal) part of the cardiac plexus lies anterior to the tracheal bifurcation, superior to the point of division of the pulmonary trunk and posterior to the aortic arch. It is formed by the cardiac branches of the cervical and upper thoracic sympathetic ganglia, the vagus and recurrent laryngeal nerves. The only cardiac nerves that do not join it are those that join the superficial part of the plexus. The deep plexus consists of right and left halves; the right typically surrounds the brachiocephalic trunk, and the left surrounds the aortic arch. The more dorsal (deep) aspect is larger than its more ventral (superficial) aspect on both sides. Branches from the right half of the deep part of the cardiac plexus pass both anterior and posterior to the right pulmonary artery. Fibres passing anterior to the pulmonary artery are more numerous; they supply a few filaments to the right anterior pulmonary plexus before continuing

onwards to form part of the right coronary plexus. Fibres passing posterior to the pulmonary artery supply a few filaments to the right atrium and then continue into the left coronary plexus. The left half of the deep part of the cardiac plexus is connected to the superficial part; it supplies filaments to the left atrium and left anterior pulmonary plexus, and forms much of the left coronary plexus.

Left coronary plexus The left coronary plexus is larger than the right and is formed mainly by the prolongation of the left half of the deep part of the cardiac plexus and a few fibres from the right half. It accompanies the left coronary artery to supply the left atrium and ventricle.

Right coronary plexus The right coronary plexus is formed from both superficial and deep parts of the cardiac plexus. It accompanies the right coronary artery to supply the right atrium and ventricle.

Atrial plexuses The atrial plexuses are derivatives of the right and left continuations of the cardiac plexus along the coronary arteries. Their fibres are distributed to the corresponding atria, overlapping those from the coronary plexuses.



Bonus e-book images and table

Fig. 57.5 Acute cardiac tamponade due to ruptured aortic dissection.

Fig. 57.6 A, The subxiphoid approach in pericardiocentesis. **B**, A mid-sagittal section of a cadaver. **C**, A cross-section of the lower part of the thorax to show the relationships of the pericardial cavity with adjacent structures. **D**, Bedside cardiac ultrasound performed in a 62-year-old male, demonstrating a large pericardial effusion consistent with cardiac tamponade.

Fig. 57.7 The typical appearance of a pericardial cyst.

Fig. 57.8 Congenital absence of the pericardium.

Fig. 57.11 The fundamental differences between describing the heart according to the conventional **(A)** or the attitudinally correct **(B)** orientation.

Fig. 57.15 An inferior view of a cadaveric heart, demonstrating an exaggerated Eustachian valve occupying a significant portion of the lumen of the inferior vena cava.

Fig. 57.16 Chiari's network.

Fig. 57.31 A histological section through one of the coronary aortic sinuses to demonstrate the way in which ventricular muscle supports the transition from the fibroelastic wall of the sinus to the tissues of the leaflet.

Fig. 57.34 A, Tricuspid atresia at 21 weeks' gestation. **B**, An atrioventricular septal defect at 20 weeks' gestation. **C**, Hypoplastic left heart syndrome at 32 weeks' gestation. **D**, Hypoplastic left heart syndrome at 32 weeks' gestation. **E**, Tetralogy of Fallot at 33 weeks' gestation.

F, Transposition of the great arteries at 21 weeks' gestation.

Fig. 57.35 A, Cardiac magnetic resonance angiography in a patient with a normal heart. **B**, A cardiac magnetic resonance angiogram from a patient with transposition of the great arteries.

Fig. 57.38 Measurements of the width and height of the sinu-atrial node and the distances of nodal tissue to epicardium and endocardium, measured in cadaveric tissue.

Fig. 57.39 Histological sections of the nodal body.

Fig. 57.40 Histological sections showing the features of **(A)** the atrioventricular (AV) node and **(B)** the penetrating atrioventricular bundle.

Fig. 57.43 The right coronary sinus, revealing a coronary orifice for the right coronary artery and a coronary orifice for the conal artery.

Fig. 57.44 A variation of the anulus of Vieussens.

Fig. 57.46 Normal ECG-gated CT anatomy of the right coronary artery and its branches.

Fig. 57.47 Normal ECG-gated multidetector row CT anatomy of the right coronary artery and its branches.

Fig. 57.48 Normal ECG-gated multidetector row CT anatomy of the right coronary artery and its branches.

Fig. 57.49 Normal ECG-gated multidetector row CT anatomy of the left coronary artery and its branches.

Fig. 57.50 An anterior oblique volume-rendered image showing the left anterior

ventricular artery coursing along the anterior interventricular groove, and the left circumflex artery coursing in the left atrioventricular groove.

Fig. 57.51 A posterior oblique volume-rendered image showing the left anterior ventricular artery coursing along the anterior interventricular groove, and the left circumflex artery coursing in the left atrioventricular groove.

Fig. 57.53 A dissection of a cadaveric heart in which a large part of the anterior interventricular artery is covered by a myocardial bridge.

Fig. 57.54 A left coronary angiogram in left lateral projection, showing a large, tortuous vessel arising from the proximal circumflex branch of the left coronary artery and anastomosing with the bronchial artery, which goes on to supply the lower lobe of the left lung.

Fig. 57.55 A right coronary angiogram showing several branches of the right coronary artery.

Fig. 57.56 A left coronary angiogram showing several branches of the left coronary artery.

Fig. 57.57 A, A left coronary angiogram showing a stent placement within the anterior interventricular artery. **B**, After stent placement (and once the contrast medium has filled the coronary arterial tree), the anterior interventricular artery shows no evidence of stenosis.

Fig. 57.58 A coronary artery fistula in a 54-year-old woman with palpitations.

Table 57.1 Patterns of cardiac collateral arteries.

REFERENCES

- Anderson R, Smerup M, Sanchez-Quintana D et al 2009a The three-dimensional arrangement of the myocytes in the ventricular walls. *Clin Anat* 22:64–76.
- Anderson RH, Boyett MR, Dobrzynski H et al 2013a The anatomy of the conduction system: implications for the clinical cardiologist. *J Cardiovasc Transl Res* 6:187–96.
- Anderson RH, Loukas M 2009 The importance of attitudinally appropriate description of cardiac anatomy. *Clin Anat* 22:47–51.
- Anderson RH, Spicer DE, Hlavacek AM et al 2013b Wilcox's Surgical Anatomy of the Heart, 4th ed. Cambridge: Cambridge University Press.
- Azancot A, Caudell TP, Allen HD et al 1983 Analysis of ventricular shape by echocardiography in normal fetuses, newborns, and infants. *Circulation* 68:1201–11.
- Clemente A, Del Borrello M, Greco P et al 2010 Anomalous origin of the coronary arteries in children: diagnostic role of three-dimensional coronary MR angiography. *Clin Imaging* 34:337–43.
- de Jonge LL, van Osch-Gevers L, Willemsen SP et al 2011 Growth, obesity, and cardiac structures in early childhood: the Generation R study. *Hypertension* 57:934–40.
- Ho SY, McCarthy KP, Faletra FF 2011 Anatomy of the left atrium for interventional echocardiography. *Eur J Echocardiogr* 12:11–15.
- Hutchison SJ 2009 Pericardial Diseases: Clinical Diagnostic Imaging Atlas. Philadelphia: Saunders; pp. 8–15.
- Kaiser T, Kellenberger CJ, Albisetti M et al 2008 Normal values for aortic diameters in children and adolescents – assessment in vivo by contrast-enhanced CMR-angiography. *J Cardiovasc Magn Reson* 10:56.
- Karagol BS, Orun UA, Zenciroglu A et al 2012 The diameter of coronary arteries in healthy newborns at birth, 1 and 6 months of ages. *J Matern Fetal Neonatal Med* 25:2729–34.
- Kortelainen ML 1997 Adiposity, cardiac size and precursors of coronary atherosclerosis in 5 to 15-year-old children: a retrospective study of 210 violent deaths. *Int J Obes Relat Metab Disord* 21:691–7.
- Loukas M, Abel N, Tubbs RS et al 2011 The cardiac lymphatic system. *Clin Anat* 24:684–91.
- Loukas M, Bilinsky E, Bilinsky S et al 2014a The anatomy of the aortic root. *Clin Anat* 27:748–56.
- Loukas M, Groat C, Khangura R et al 2009 The normal and abnormal anatomy of the coronary arteries. *Clin Anat* 22:114–28.
- Loukas M, Patel S, Cesmebasi A et al 2014b The clinical anatomy of the conal artery. *Clin Anat*. doi: 10.1002/ca.22469 (ePub ahead of print).
- Loukas M, Tubbs RS, Bright JL et al 2007 The anatomy of the tendon of the infundibulum revisited. *Folia Morphol (Warsz)* 66:33–8.
- Mani BC, Pavri BB 2014 Dual atrioventricular nodal pathways physiology: a review of relevant anatomy, electrophysiology, and electrocardiographic manifestations. *Indian Pacing Electrophysiol J* 14:12–25.
- Molenaar P, Anderson R, Sharma V et al 2011 Computer three-dimensional anatomical reconstruction of the human sinus node and a novel paranodal area. *Anat Rec (Hoboken)* 294:970–9.
- Mollova M, Bersell K, Walsh S et al 2013 Cardiomyocyte proliferation contributes to heart growth in young humans. *Proc Nat Acad Sci U S A* 110:1446–51.
- Monfredi O, Dobrzynski H, Mondal T et al 2010 The anatomy and physiology of the sinoatrial node – a contemporary review. *Pacing Clin Electrophysiol* 33:1392–406.
- Nidorf SM, Picard MH, Triulzi MO et al 1992 New perspectives in the assessment of cardiac chamber dimensions during development and adulthood. *J Am Coll Cardiol* 19:983–8.
- Ogawa K, Hishitani T, Hoshino K 2013 Absence of the coronary sinus with coronary venous drainage into the main pulmonary artery. *Cardiol Young* 23:759–62.
- Ozdemir O, Hizli S, Abaci A et al 2010 Echocardiographic measurement of epicardial adipose tissue in obese children. *Pediatr Cardiol* 31:853–60.
- Pesonen E, Hirvonen J, Karkola K et al 1991 Dimensions of the coronary arteries in children. *Ann Med* 23:85–8.
- Poutanen T, Tikanoja T, Sairanen H et al 2006 Normal mitral and aortic valve areas assessed by three- and two-dimensional echocardiography in 168 children and young adults. *Pediatr Cardiol* 27:217–25.
- Sánchez-Quintana D, Pizarro G, López-Mínguez JR et al 2013 Standardized review of atrial anatomy for cardiac electrophysiologists. *J Cardiovasc Transl Res* 6:124–44.
- Spicer DE, Anderson RH 2013 Methodological review of ventricular anatomy – the basis for understanding congenital cardiac malformations. *J Cardiovasc Transl Res* 6:145–54.
- Uysal F, Bostan OM, Semizel E et al 2014 Congenital anomalies of coronary arteries in children: the evaluation of 22 patients. *Pediatr Cardiol* 35:778–84.